climate risk in vulnerable mountain regions of the world

Day 2 (27 May)

Jakob Steiner

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fidelsteiner.github.io

Course content

22 May (1400 – 1700)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

27 May (0900 – 1500)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

28 May (0900 – 1500)

Science to policy

Serious game

29 May (0900 – 1500)

Serious game – course wrap up

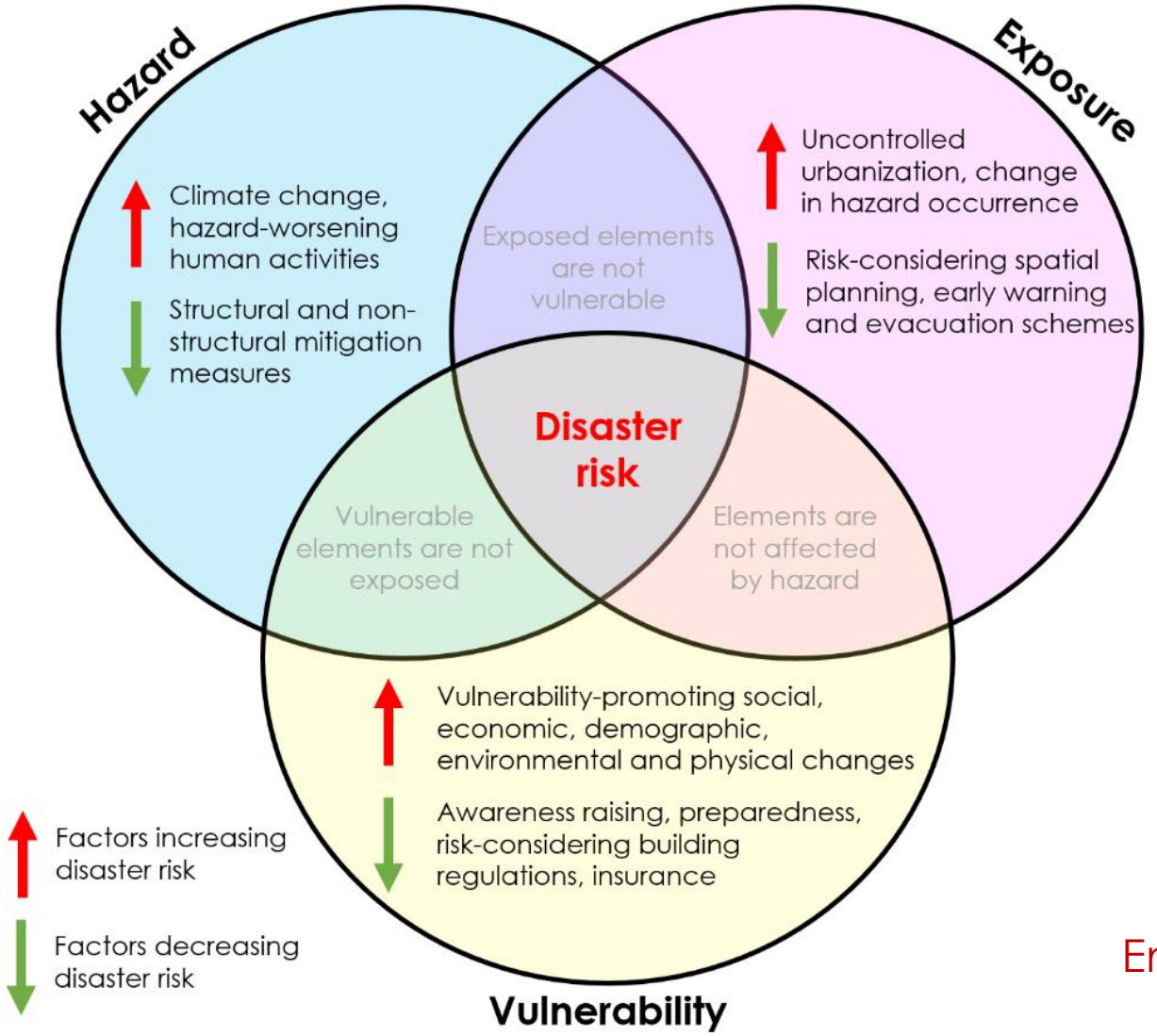
Breakout discussions

In your view and based on your experience

- what are likely the most pressing issues that are being discussed at the global stage, when it comes to addressing risks in mountain regions? This can be climate related or with respect to socio-economic challenges.

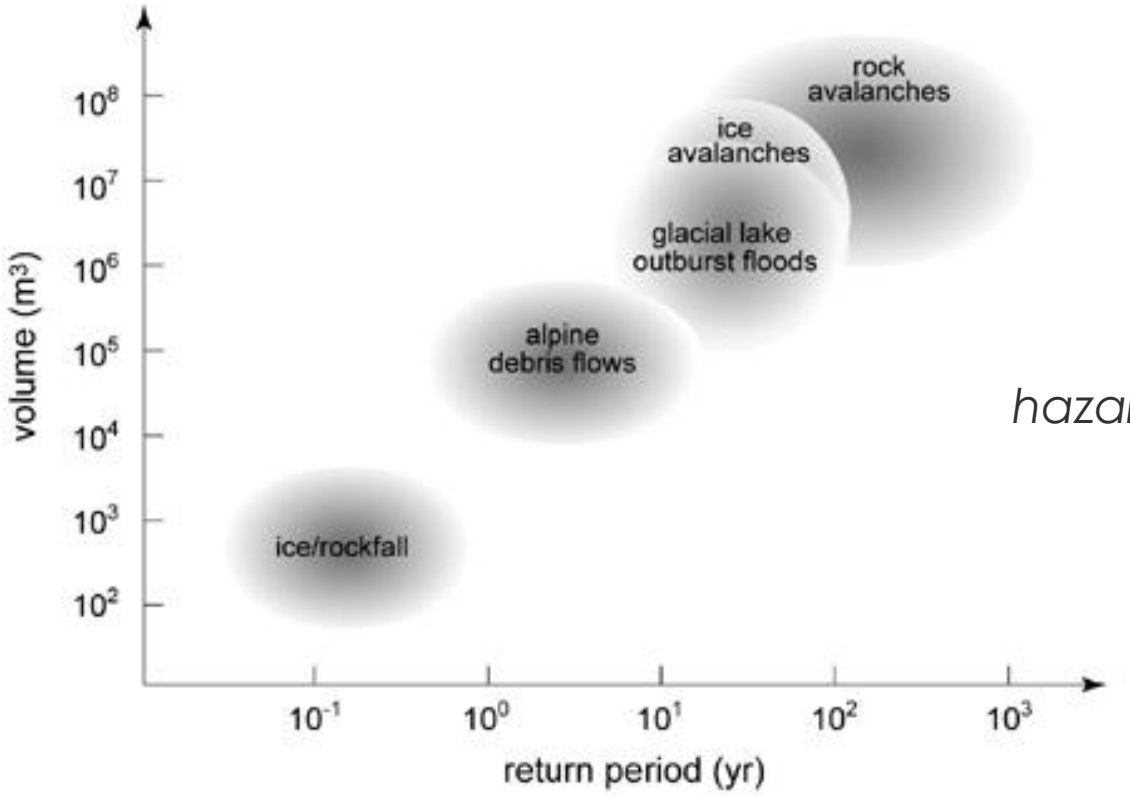
- thinking about the future, which risks do you think will change or may only appear in future with our knowledge on anticipated climatic and socioeconomic changes?

Risk in mountain contexts



Emmer, Steiner and Kalvoda (2024)

Return periods vs. volumes



hazard = f(probability, intensity)

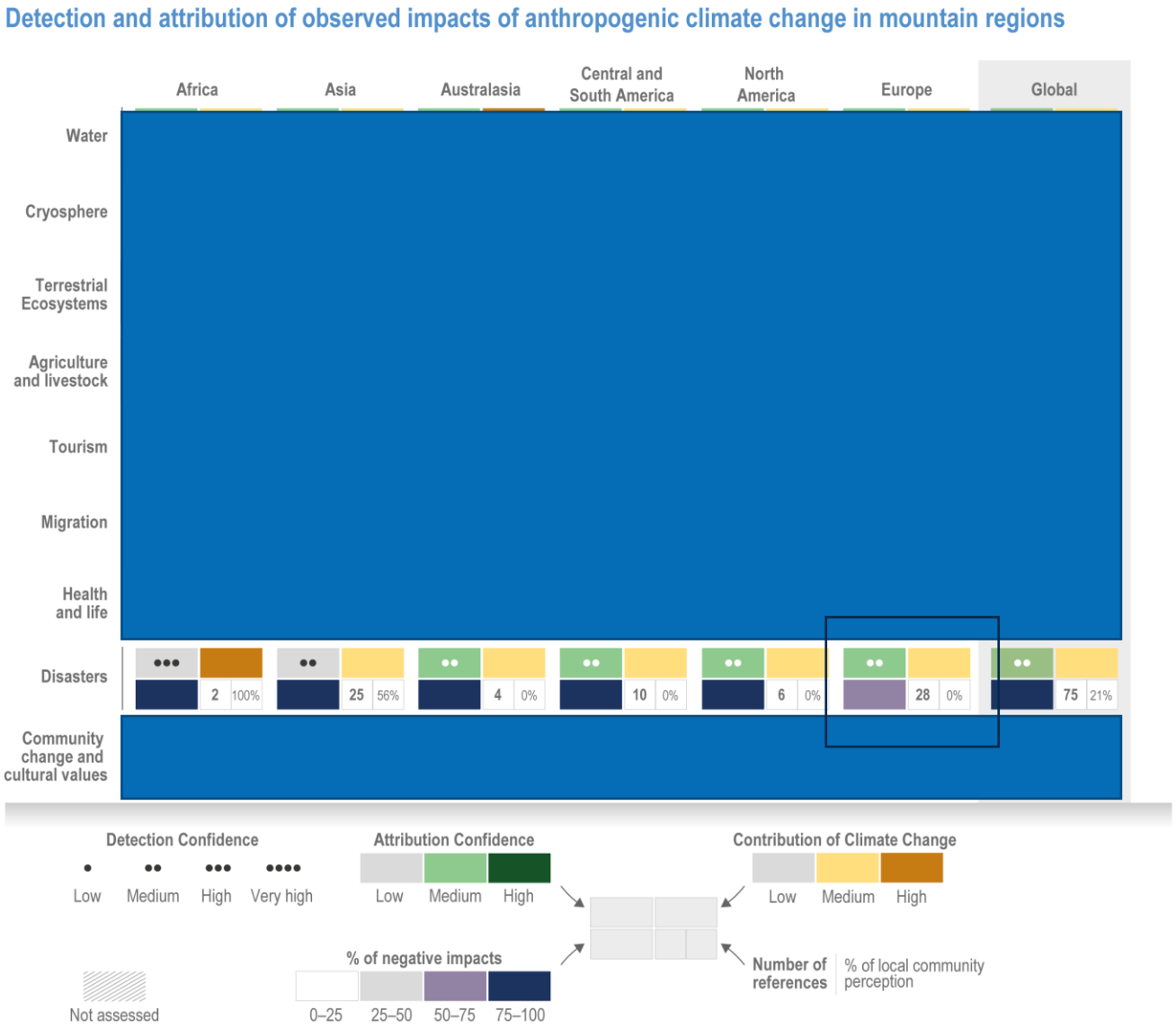
Geology Today, Volume: 28, Issue: 3, Pages: 98-104, First published: 25 May 2012, DOI: (10.1111/j.1365-2451.2012.00836.x)

IPCC AR6 WGII

Summary of research on mountains as captured by the IPCC report (AR6)

Compilation of the essence of our current state of knowledge

Approved by governments (IPCC process)

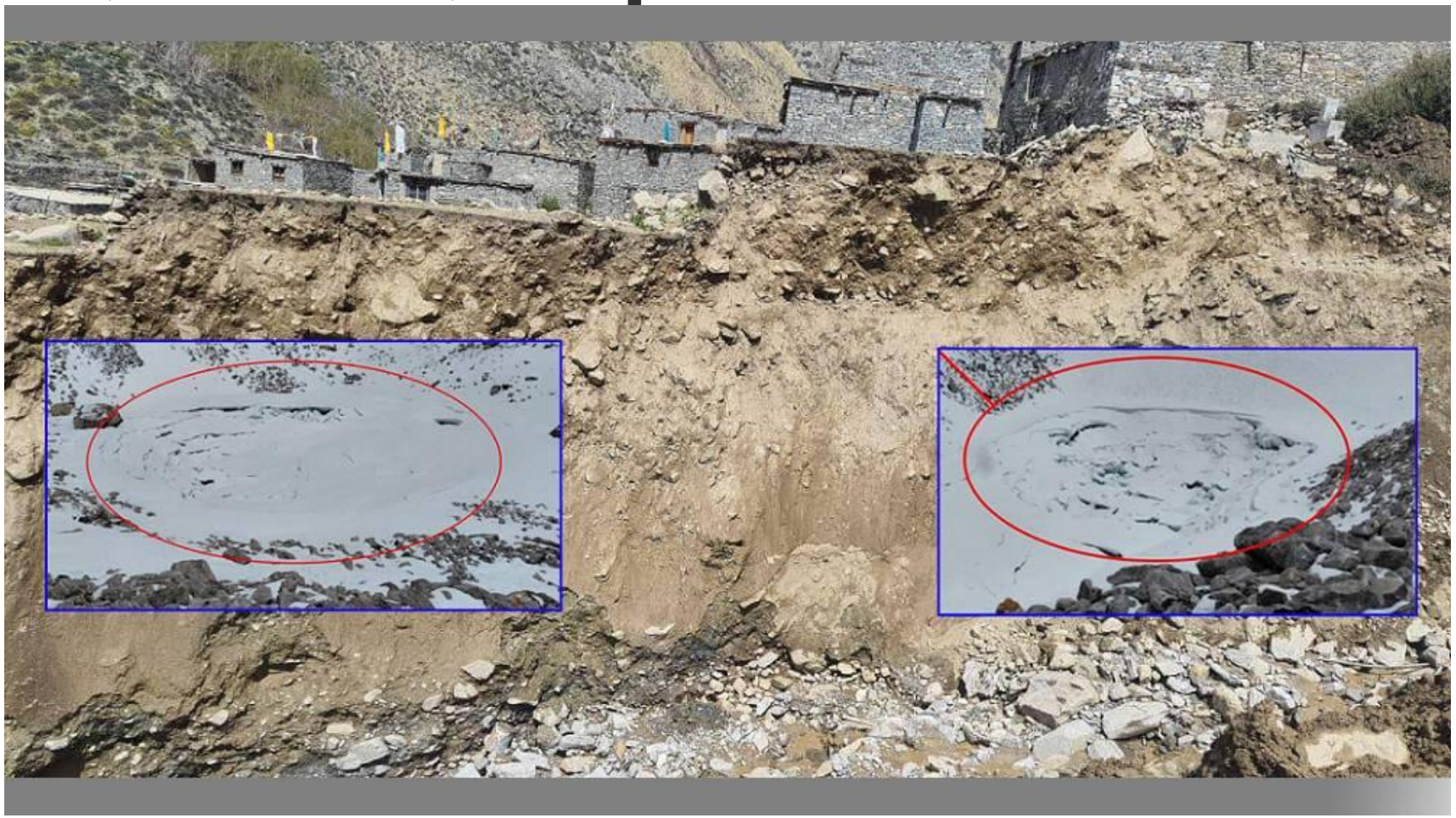


A few global examples

	Glacier collapse	GLOF	GLOF/Debris flow	Drought	Ageing
Location	Blatten (Switzerland)	Til (Nepal)	Taghanak (Afghanistan)	Kabul (Afghanistan)	Alps (Italy)
Risk type	Slow to fast onset (2013 - May 2025)	Fast onset (May 2025)	Fast onset (June 2025)	Slow onset (2010 -)	Slow onset (1980s? -)
Population	290	30	unknown	5 million	ca 14 million
Elevation	1500 m	4100 m	2200 m	1800 m	0 – 2100 m
Fatalities	1	0	unknown	unknown	-

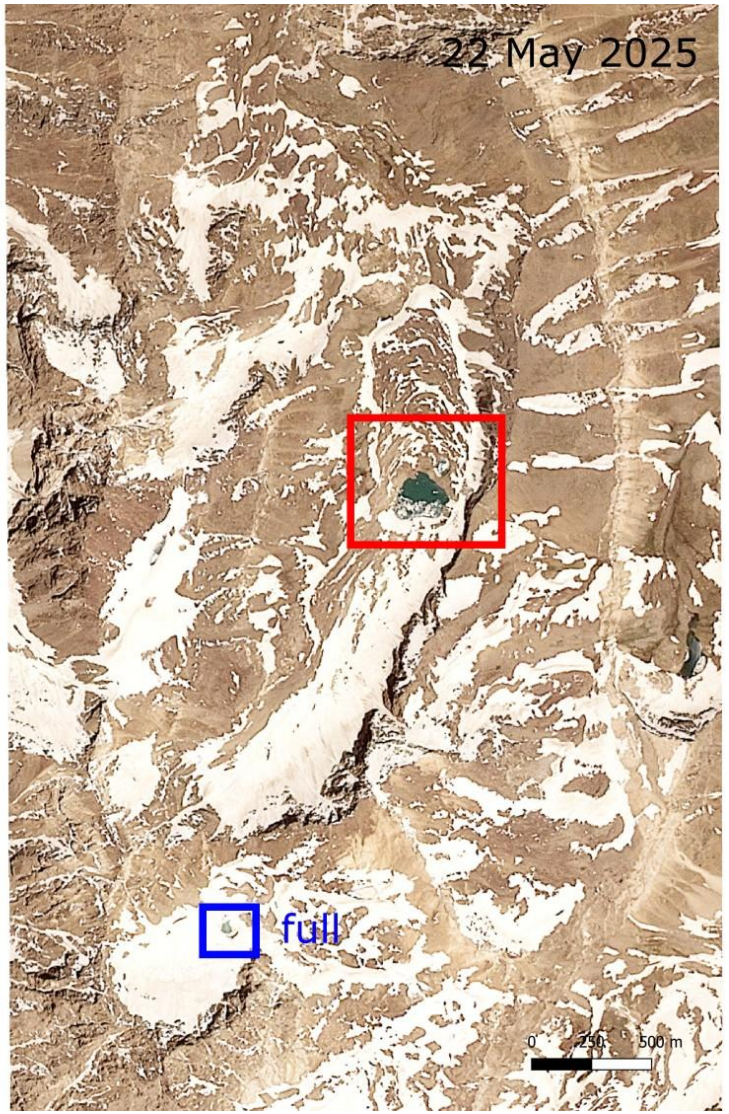
Blatten, Lötschental, Switzerland

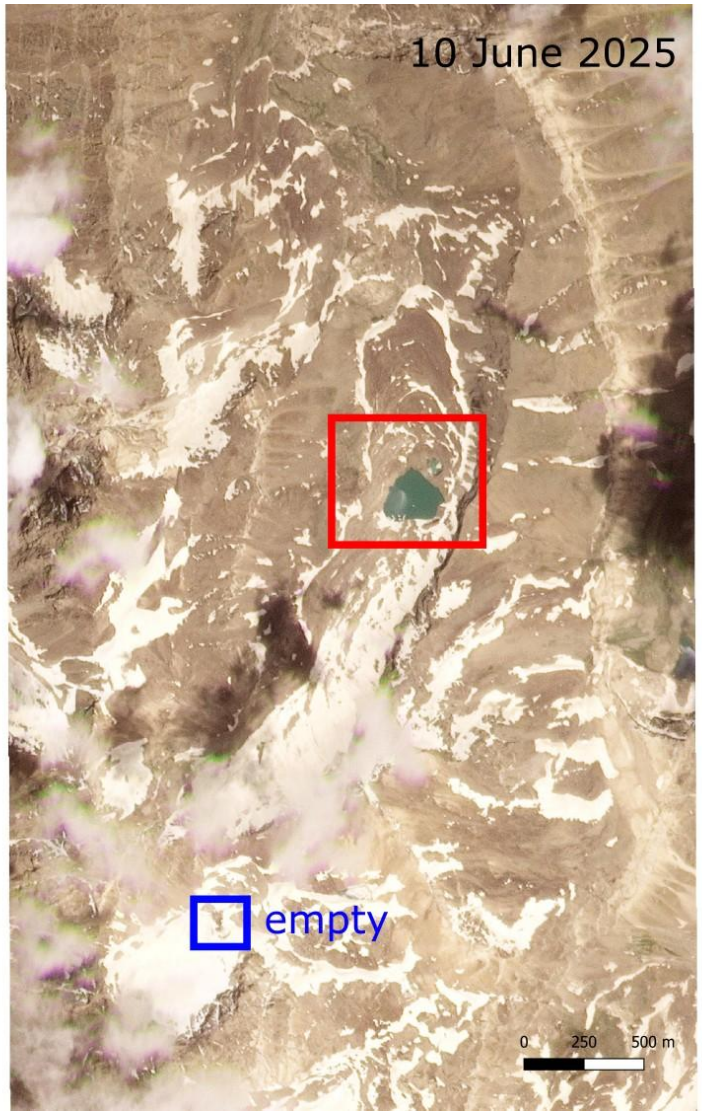
Til, Humla, Nepal

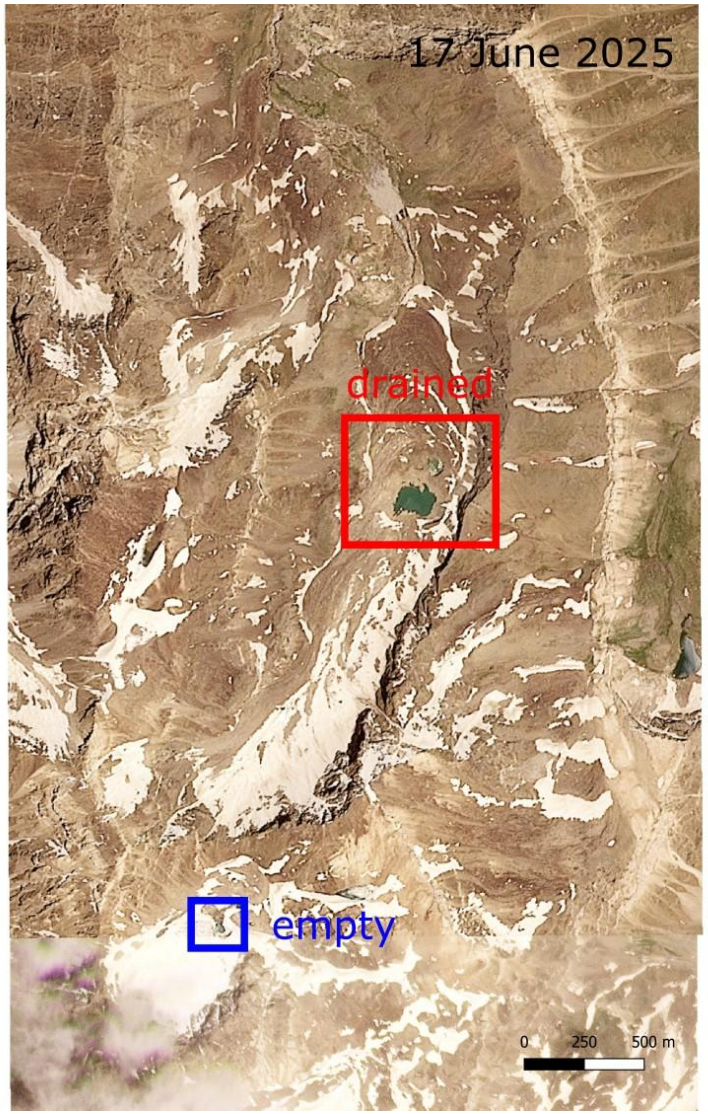


https://en.nepalkhabar.com/news/detail/13863/

Taghanak, Andarab, Afghanistan



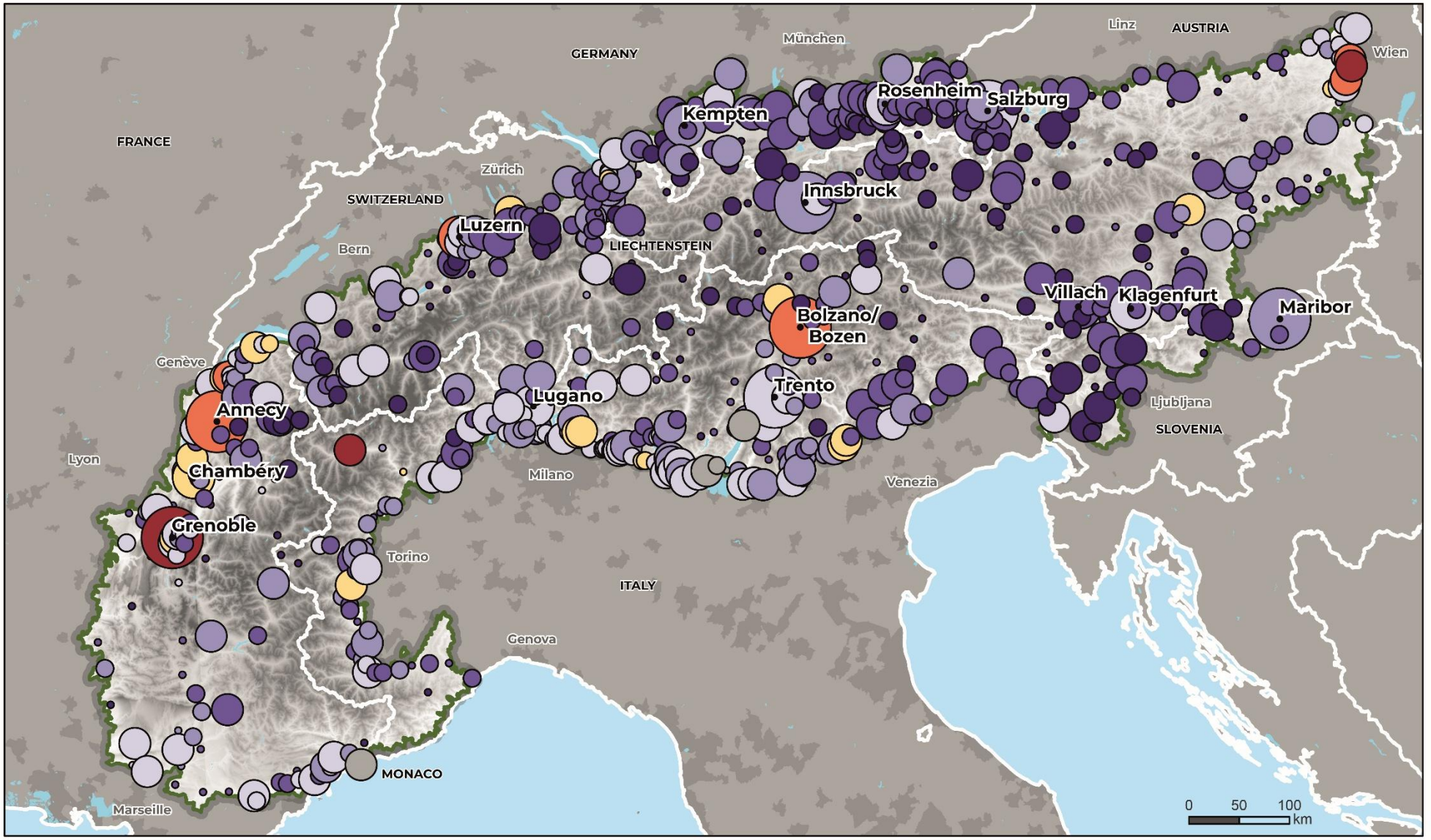




Kabul, Afghanistan

Alps

https://alpinetowns.alpconv.org/map-download/



Water use index

Population of Alpine towns (2015)

- > 100.000
- > 50.000 - 100.000
- > 10.000 - 50.000
- > 5.000 - 10.000
- 3.000 - 5.000

Water extraction from available water in %

- ≤ 1
- > 1 - 5
- > 5 - 10
- > 10 - 20
- > 20 - 40 (water scarcity)
- > 40 - 60 (water stress)
- > 60 (heavy water stress)
- no data

FAU

9th Report on the State of the Alps
Alpine Towns

Data: AlpES 2021 & FAU, Alps 2050, 2018
Layout: © EuroGeographics for the administrative boundaries; IGEAT for the Morphological Urban Areas; Data funded under GMES preparatory action 2009 on Reference Data Access by the European Commission, DG Enterprise and Industry for the Alpine raster data.
Concept & Cartography: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, **2022.**

Alpine Convention Perimeter
Urbanised areas outside the Alps (MUA)

Alps

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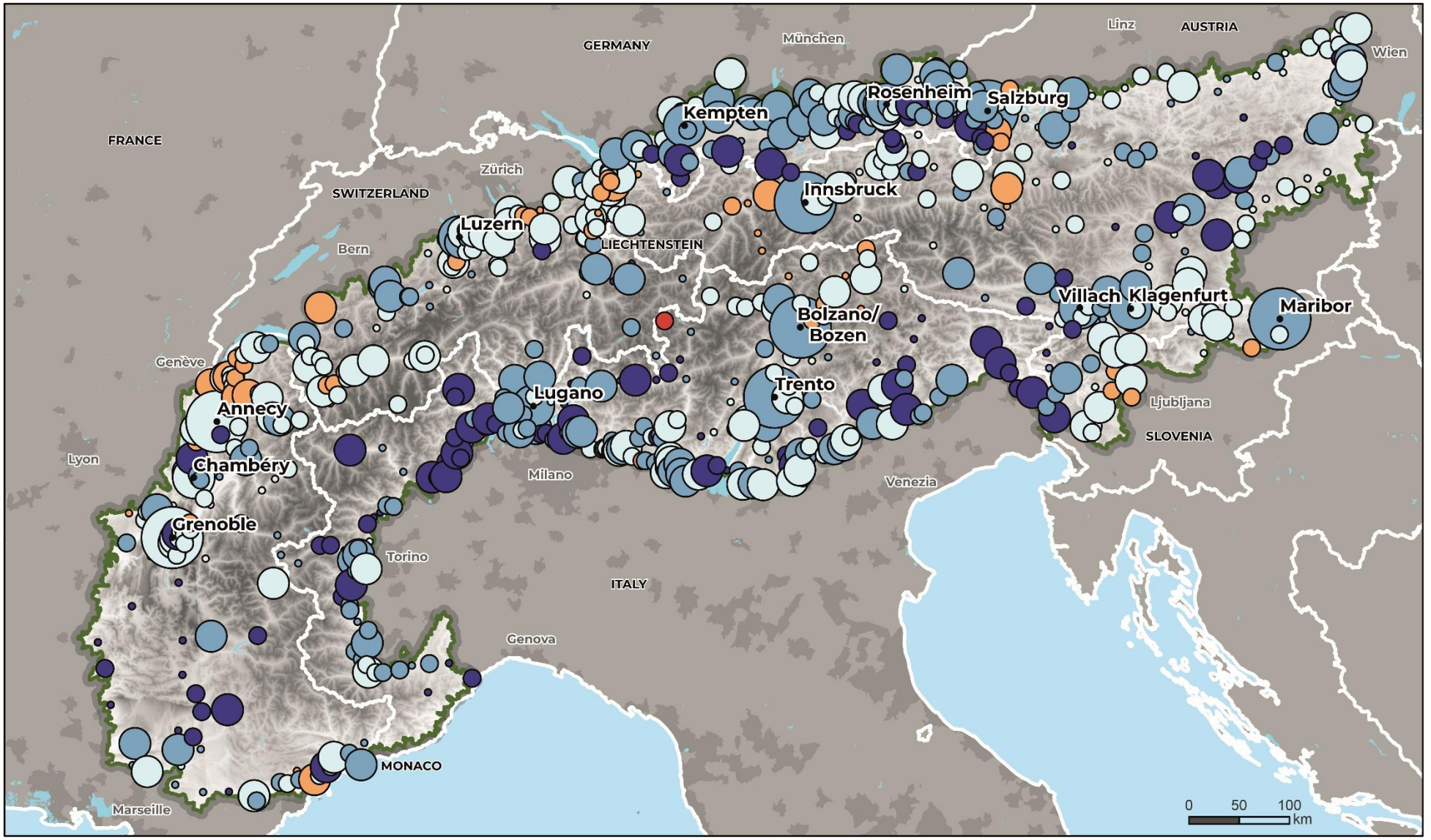

Ageing Index

Population of Alpine towns (2015)

- > 100.000
- > 50.000 - 100.000
- > 10.000 - 50.000
- > 5.000 - 10.000
- 3.000 - 5.000

Persons over 65 / Persons between 0-14 *100 (2015)

- ≤ 50
- > 50 - 100
- > 100 - 150
- > 150 - 200
- > 200

9th Report on the State of the Alps Alpine Towns

Data: Eurac Research & FAU, Alps 2050, 2018; National Statistical Offices, 2018.
Layout: © EuroGeographics for the administrative boundaries; IGEAT for the Morphological Urban Areas; Data funded under GMES preparatory action 2009 on Reference Data Access by the European Commission, DG Enterprise and Industry for the Alpine raster data.
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Mountain hazards – fast onset

Landslides

Debris flows

(Snow/Ice/Rock Ice-)Avalanches

(Glacial/Landslide dammed) Lake Outburst Floods (GLOFs/LLOFs)

Rockfall

Blizzards

Pluvial floods/intense rainfall

COVID

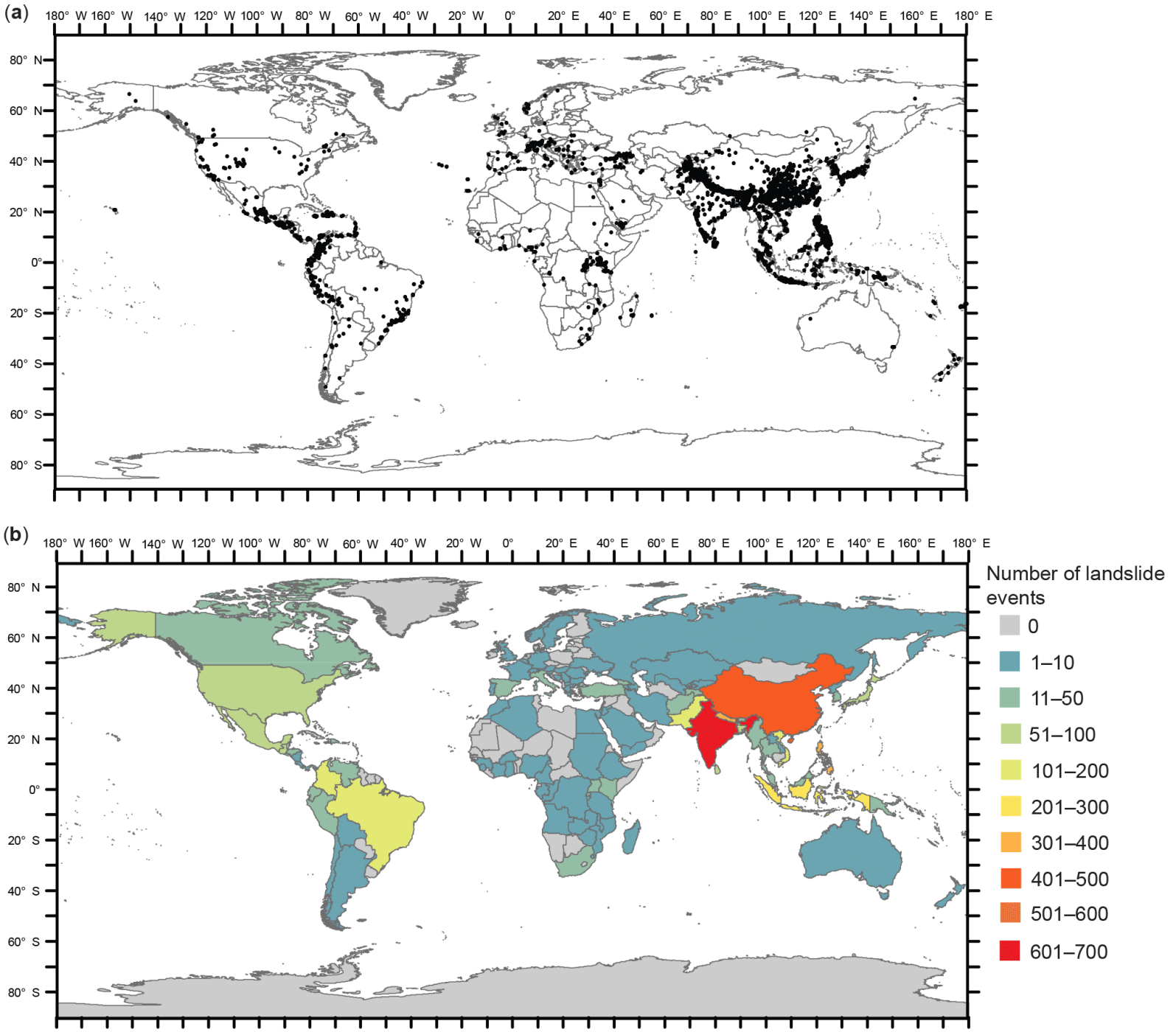
Landslides

2004 – 2016 a total of 4862 (non seismic-)landslides killed 55997 people

Rainfall a dominant driver but no clear signal of climate change

Indication that man made landslides are increasing

Froude and Petley (2018)



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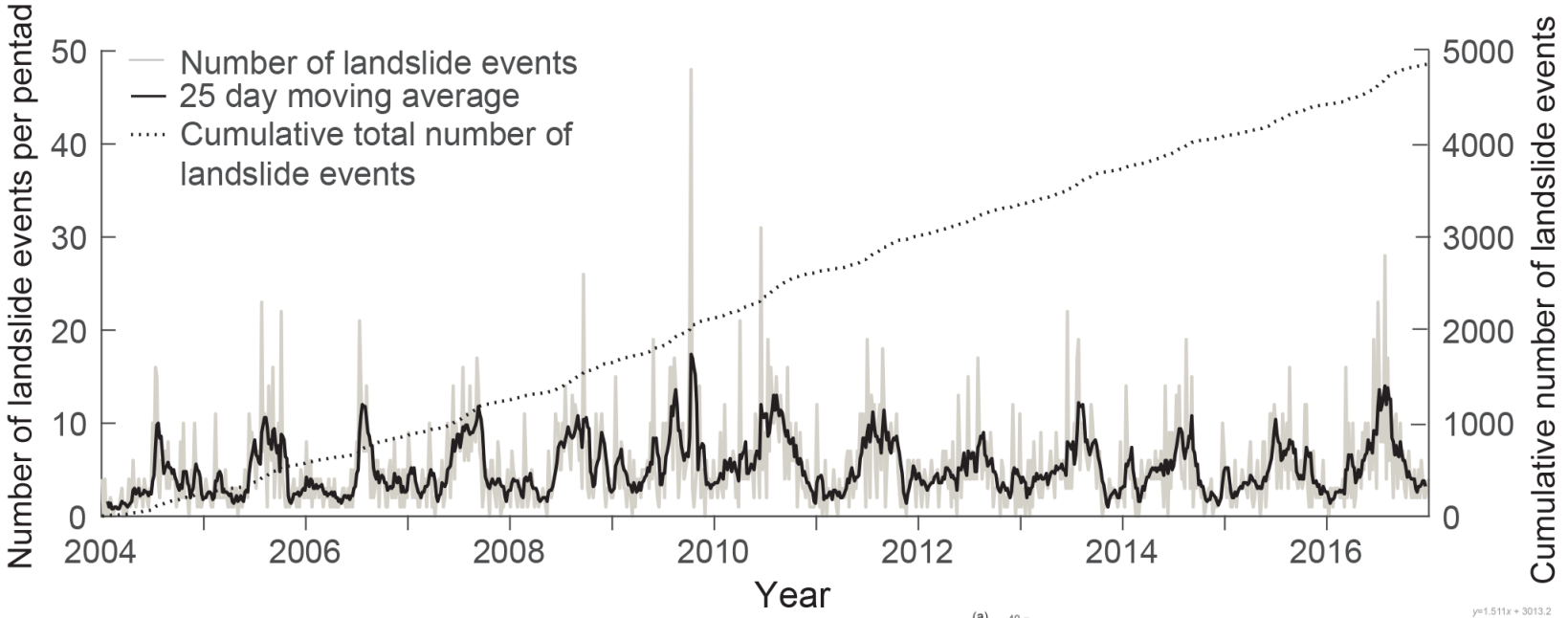

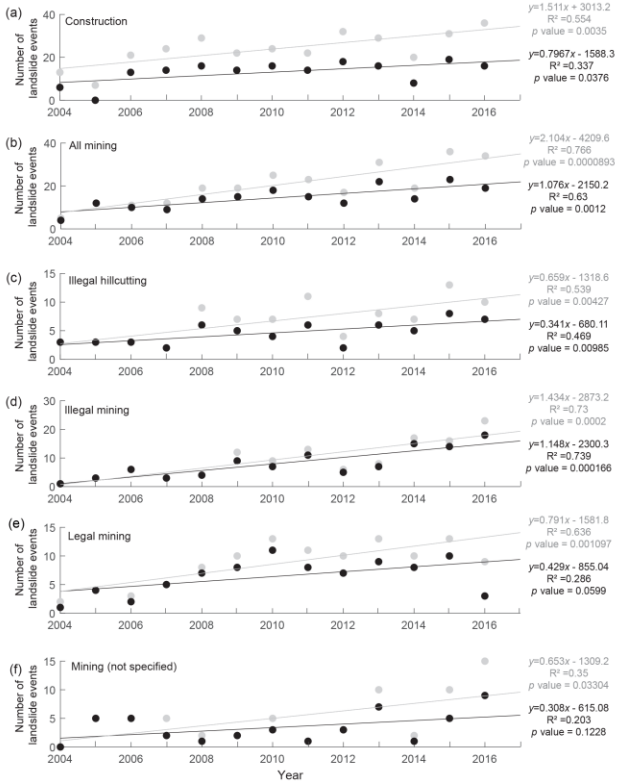

Froude and Petley (2018)

Landslides

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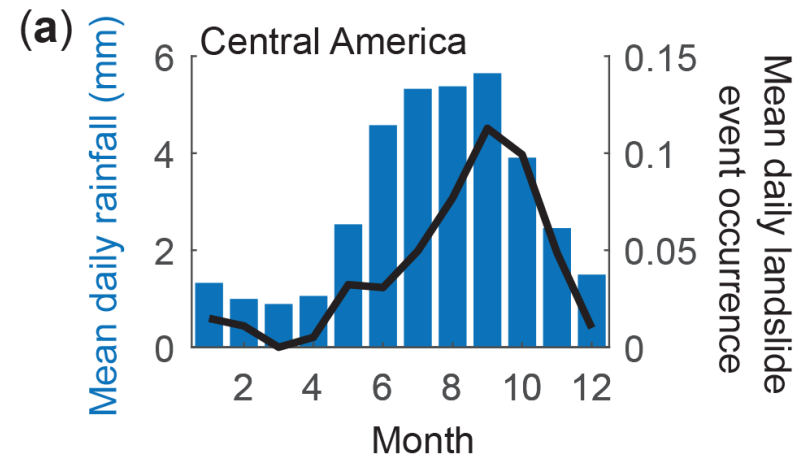
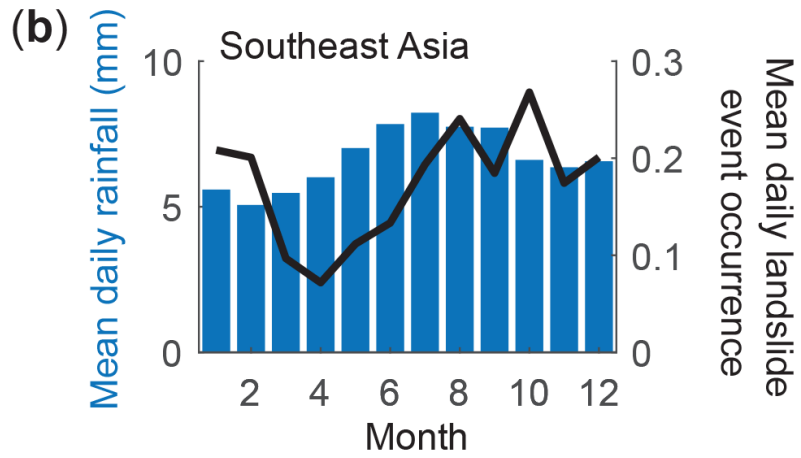
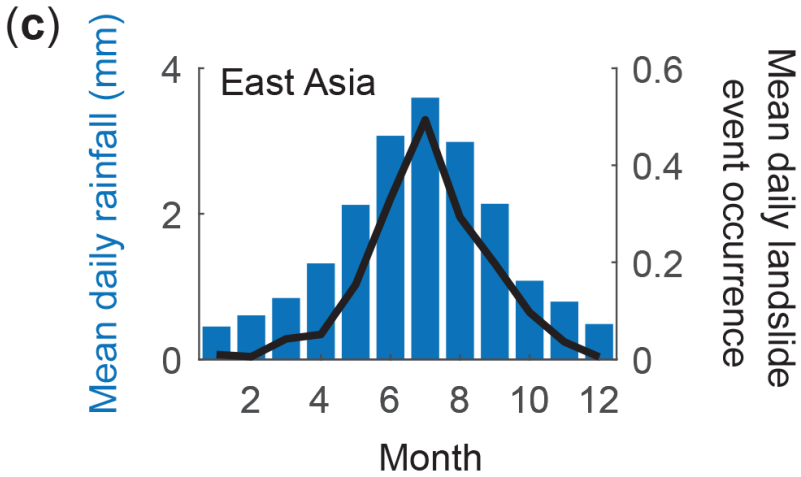
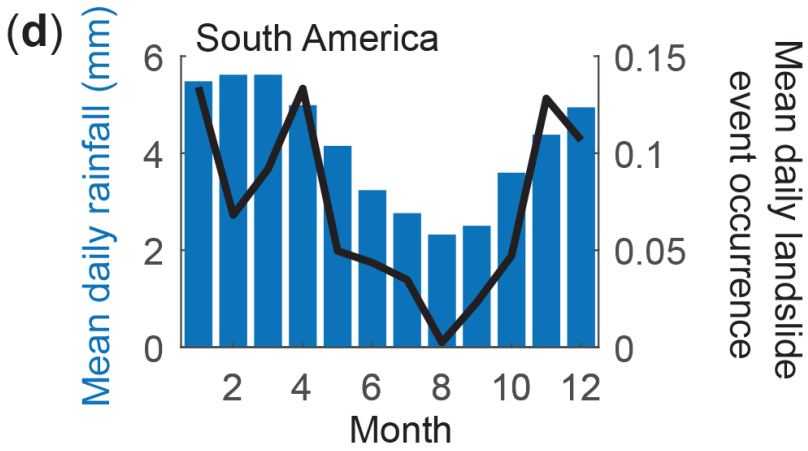
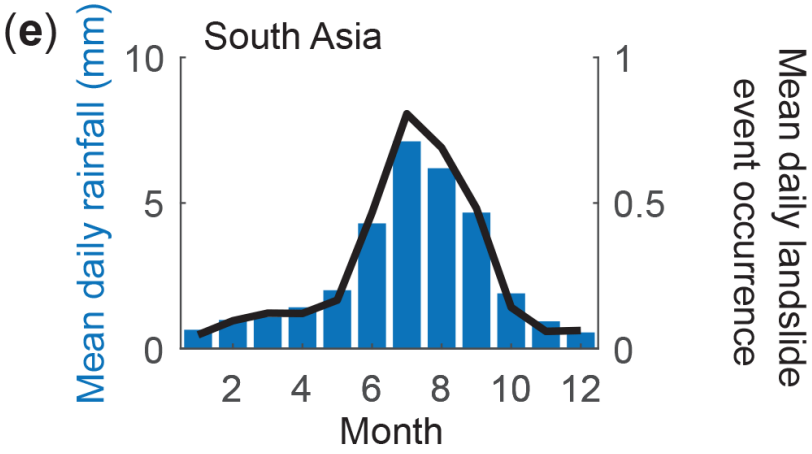
Froude and Petley (2018)



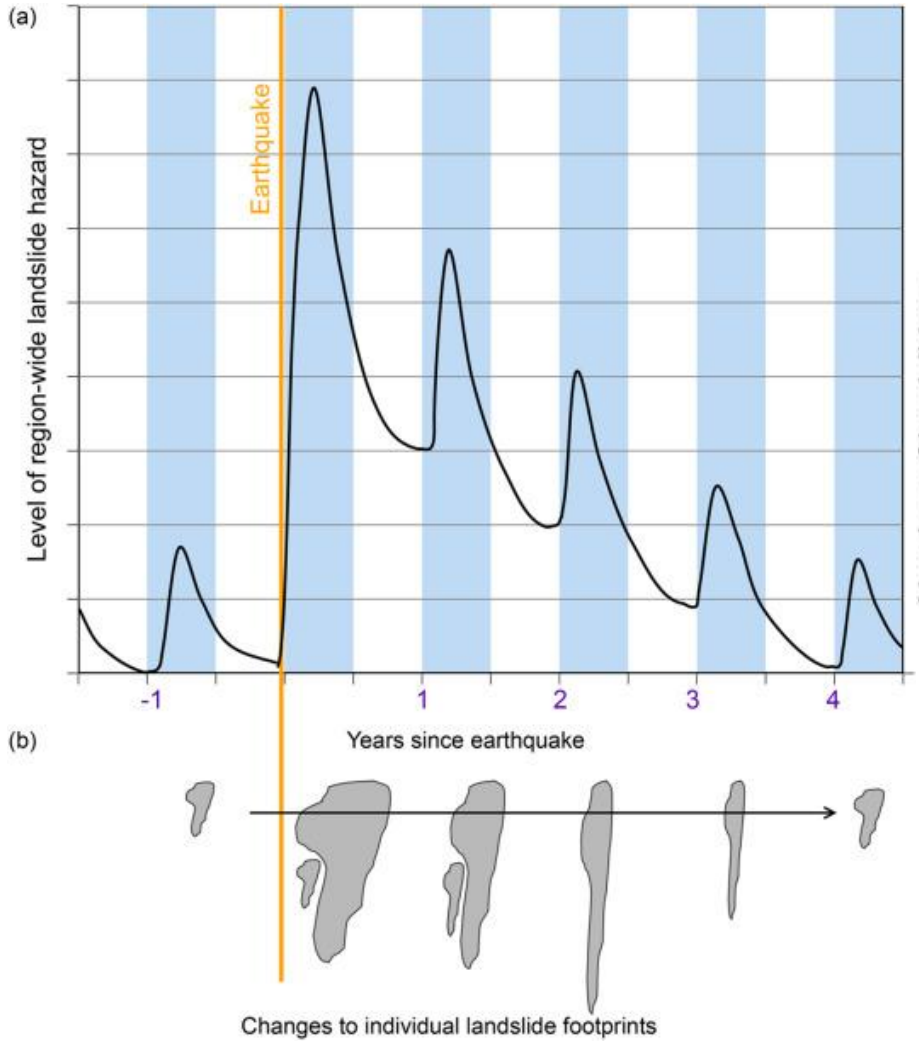


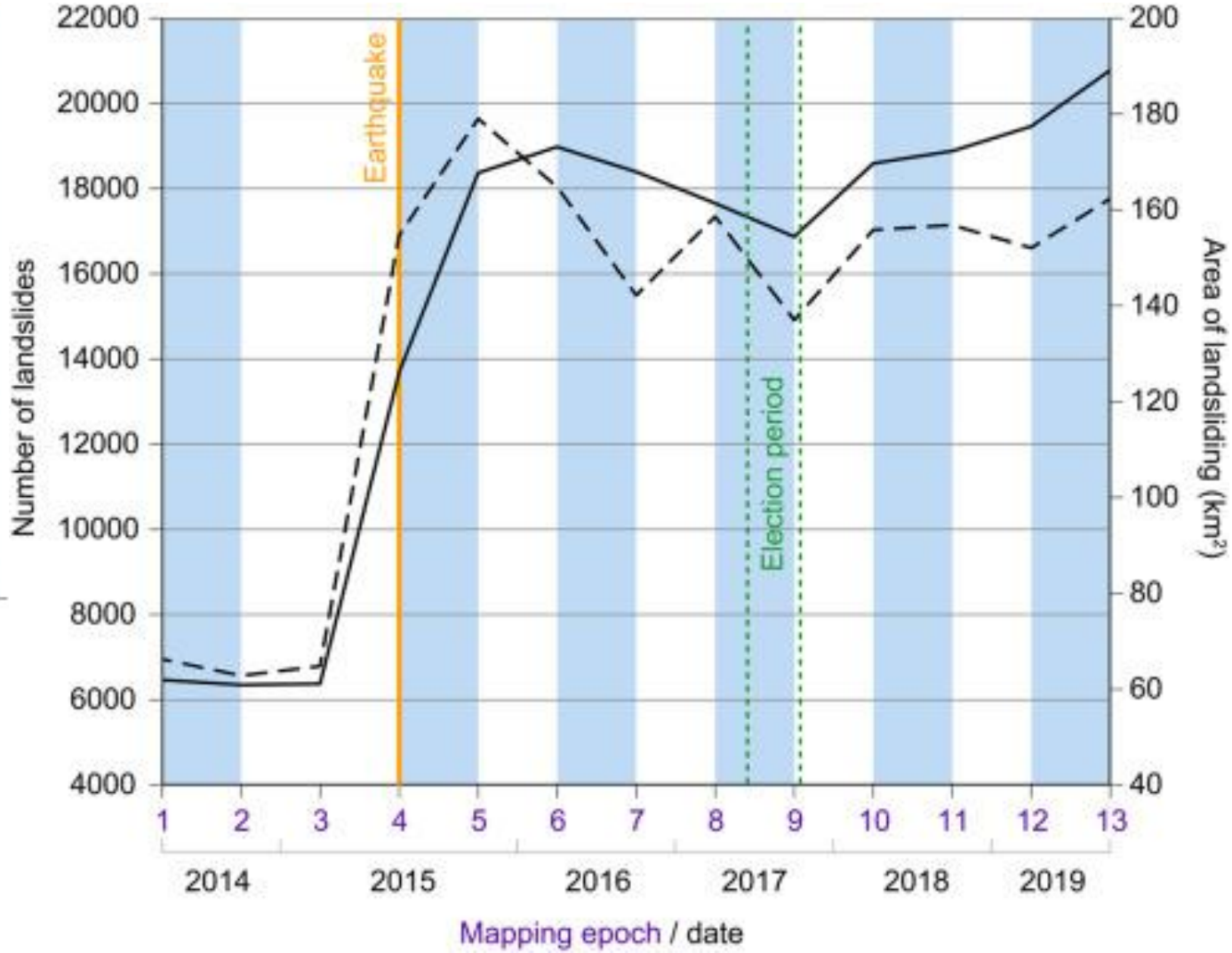






Landslides – expectation vs. reality



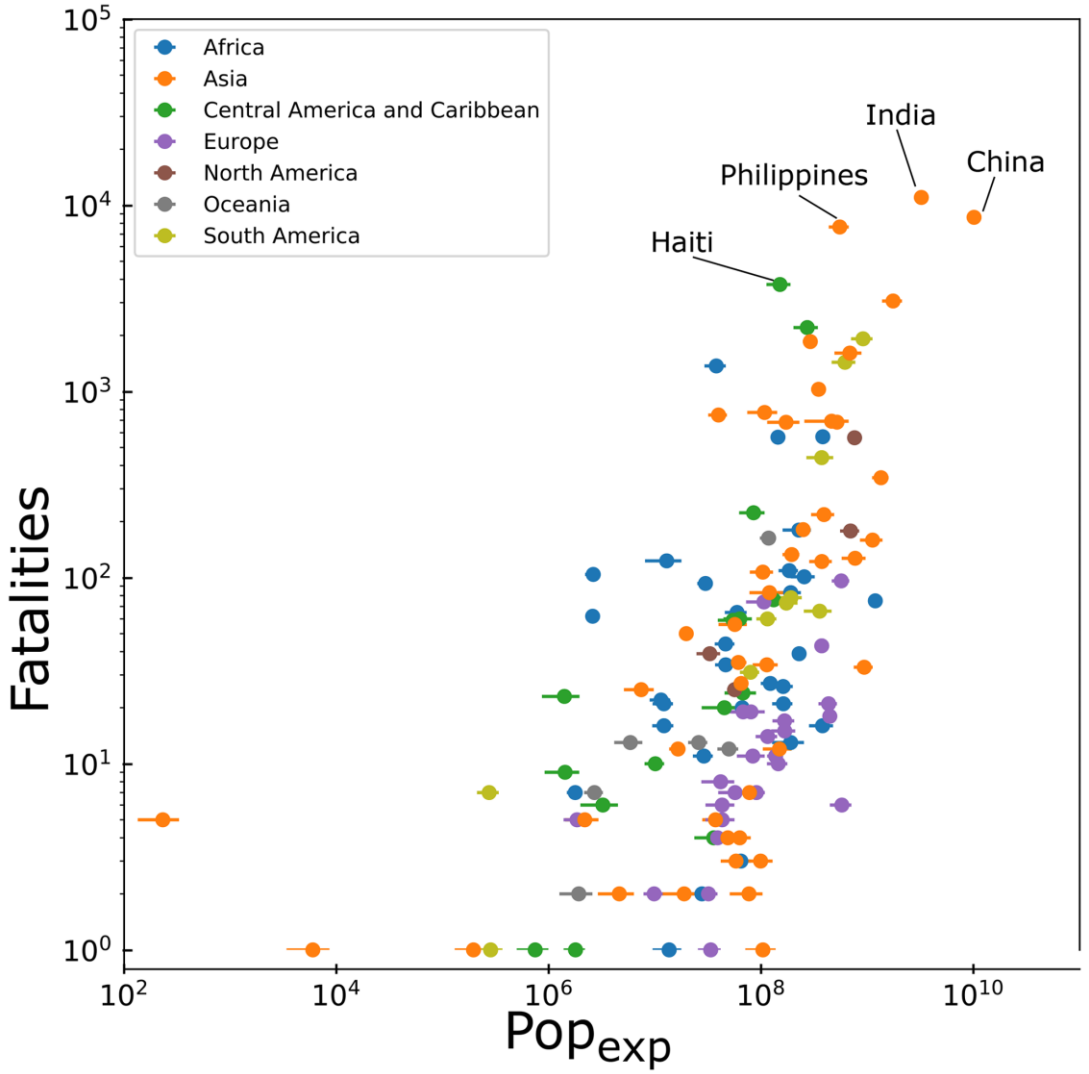


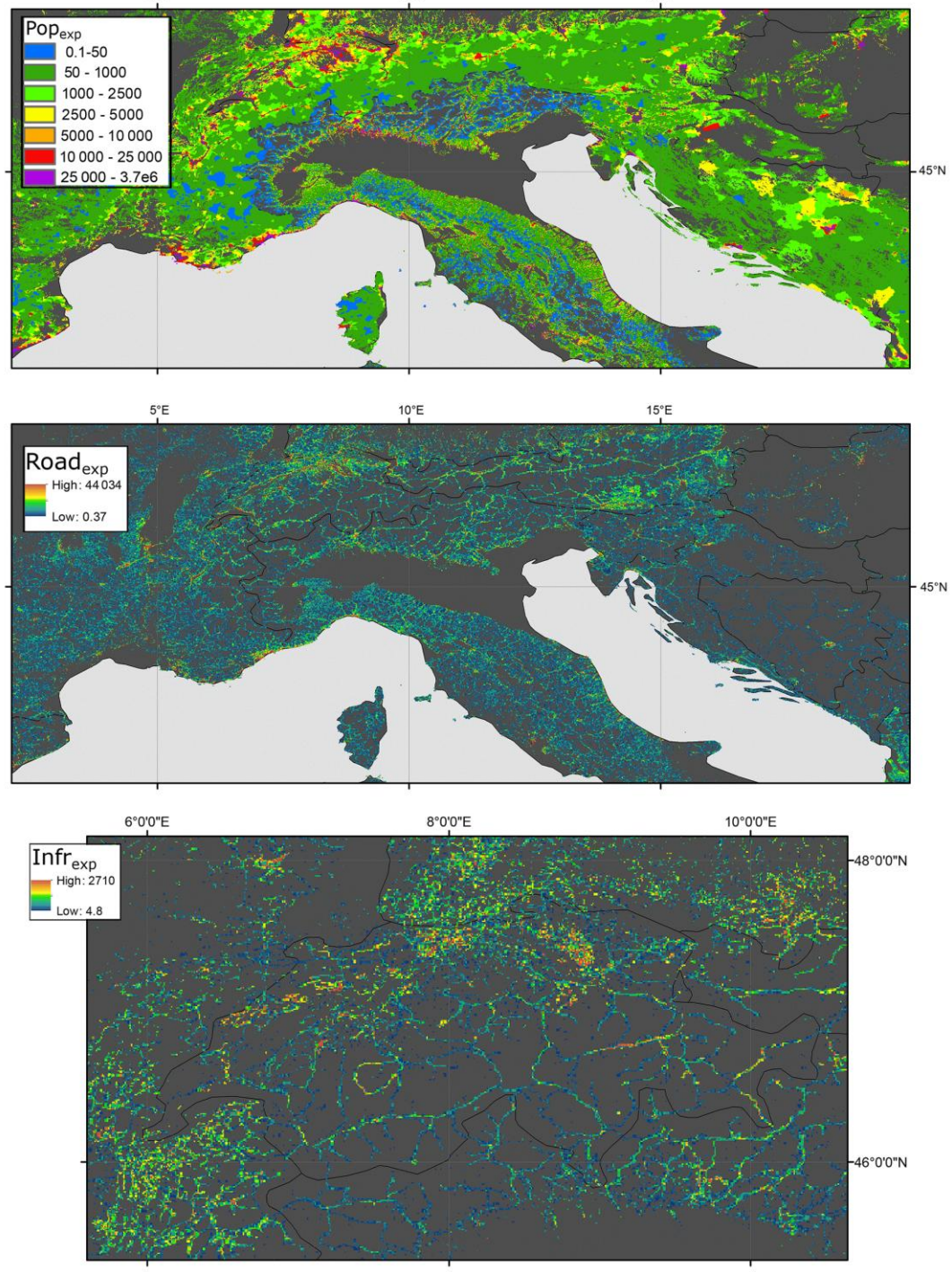
Rosser et al. (2019)

Landslides

Emmer, Steiner and Kalvoda (2024)

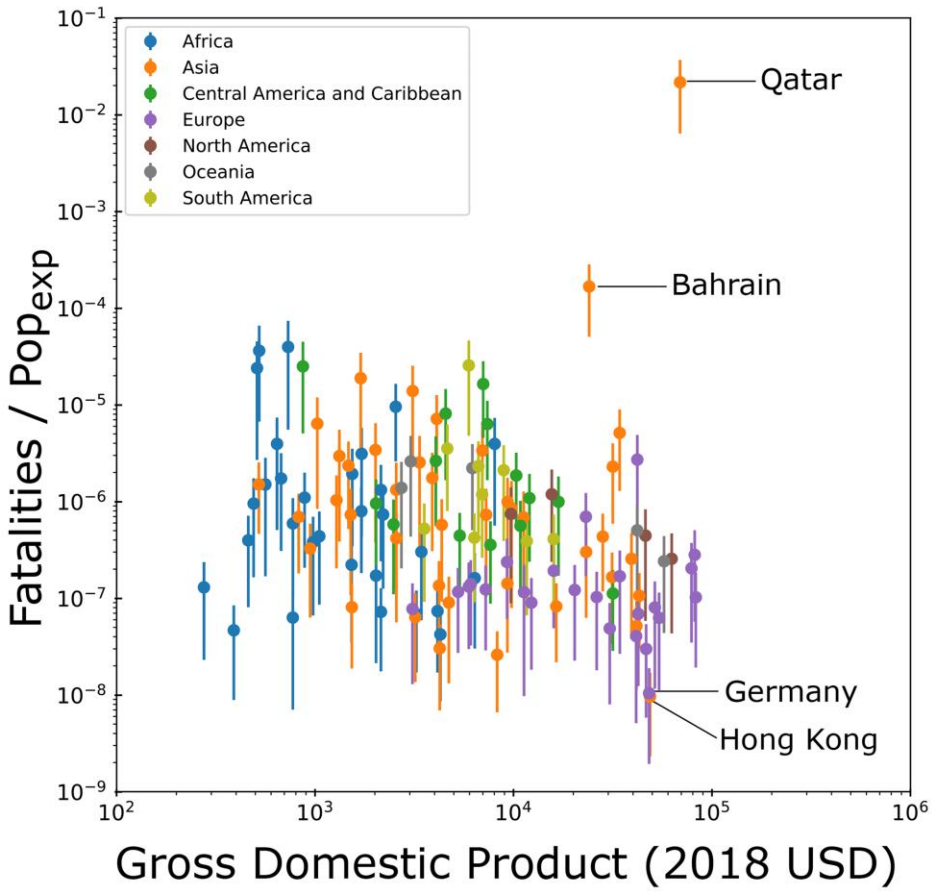
Landslide exposure

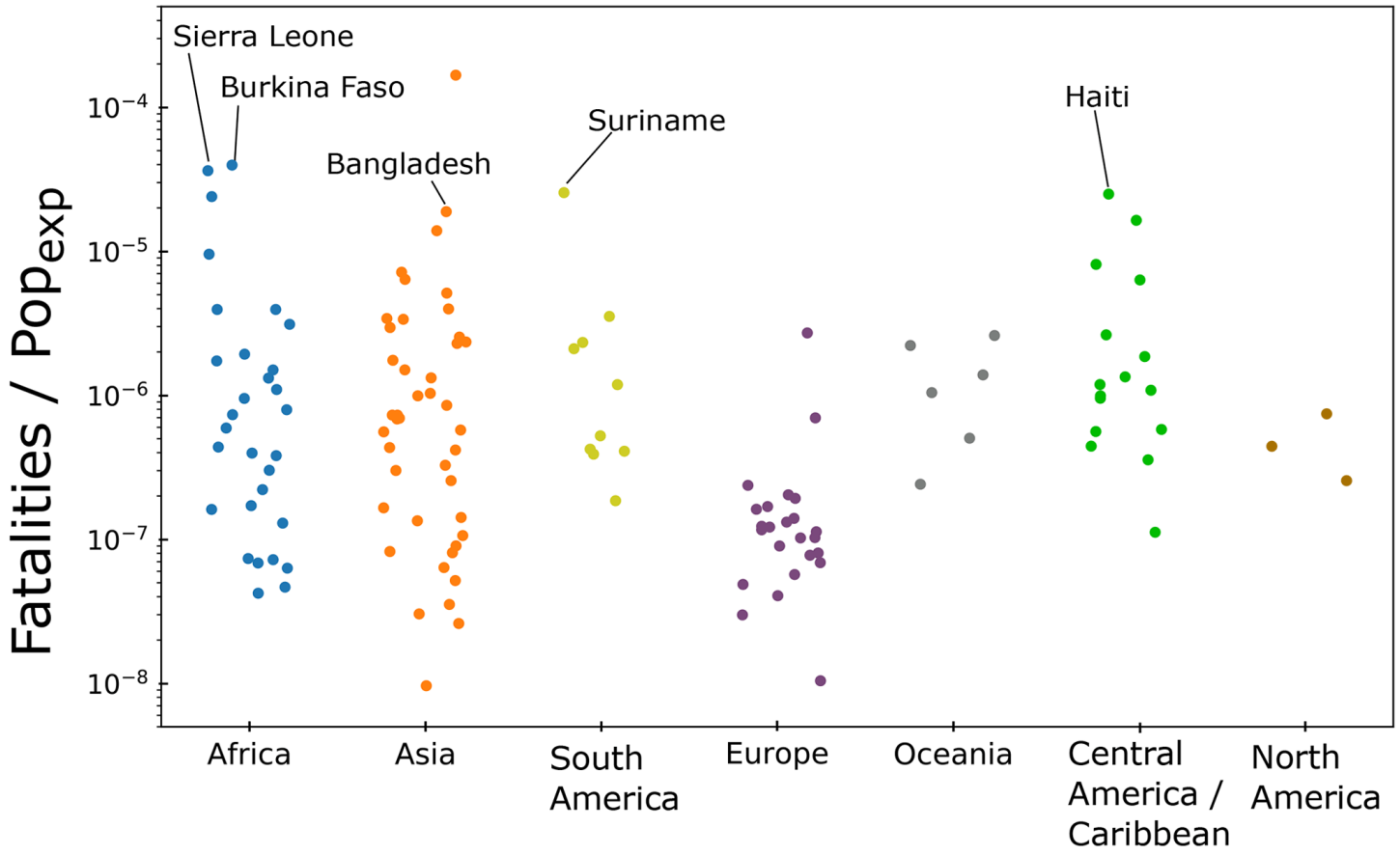




Emberson et al. (2020)

Landslide exposure





Emberson et al. (2020)

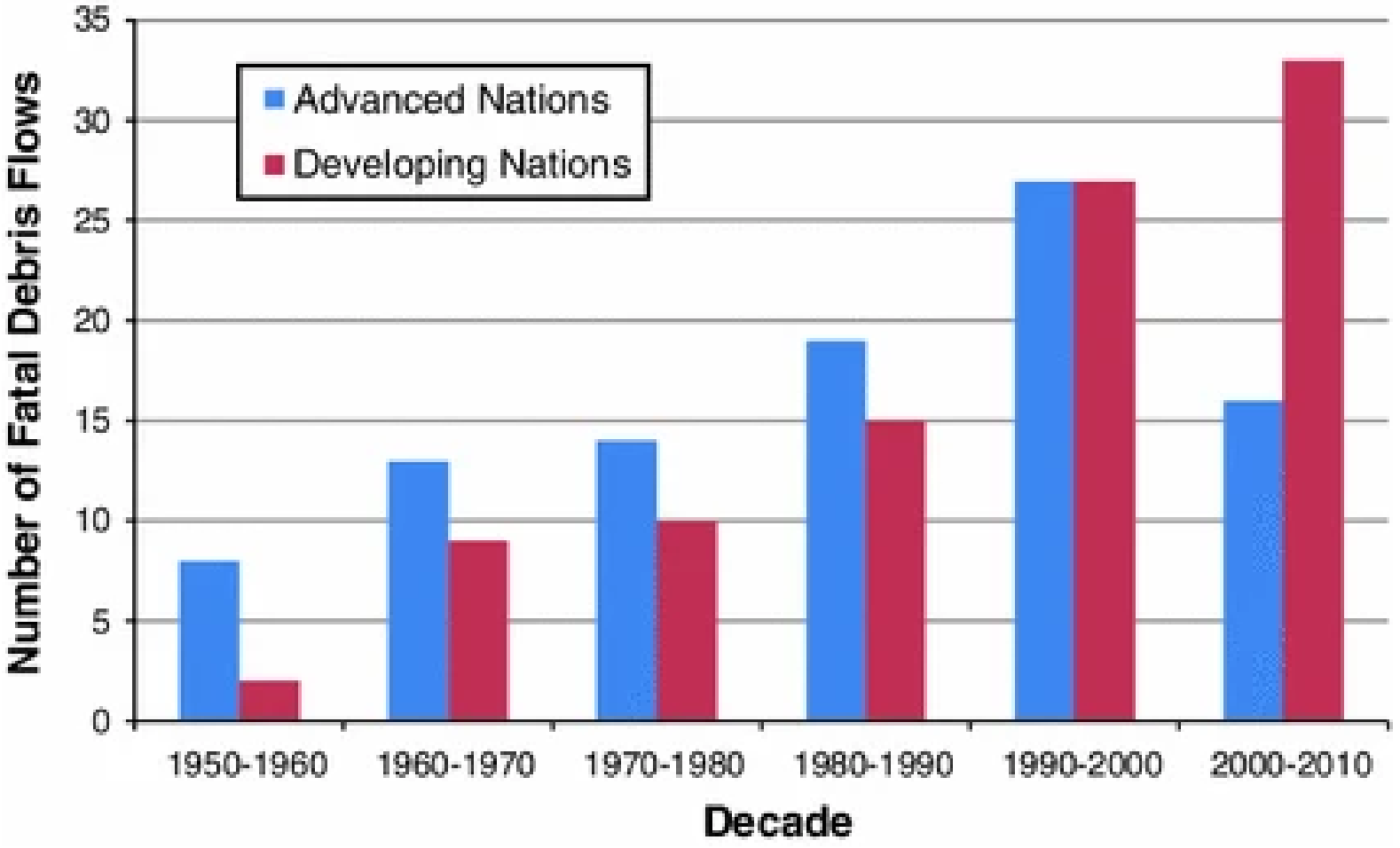
Debris flows

Debris flows

Debris flows

Between 1950 and 2011, 213 events caused 77 779 fatalities (365 event^{-1}, ~1300 yr^{-1})

Lacking comprehensive compilation

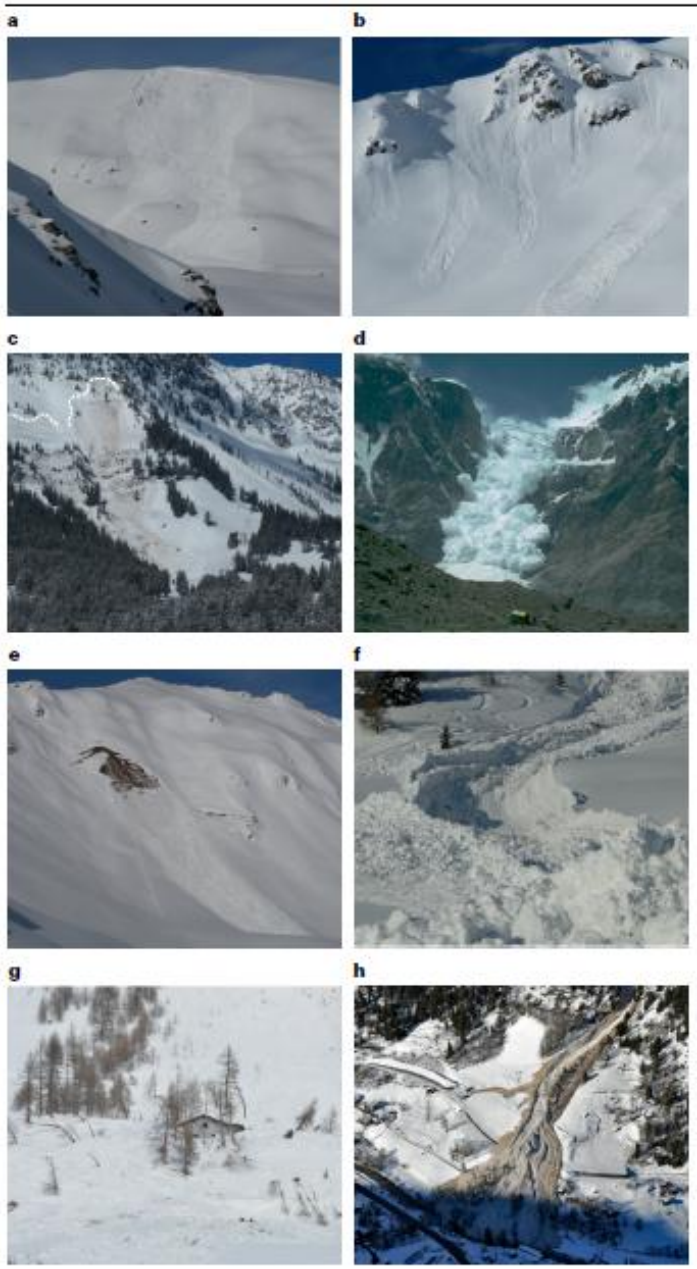


Dowling and Santi (2014)

Avalanches

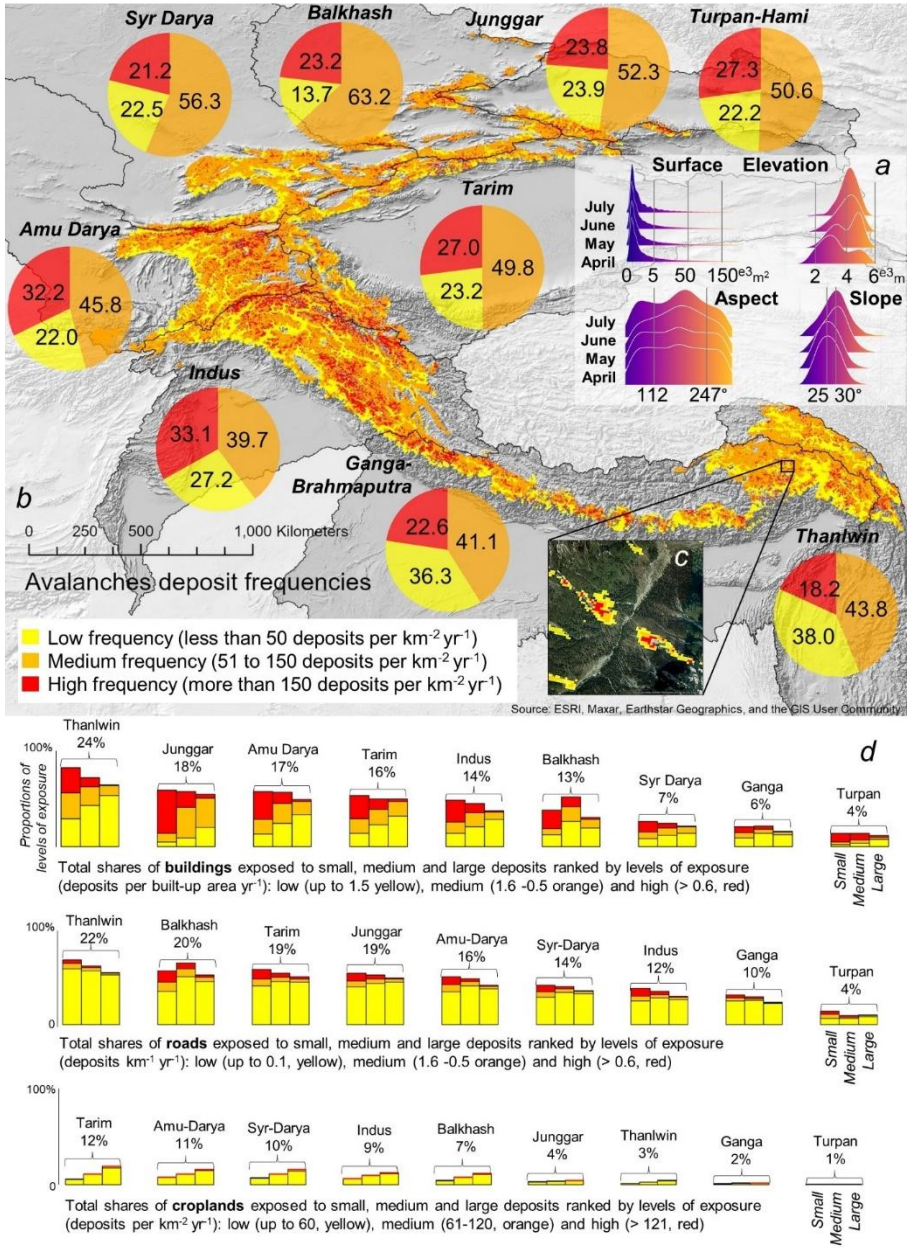
No global assessment but ca. 100 fatalities in Europe (mainly while doing sports) and 100 fatalities in High Mountain Asia (predominately non-sports related)

Climate change change signal but hard to trace



Eckert et al. (2024)

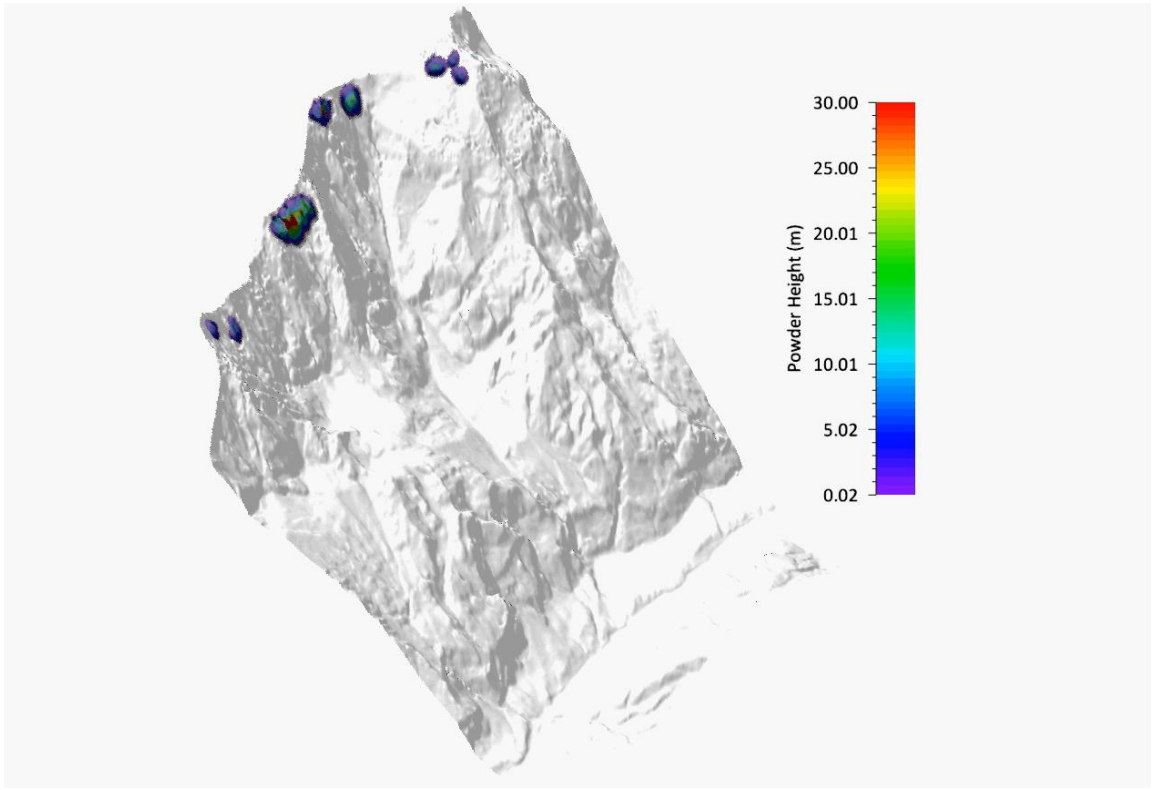
Acharya et al. (2023)



Caiserman et al. (2026)

Indigenous risk management

Acharya et al. (2023)

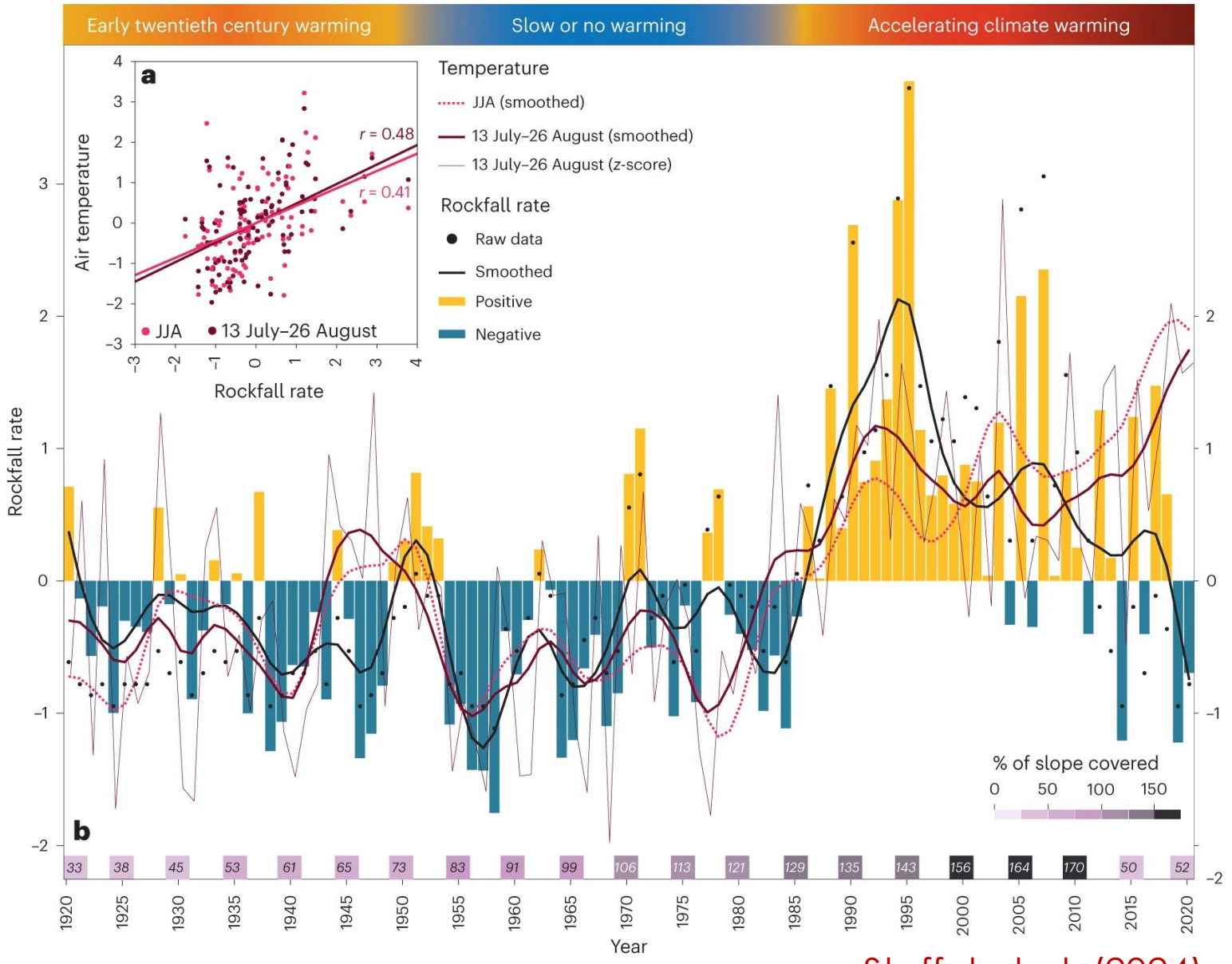


Zhuang et al. (2025)

Rockfalls

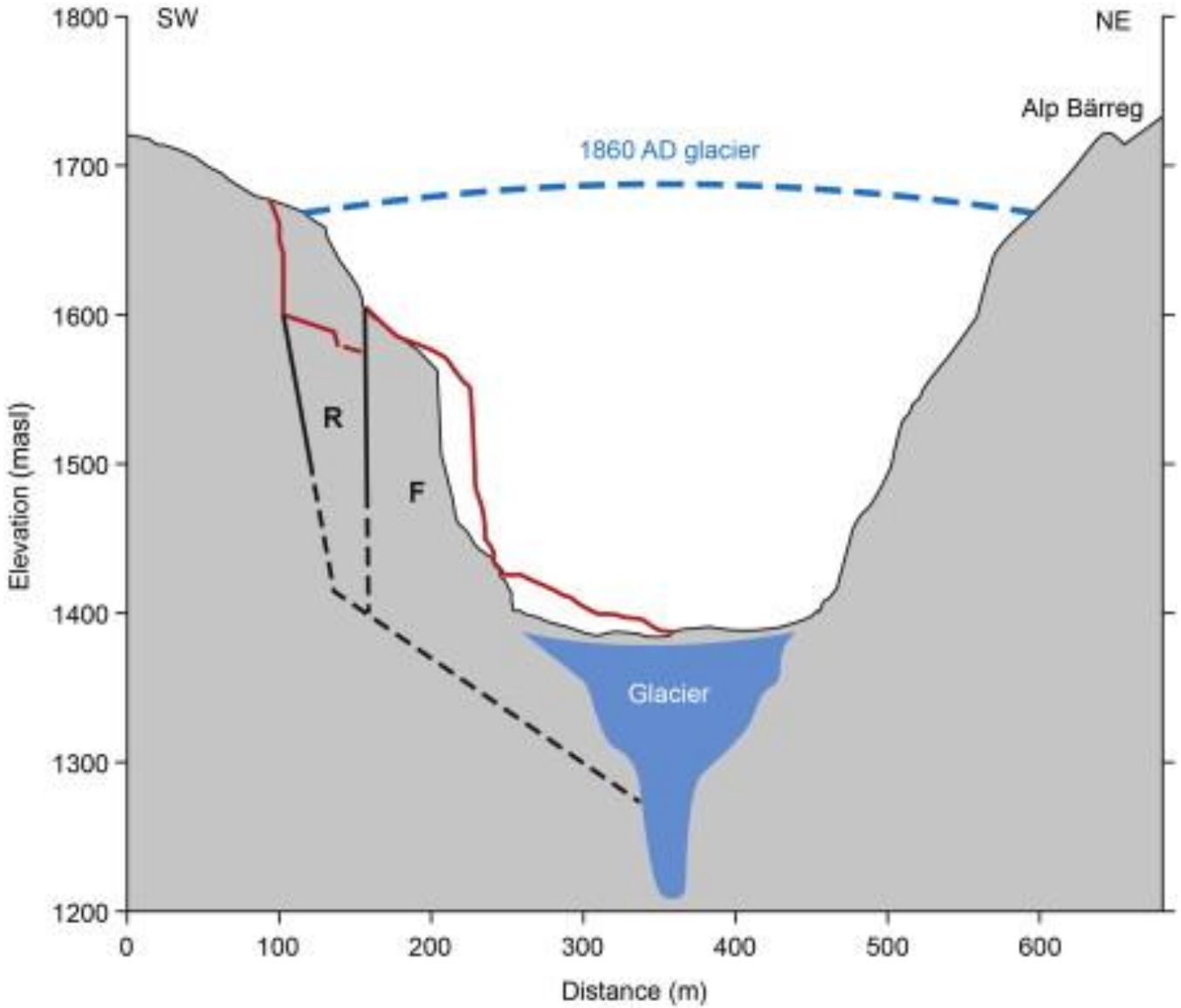
Research primarily in Switzerland and Germany

Clear indication that mountains are crumbling as temperatures rise



Stoffel et al. (2024)

Rockfalls



Deline et al. (2015)

Glacial lake outburst floods

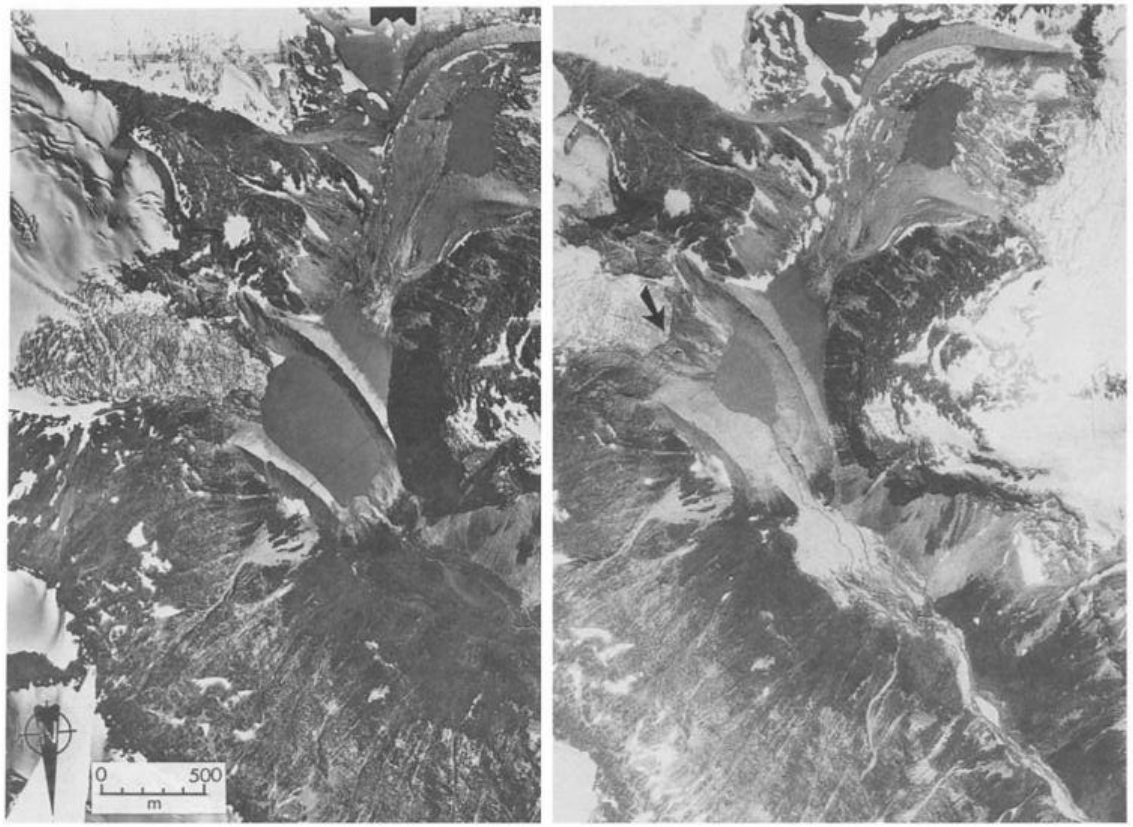
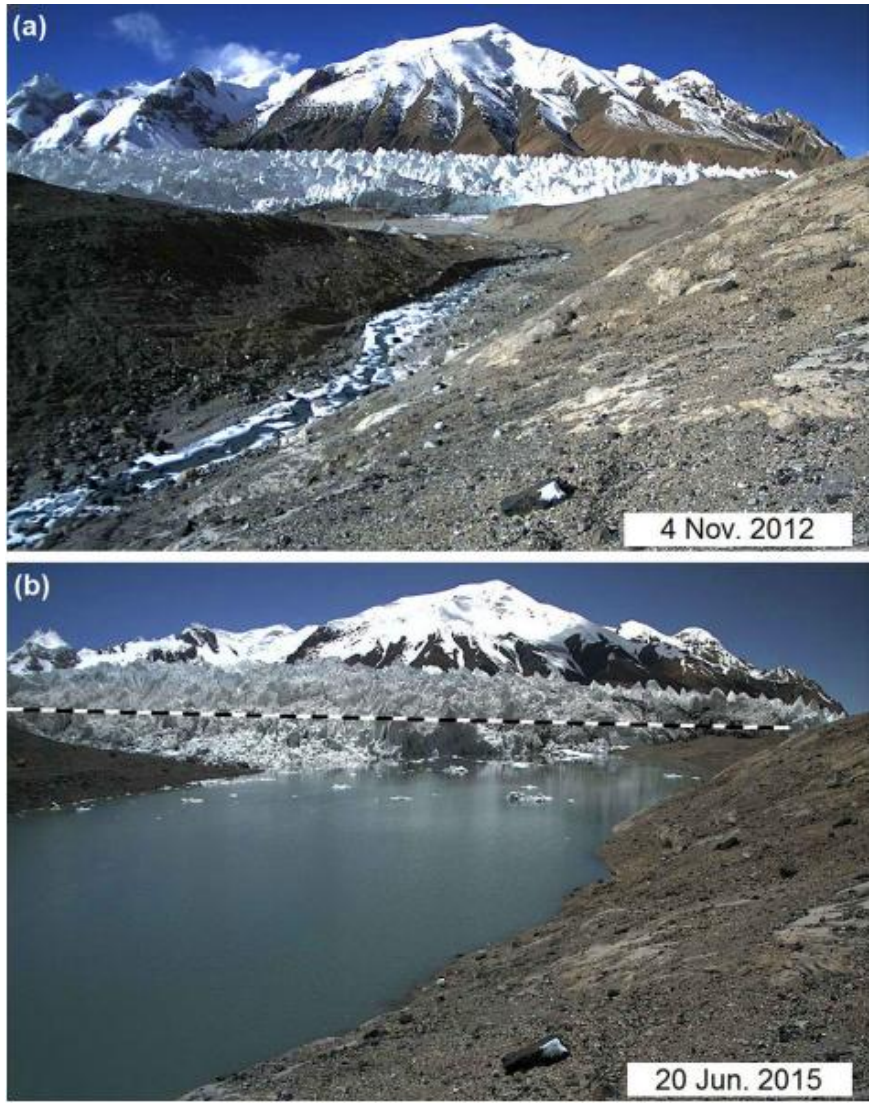
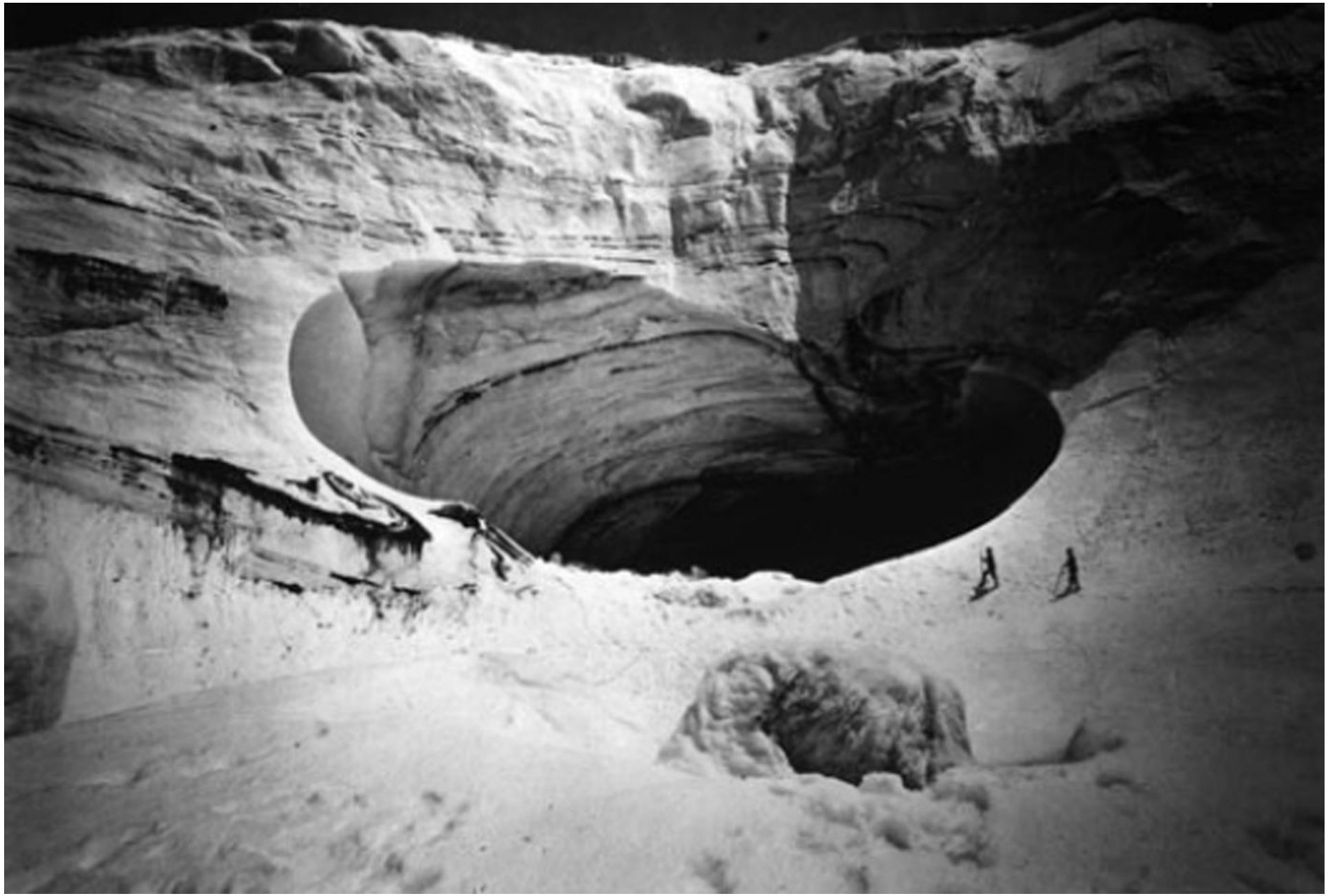


Fig. 12. Aerial photographs of Nostetuko Lake before (left, BC79069-190) and after (right, BC86048-147) the outburst flood of July 19, 1983. Failure occurred when toe of Cumberland Glacier (arrow) collapsed into the lake and generated waves that overtopped the moraine.

British Columbia
Canada



Kyagar Lake, Xinjiang, China, Geopraevent



Tête Rousse (Mont Blanc Massif) in 1892; Vincent et al., 2017

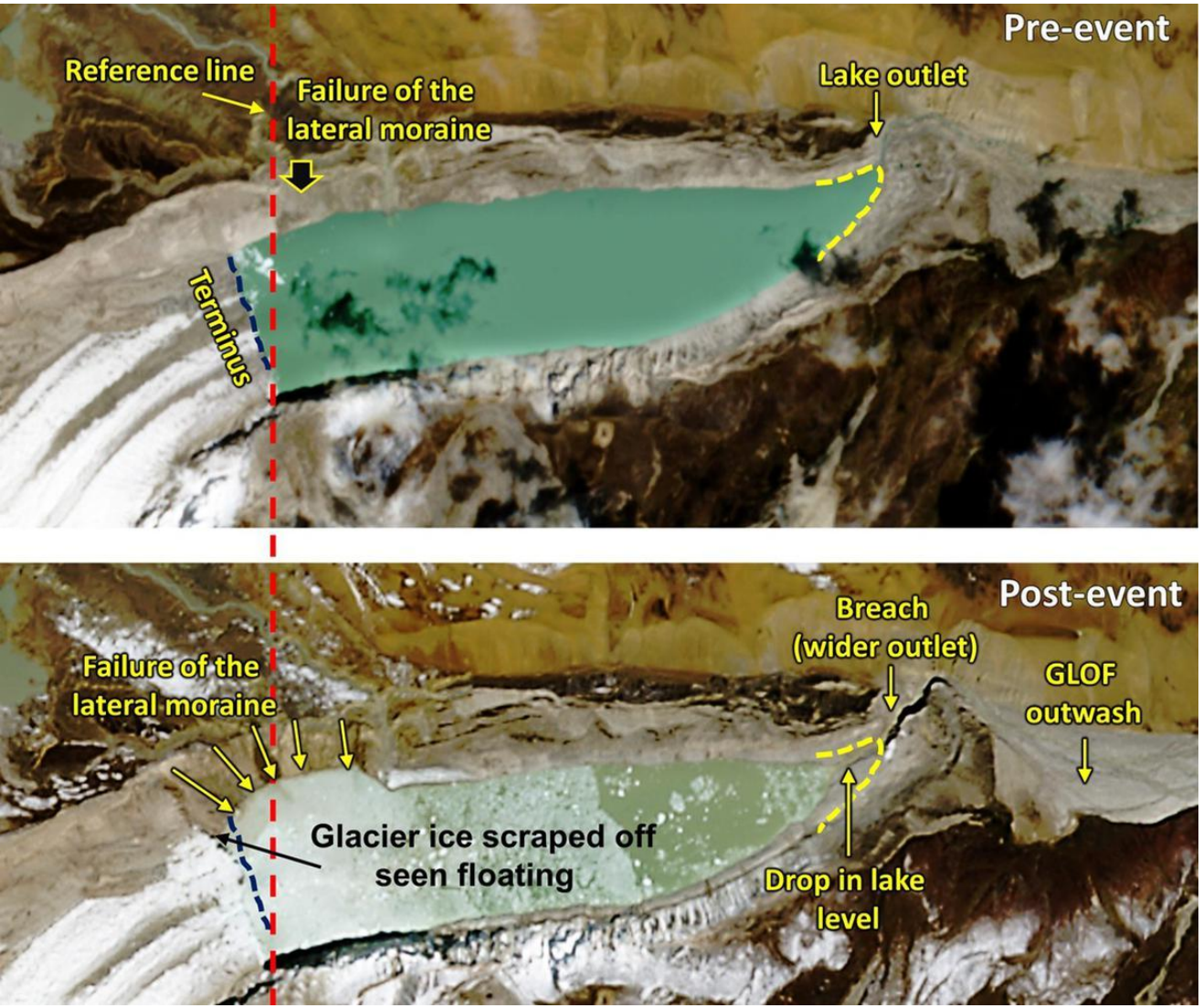


Figure 11: Schematic sketch showing a typical glacial lake outburst chain resulting from an initial mass movement (from, Worni et al., 2014). (1) A mass movement (ice, rock or debris) enters a lake, producing (2) a displacement wave that (3) overtops and (4) incises and erodes the dam area. (5) A flood then travels downstream where (6) populated areas and infrastructure are exposed. Note that displacement waves can be catastrophic with or without erosion of the dam area, and as such, can threaten also apparently stable bedrock dammed lakes.

Allen et al., 2017, GAPHAZ

Glacial lake outburst floods

~ 1 to 10 times yr^{-1} with overall not more than 20 000 people killed (ca 11000 in two events alone that were caused by many other processes)

No indication of an increase with climate change (for some even a decrease due to glaciers disappearing)

Historical a lot of attention due to the massive nature of these events ('biblical floods')

Many GLOFs misreported due to them getting more attention than other hazards → risk of mal-adaptation.

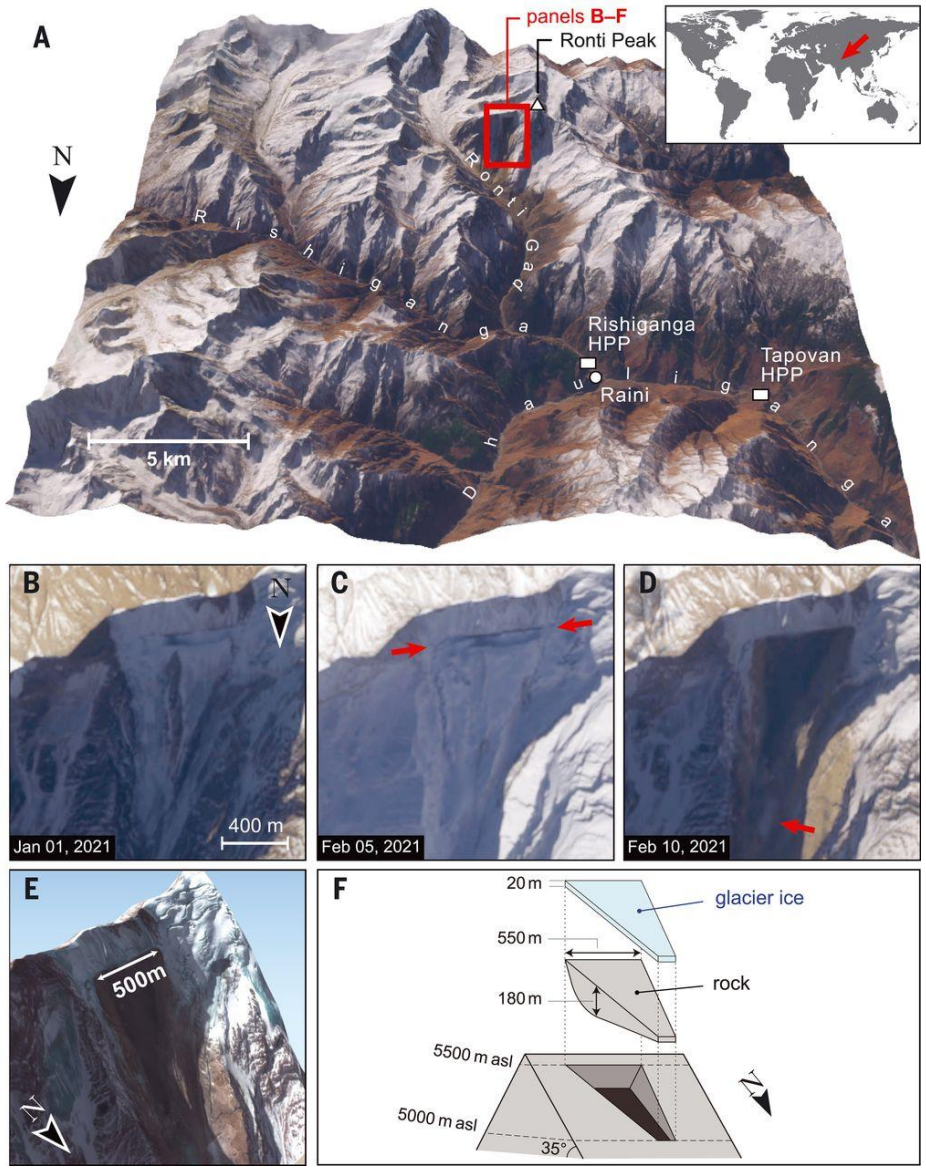
The new avalanches - multihazards

When ice, snow and rock mix, a deadly mix is created.

Increasing number of high impact events – not necessarily because more events happen, but also because more infrastructure at elevation is exposed!

Power dynamics depending on who the victims are

How to accurately report?



Shugar et al. (2022)

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Mountain hazards – slow onset

(Snow) droughts

Snow to Rain transition

Permafrost creep

Subsidence

Road construction/infrastructure development

Hydropower

Urbanisation

Tourism

Mountain (snow) droughts

Drought is a ‘deficiency in water’, i.e. demands exceeds supply over a longer period of time

Meteorological drought: less than average precipitation over a prolonged period

Hydrological drought: water reserves fall short (lakes, rivers, behind dams)

Agricultural drought: stressed crops due to lack of moisture

Snow drought: well below average snow pack leading to lack of snow melt contribution to hydrology

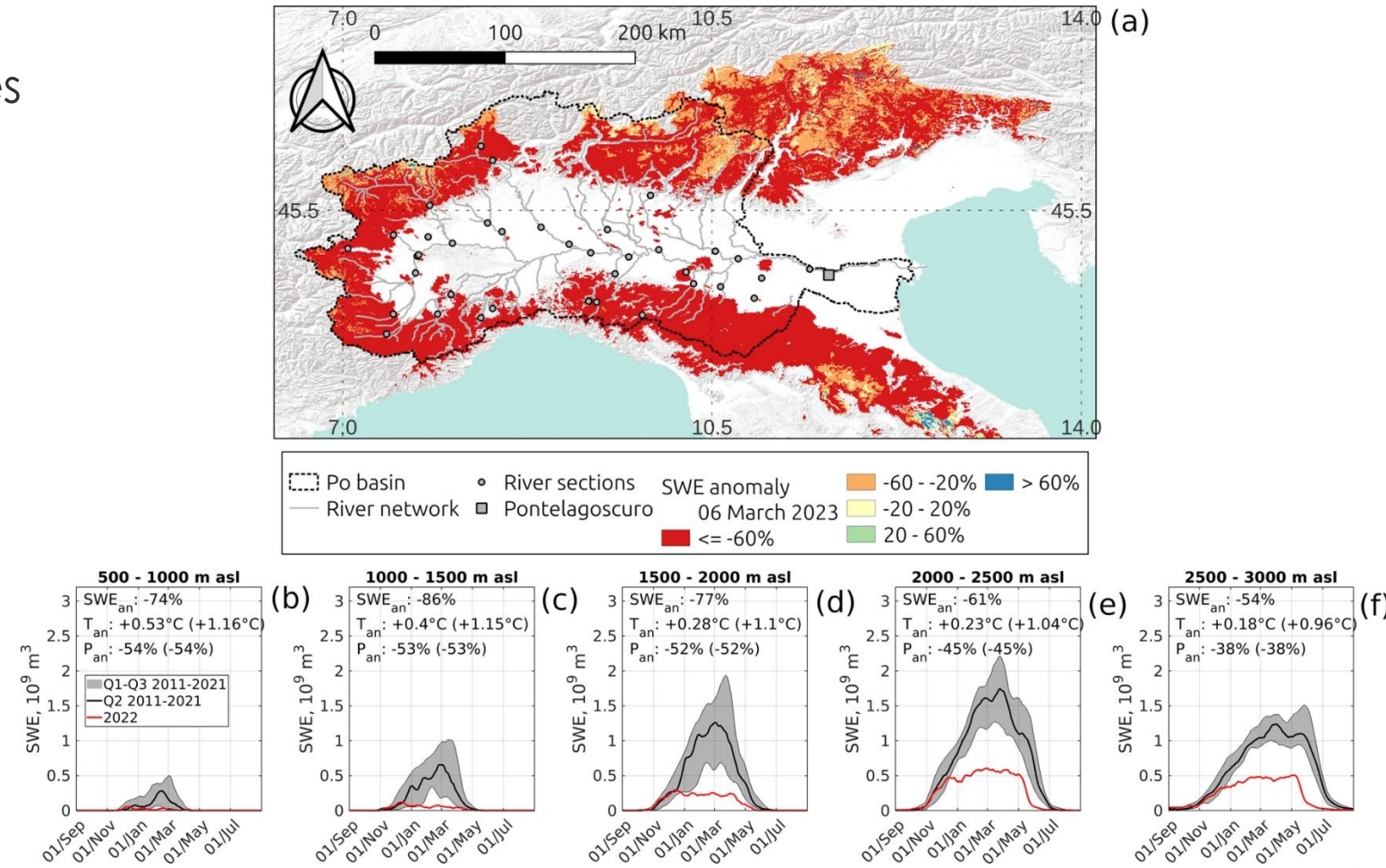
Groundwater drought: lack of groundwater supply

Mountain (snow) droughts

Snow drought happens when snow pack in winter is so small that spring/summer streamflow becomes inefficient

Example for a hazard that propagates downstream

Allows to anticipate months in advance



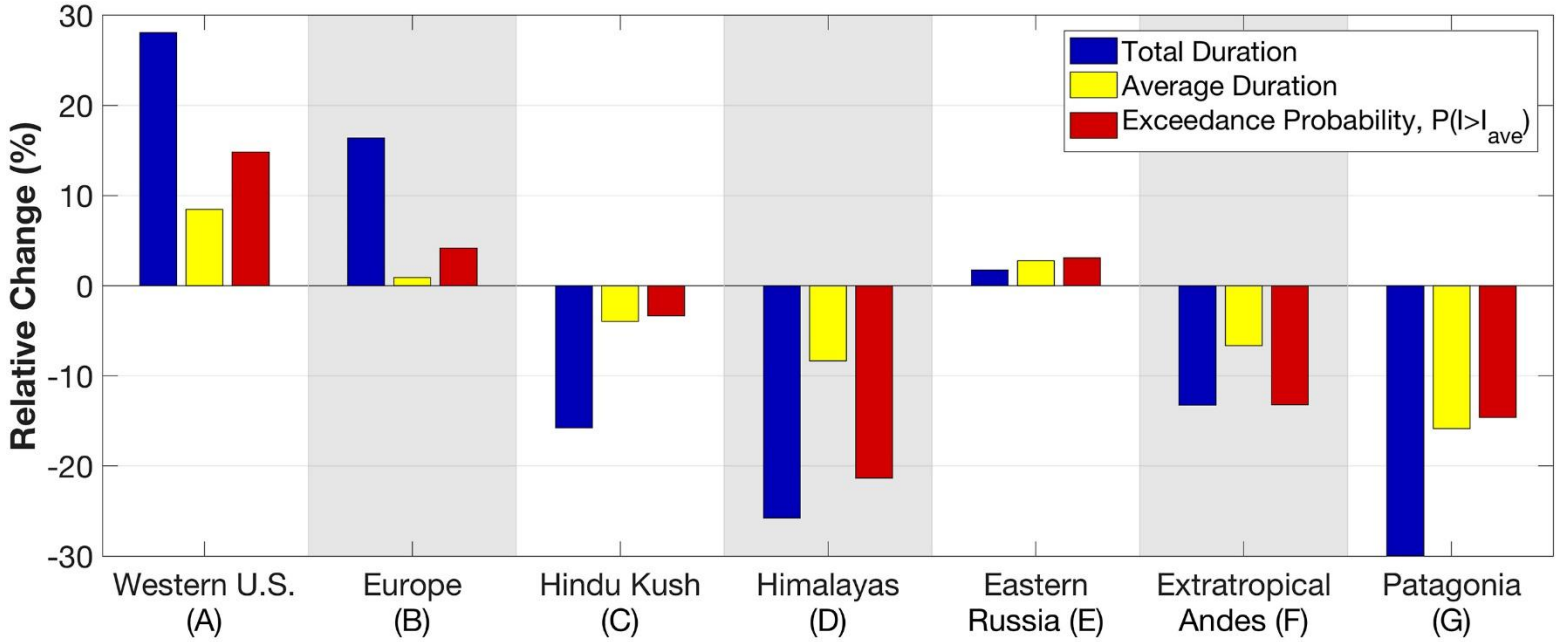
Avanzi et al. (2024)

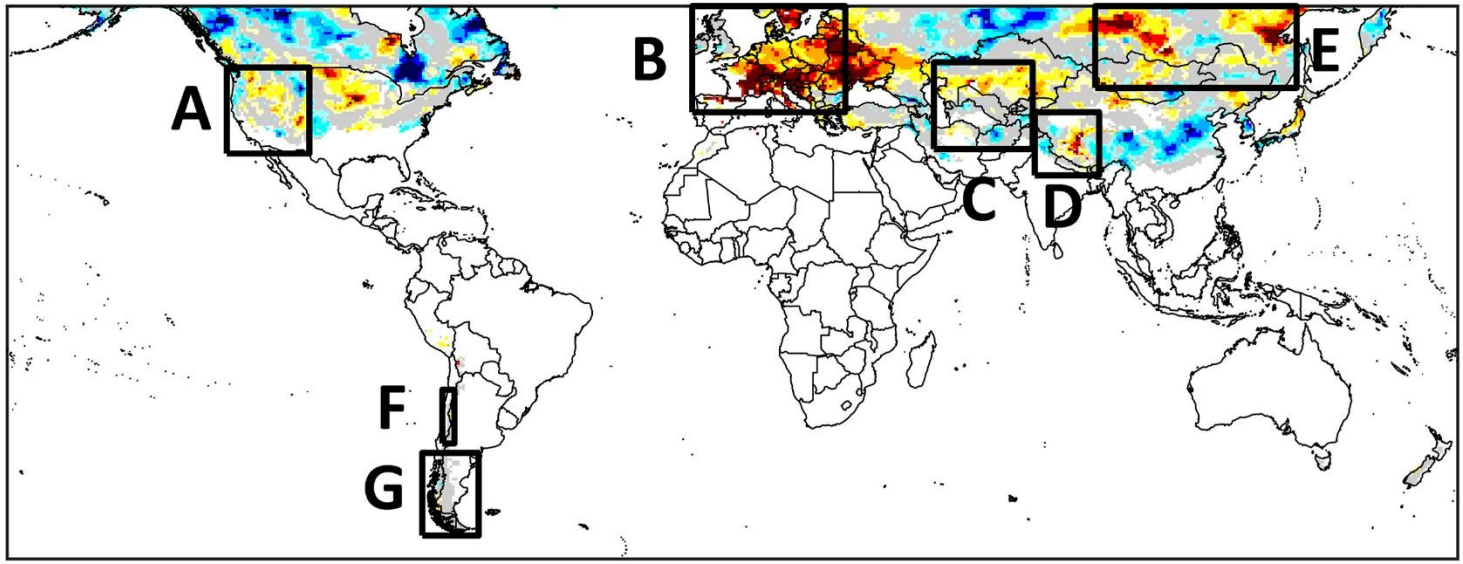
Snow droughts

Snow droughts have increased between 1980 and 2018 especially in the Western US and Europe

Afghanistan is especially vulnerable

"The 2024 snow drought will likely result in nearly an additional 110,000 children who will be severely acutely malnourished and at risk of dying. Along with recent increases due to worsening insecurity and COVID-19, there will be nearly 1 million children with severe acute malnutrition (SAM) in Afghanistan over the next year." (https://reliefweb.int/disaster/dr-2021-000022-afg)

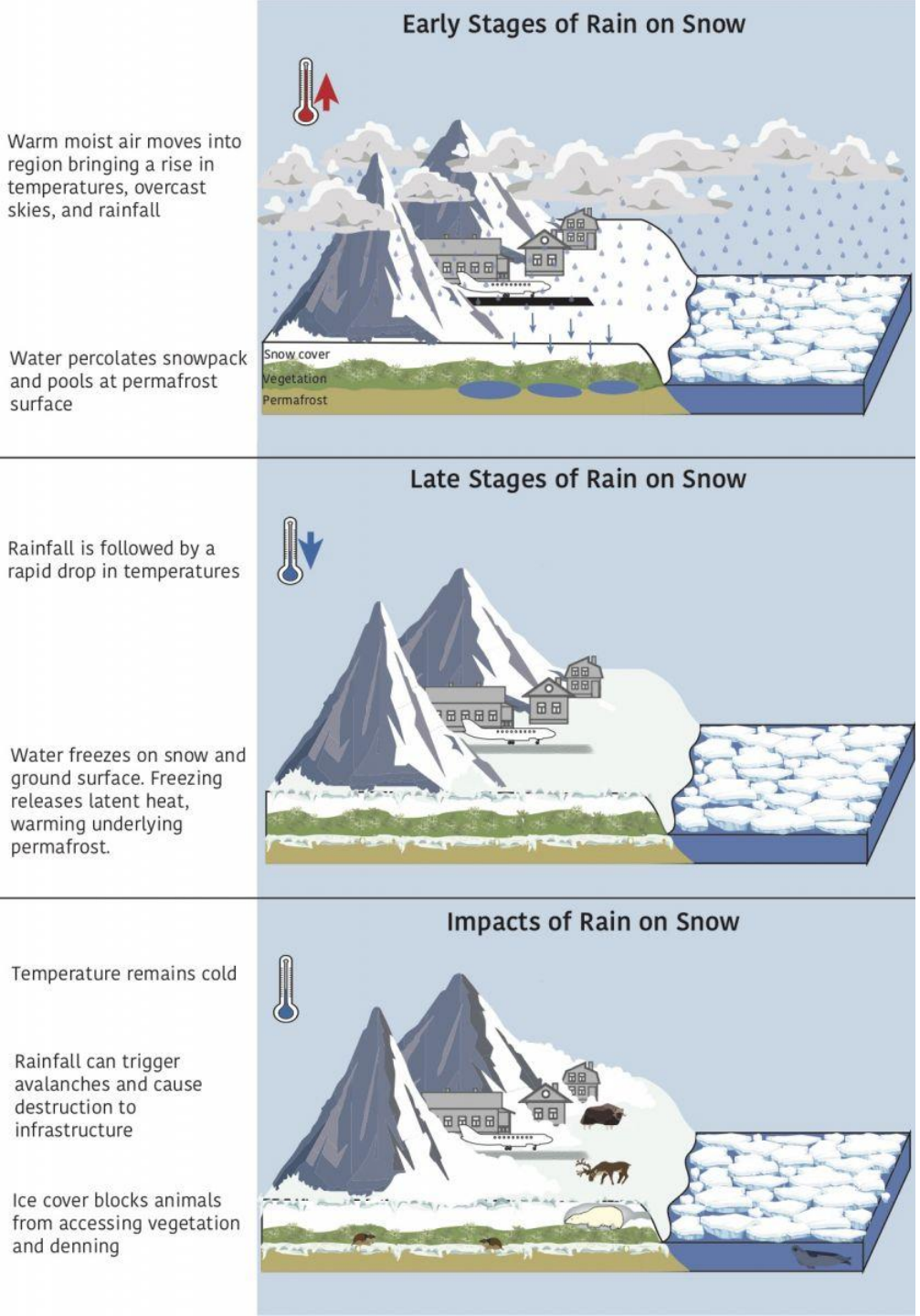


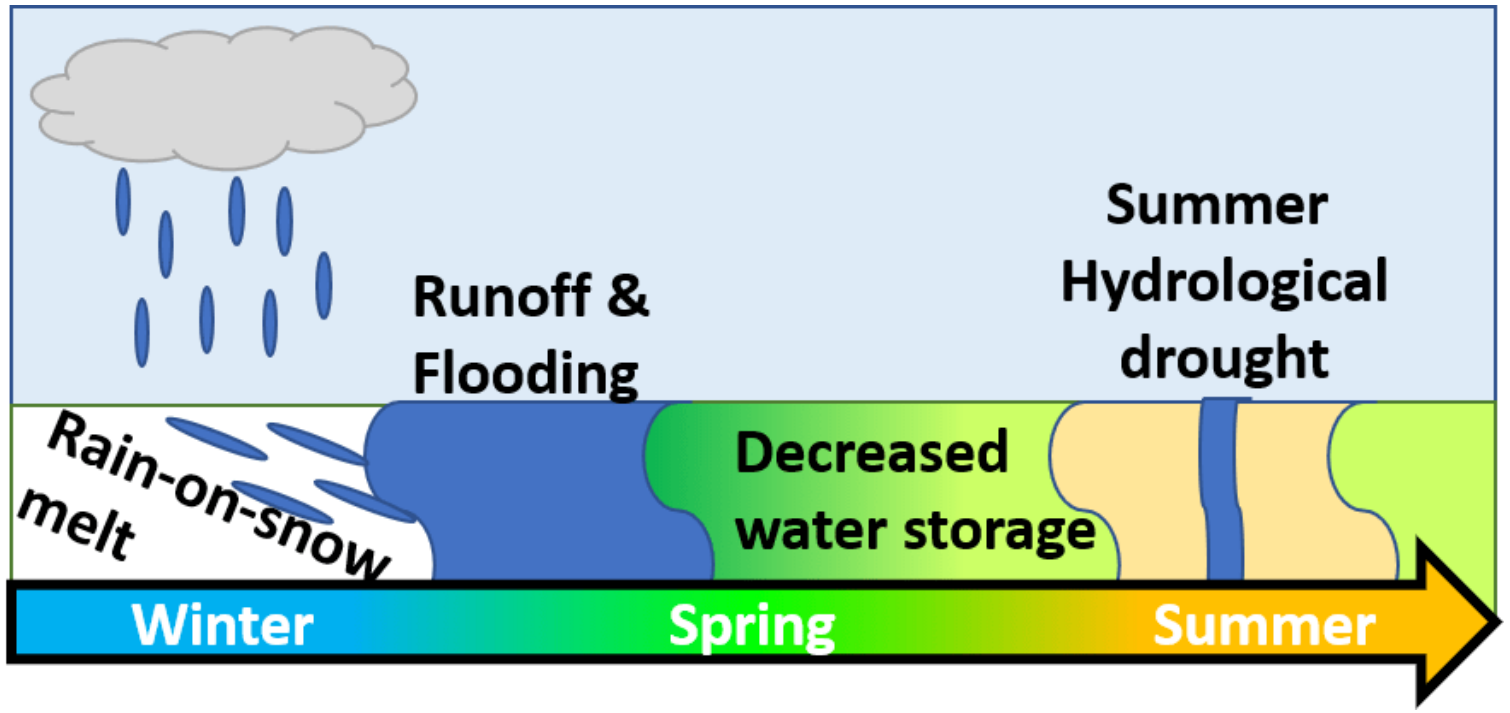


Huning et al. (2020)

Snow turning into rain

Why could that be a problem?





Potential lubricant for (rock-ice-snow-) avalanches resulting in much longer runouts

Permafrost thaw/creep

As previously constantly frozen ground warms, it looses stability

Threat to infrastructure, agriculture and downstream hydrology

Permafrost thaw/creep

As previously constantly frozen ground warms, it looses stability

Threat to infrastructure, agriculture and downstream hydrology

B Etzelmüller, UiO, Dawson, Alaska

Subsidence

Joshimath
(Uttrakhand, India)

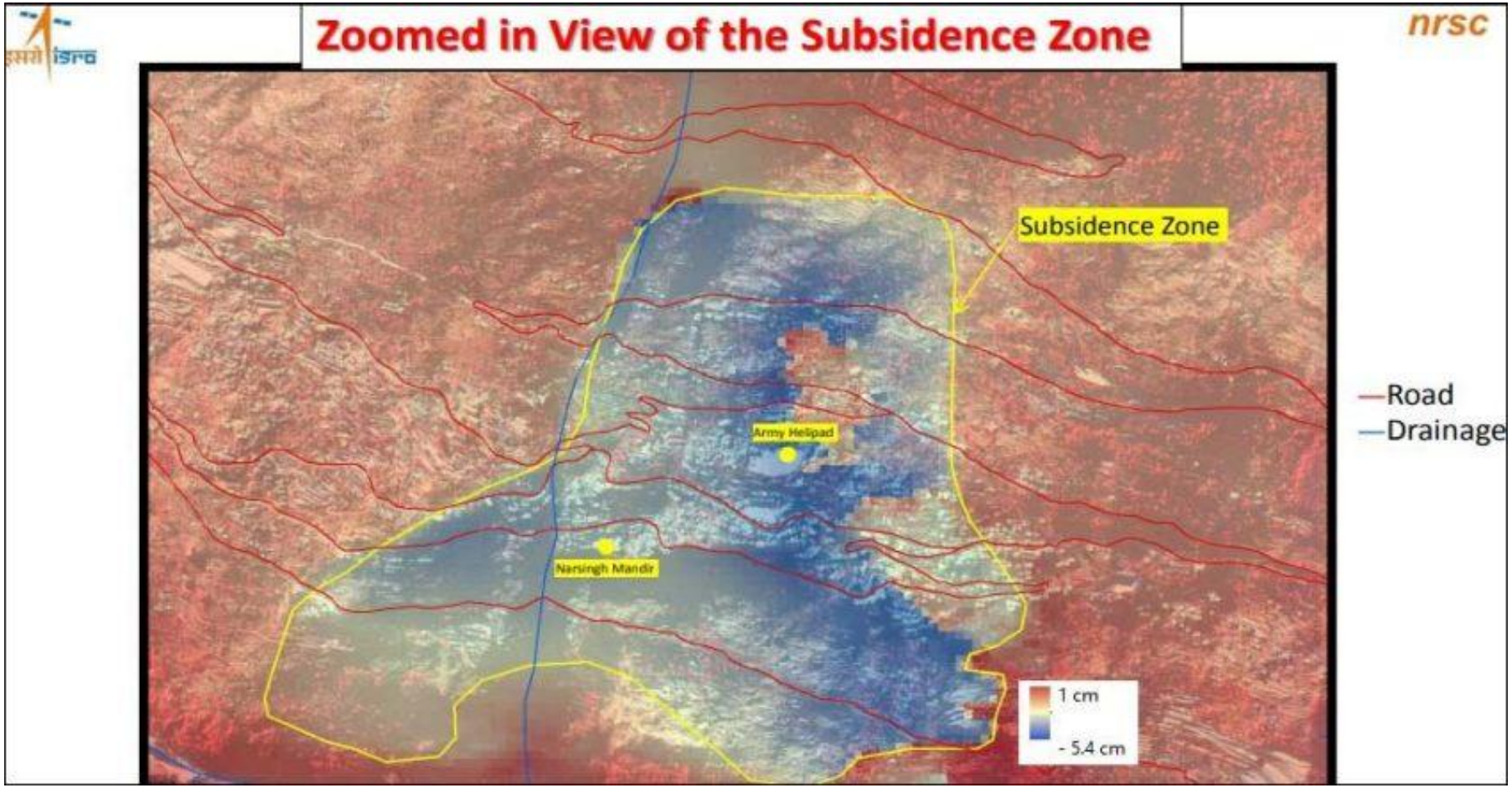
16 000 people

200 so far relocated

Intensive road construction

Chamoli disaster

Subsidence


Zoomed in View of the Subsidence Zone
nrsc
Subsidence Zone
Army Helipad
Narsingh Mandir
1 cm
- 5.4 cm
Road
Drainage

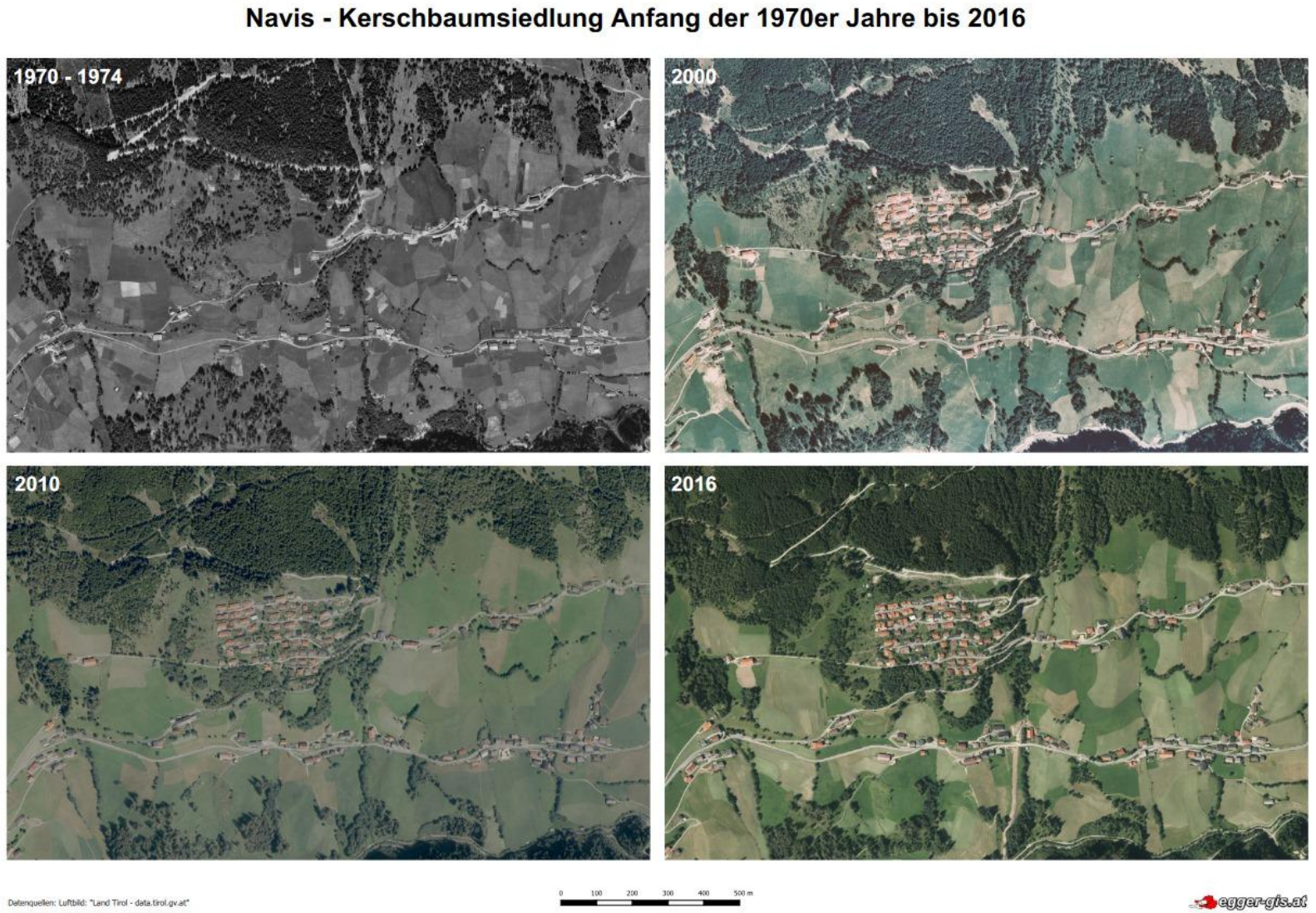
Subsidence and creep

Brenner Basis Tunnel

Subsidence and creep

Kerschbaumsiedlung, Navis, Tirol

84 houses (~250 people)



Drainage of slopes with 50 wells of 50 m depth

Total cost: >6 mio €

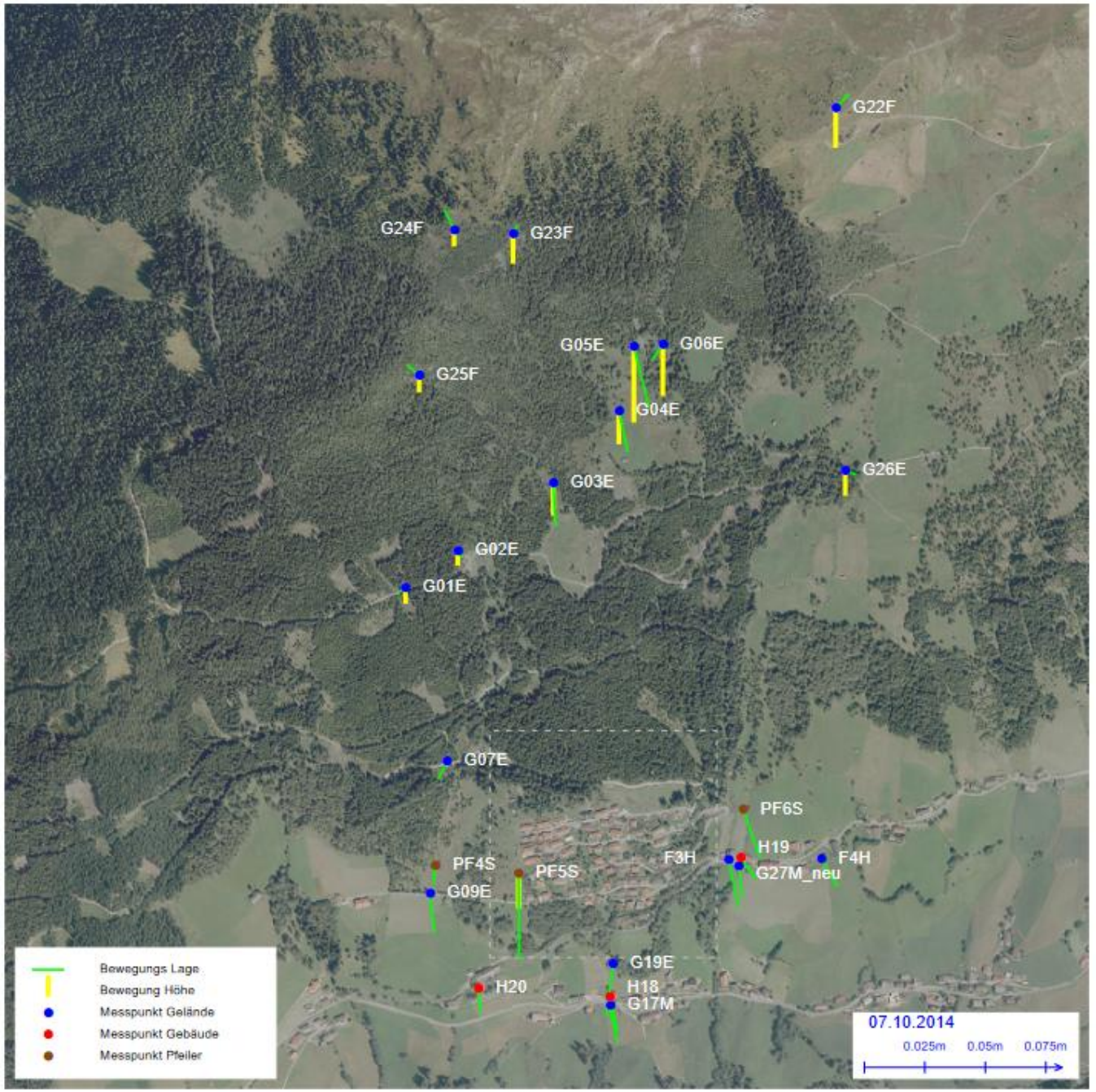


Abbildung 1: Absolutbewegung der Messpunkte seit der Nullmessung (Median aus 14 Tagen)

Course content

27 May (0900 – 1500)

10:00 - 10:30	Fast onset hazards
10:30 - 11:00	Slow onset hazards
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Multihazards, compound events, cascading and transboundary risks

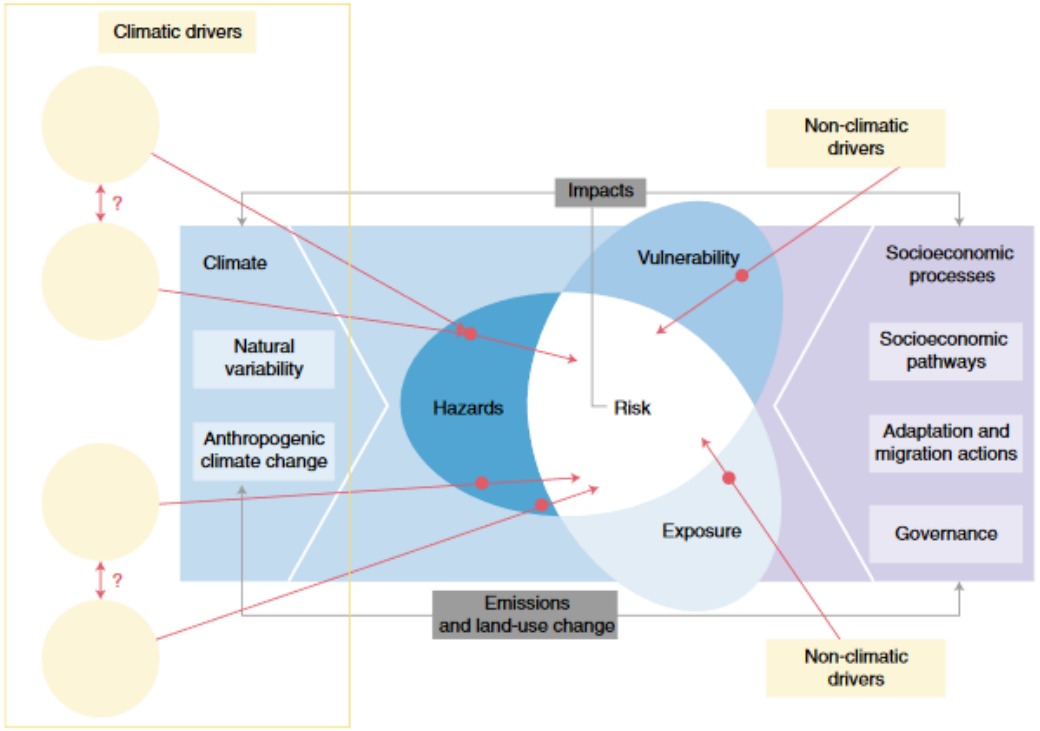
multi hazard: 'hazardous events may occur simultaneously, cascadingly or cumulatively over time, and taking into account the potential interrelated effects' (UNDRR)

multirisk: risk in a multihazard framework

cascading hazards: chain of hazards where an initial one sets off a number of others down the chain

Multihazards, compound events, cascading and transboundary risks

Compound events/drivers: The combination of processes (climate drivers and hazards) leading to a significant impact is referred to as a 'compound event'.



Zscheischler et al. (2018)

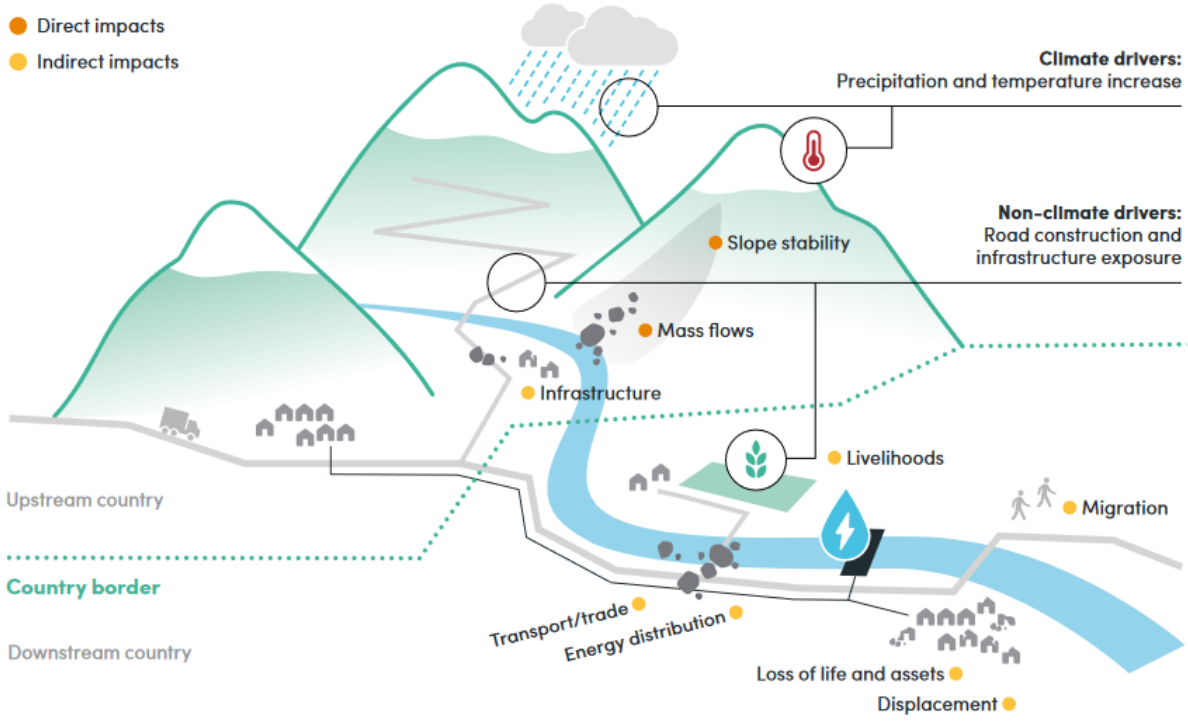
Fig. 1 | Extended risk framework. Multiple climatic drivers cause one or multiple hazards leading to societal and environmental risk. The climate drivers (which may vary from local-scale weather to large-scale climate modes, represented by yellow circles) and/or hazards may be mutually dependent. Non-climatic drivers related to vulnerability and exposure may also contribute to risk. Background risk figure adapted from ref. [96], IPCC.

Multihazards, compound events, cascading and transboundary risks

Compound events/drivers: The combination of processes (climate drivers and hazards) leading to a significant impact is referred to as a 'compound event'.

Cascading risks: risks propagating across domains (becoming transboundary when crossing borders)

Figure 4. Transboundary climate risks for shared terrestrial natural resources: the example of mass flows in cross-border river corridors



Notes: Conceptual overview on transboundary climate risks for terrestrial shared natural resources, using the example of mass flows in river corridors that cross multiple countries. The drivers of risk are related to both climate and non-climate variables, and potentially related directly or indirectly to anthropogenic drivers. The impacts can be direct (orange) and transboundary (e.g. flooding crossing borders threatening lives, damaging agricultural land and hydropower infrastructure), and indirect (yellow), the latter potentially transmitting risks across borders in the reverse direction (e.g. disruption of food or energy supply, effects on livelihoods). Different examples related to the concepts are discussed in the text.

Steiner et al. (2022)

Multihazards, compound events, cascading and transboundary risks

Multihazards, compound events, cascading and transboundary risks

Multihazards

(permafrost thaw, rain on snow, glacial lake outburst flood)

… and compound event (multiple hazards with additional pressure from a changing climate on ice, snow and rain)

Multihazards, compound events, cascading and transboundary risks

Cascading hazard

Multihazards, compound events, cascading and transboundary risks

Cascading hazard to risk

Multihazards, compound events, cascading and transboundary risks

Cascading hazard to risk

Multihazards, compound events, cascading and transboundary risks

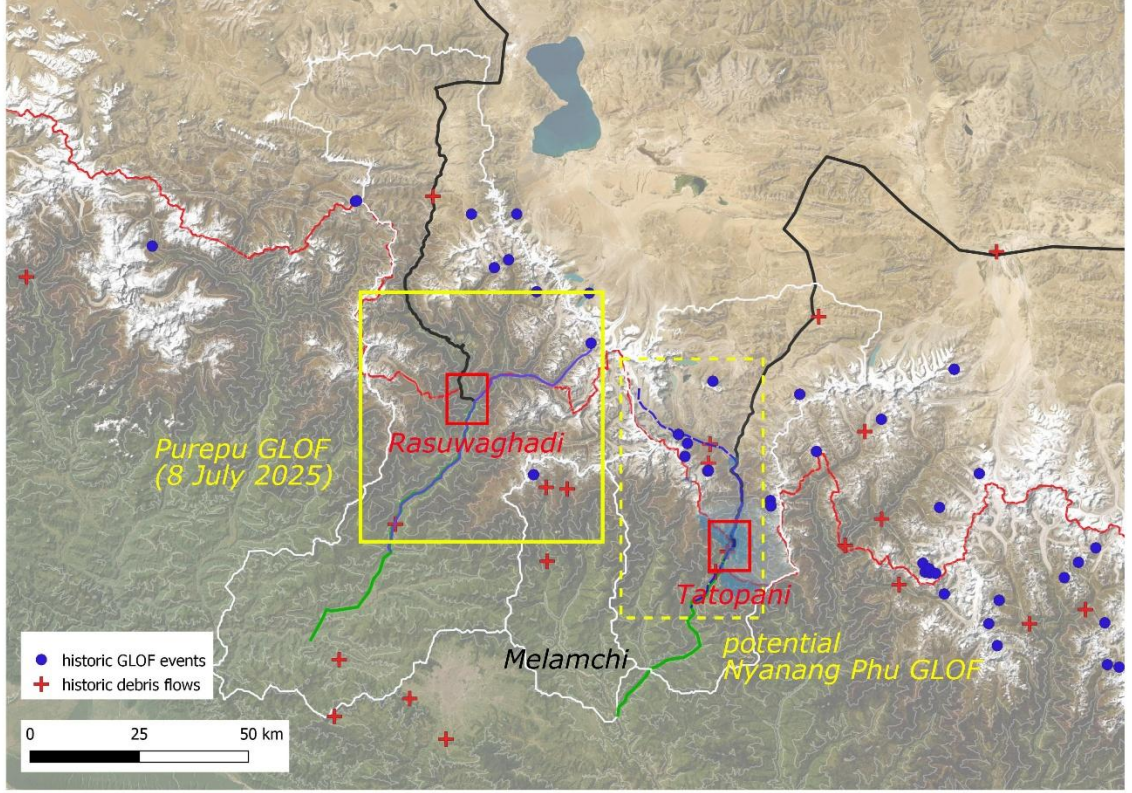
Cascading risk

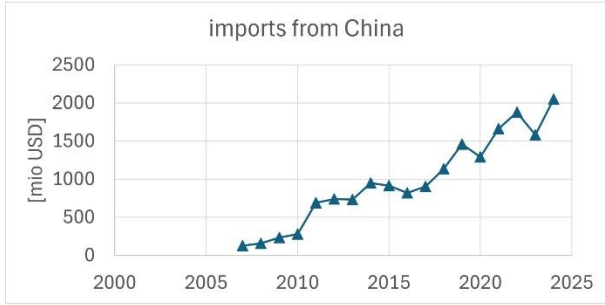
Multihazards, compound events, cascading and transboundary risks

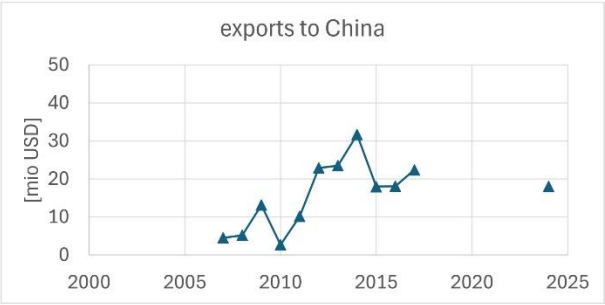
Cascading risk

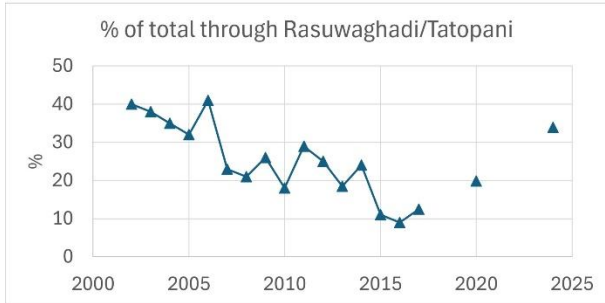
Community Interaction,
Melamchi, Nepal
May 2022

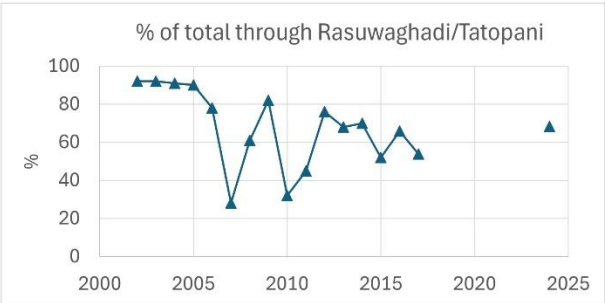
Transboundary cascading risk + compound event

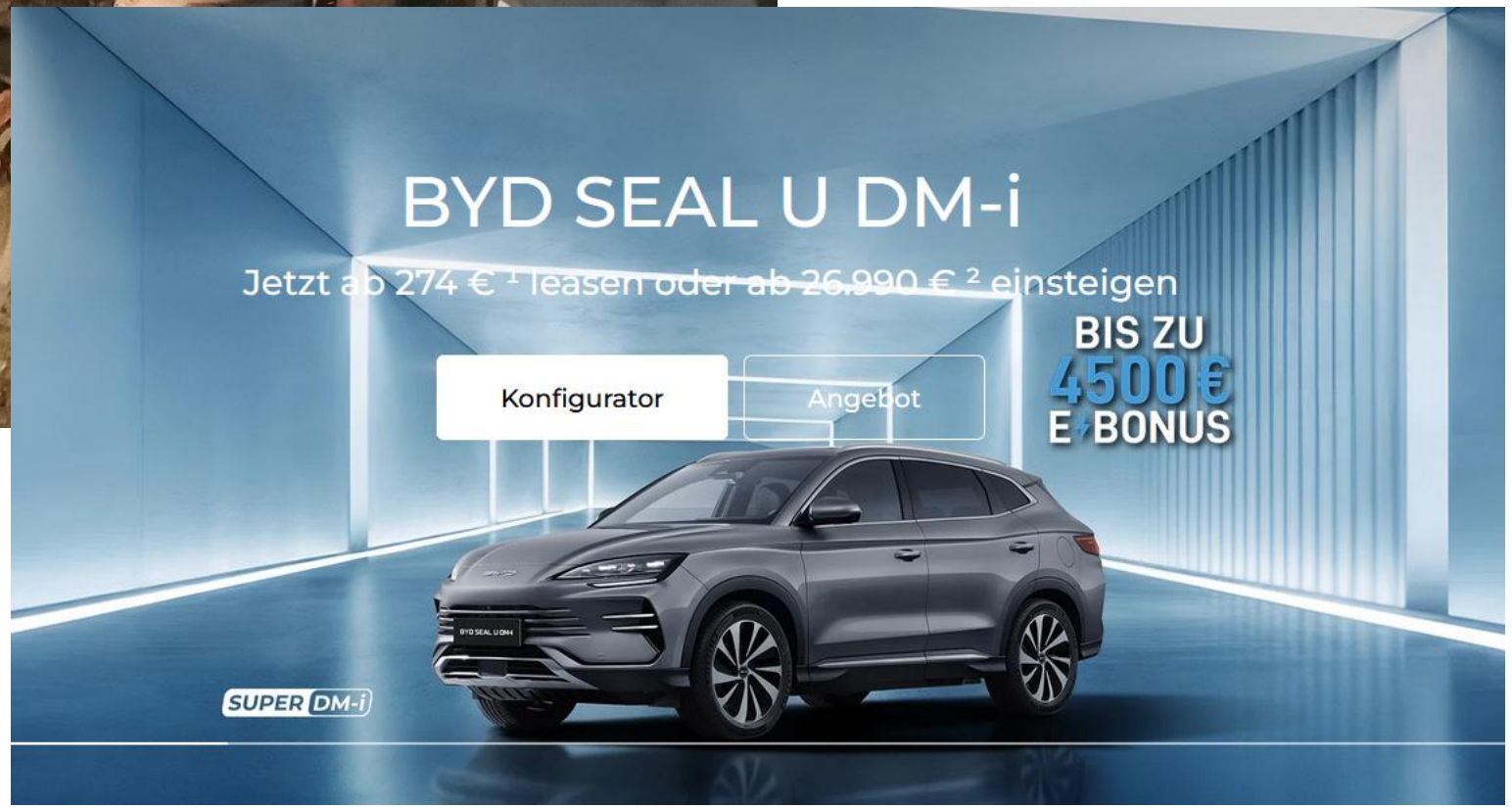











BYD SEAL U DM-i
Jetzt ab 274 € [1] leasen oder ab 26.990 € [2] einsteigen
Konfigurator
Angebot
BIS ZU
4500 €
E-BONUS
SUPER DM-i

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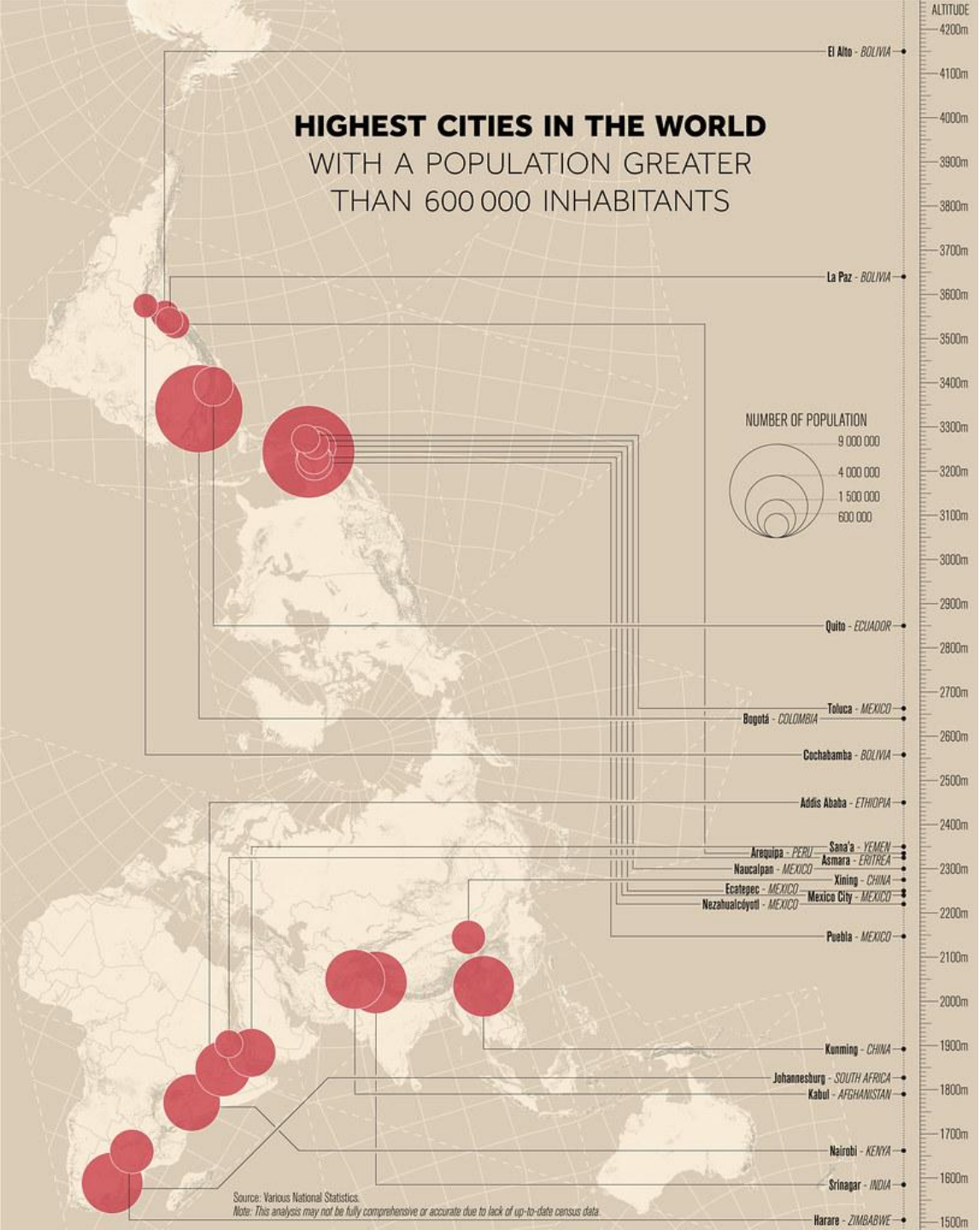
Urbanisation

Mountains are often associated with a *rural* context, resulting in risk assessments tailored for that direction.

How do we define cities/towns/urban?

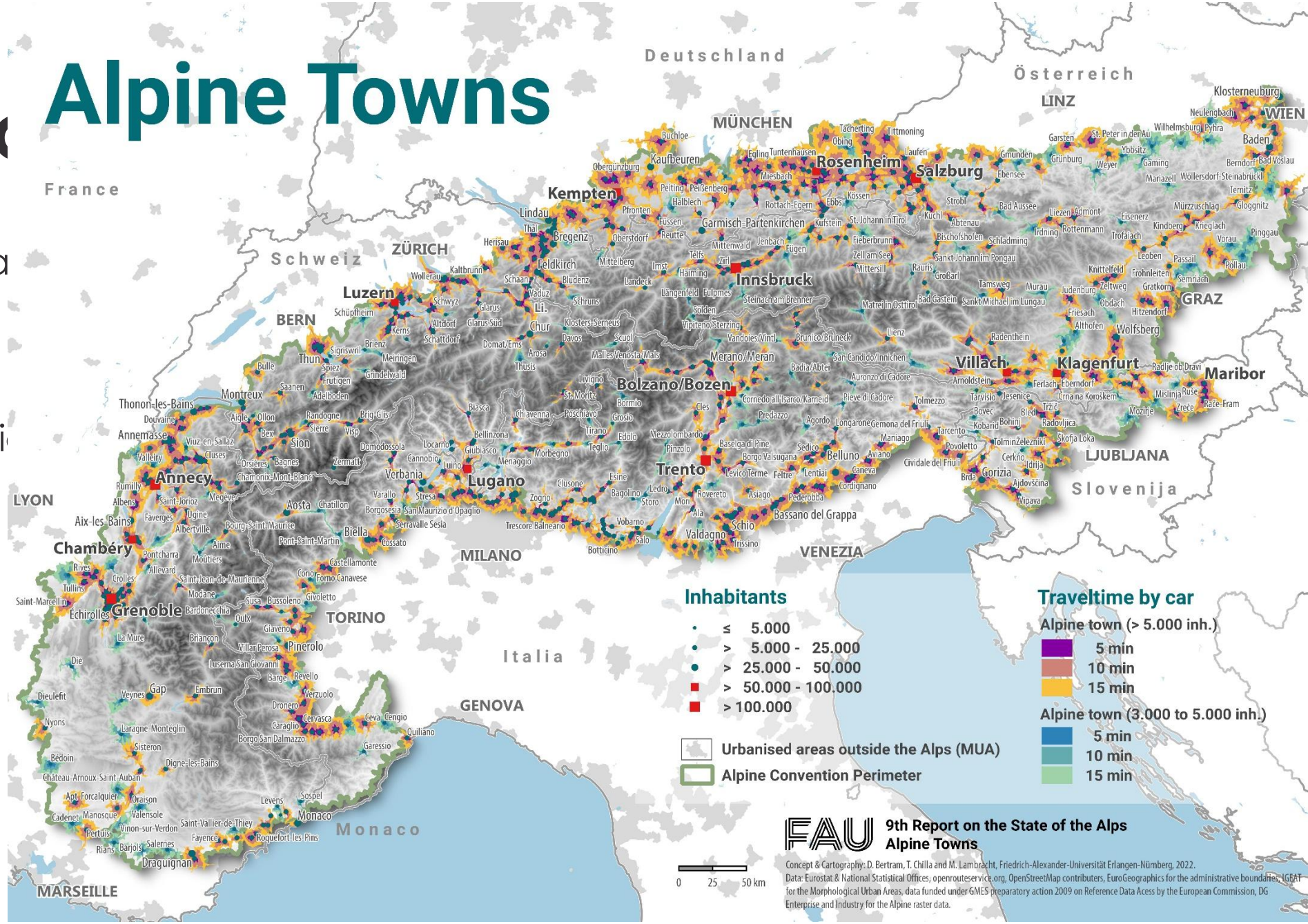


https://www.grida.no/resources/5029

Urbanisatio

Mountains are often a
that direction.

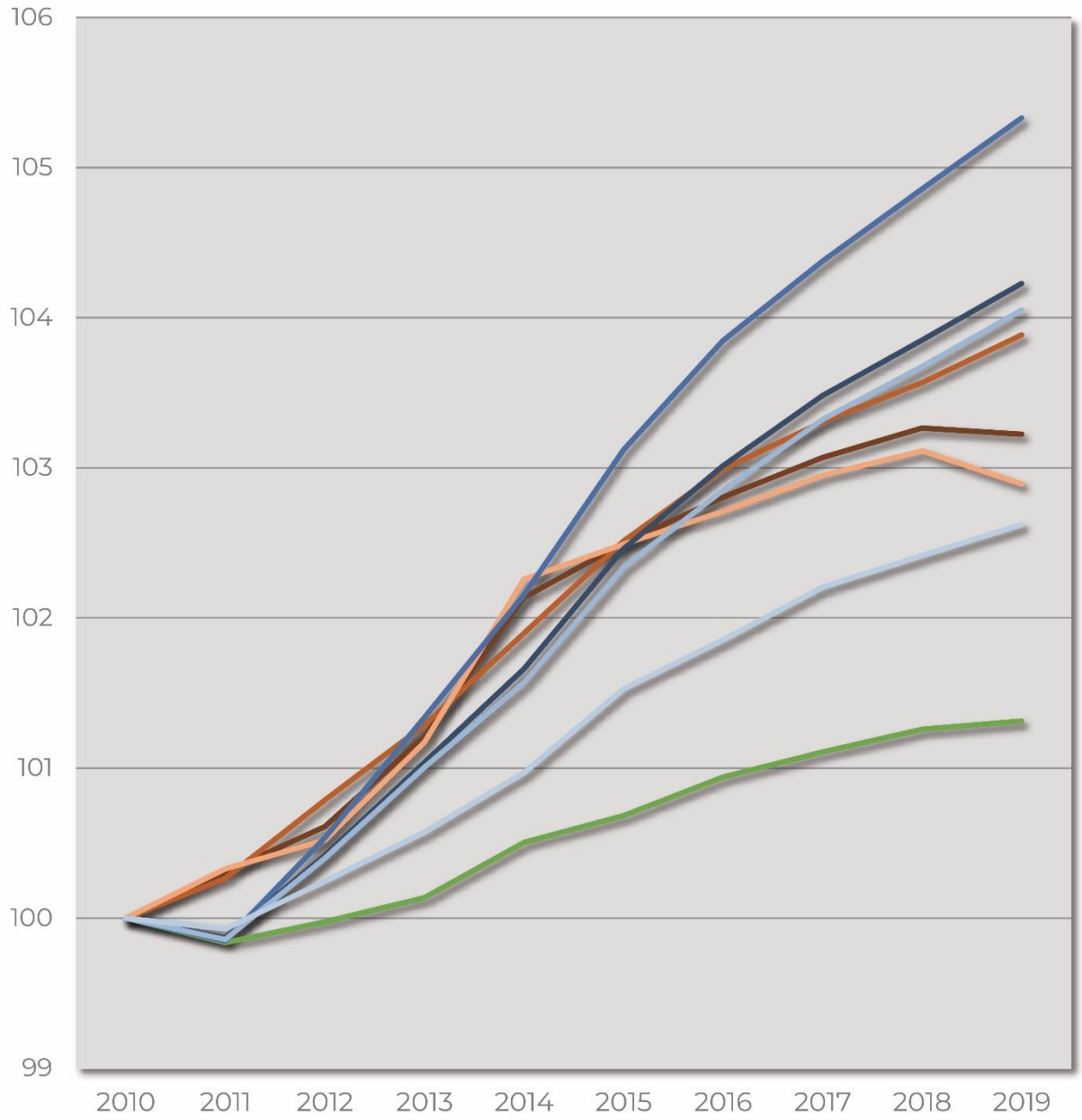
How do we define citi



https://alpinetowns.alpconv.org/

1 DOLLAR SHOP KHAPLU
0355-5225928 , 0355-5239950
0342-2780320 , 0312-2443311
1 DOLLAR
SHOP
KHAPLU

BYKA



Population change
(2010 = 100)

Alpine area
Alpine Convention perimeter

- Total
- Alpine towns (≥ 3.000/5.000 Inh., RSA 9 definition)
- Rural areas

Peri-Alpine area
EUSALP perimeter without Alpine Convention area

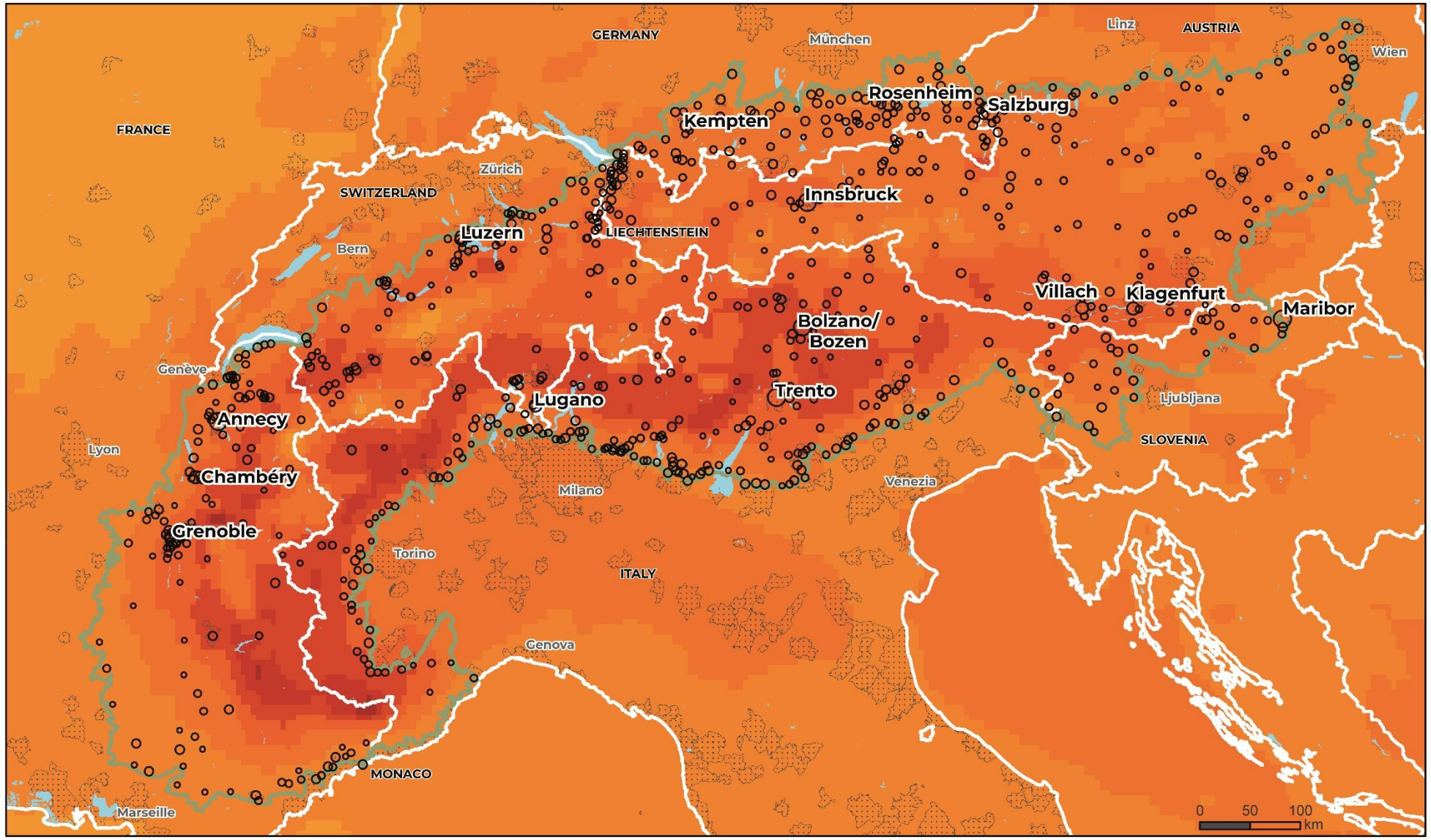
- Total
- Cities (DEGURBA 1)
- Towns and suburbs (DEGURBA 2)
- Rural areas (DEGURBA 3)

Europe

- Total EU 27

FAU **9th Report on the State of the Alps**
Alpine Towns

Data: Eurostat, 2021; National Statistical Offices, 2018 & 2021.
Concept & Visualisation: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, **2022.**



Temperature change

Population of Alpine towns (2015)

- 3.000 - 5.000
- > 5.000 - 10.000
- > 10.000 - 50.000
- > 50.000 - 100.000
- > 100.000

Changes in surface temperature between 2021-2050 (in Celsius)

0.81 - 0.9	1.41 - 1.5
0.91 - 1	1.51 - 1.6
1.01 - 1.1	1.61 - 1.7
1.11 - 1.2	1.71 - 1.8
1.21 - 1.3	1.81 - 1.9
1.31 - 1.4	1.91 - 2

FAU

9th Report on the State of the Alps
Alpine Towns

Data: Alpine Convention for projected changes in annual near surface temperature in Southern Europe 2021-2050 in Celsius (WMS).
Layout: © EuroGeographics for the administrative boundaries; IGEAT for the Morphological Urban Areas.
Concept & Cartography: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, **2022.**

Alpine Convention Perimeter

Urbanised areas outside the Alps (MUA)

Urbanisation

Mountains are often associated with a *rural* context, resulting in risk assessments tailored for that direction.

How do we define cities/towns/urban?

Why is urban space different in terms of risks?

Urbanisation

Mountains are often associated with a *rural* context, resulting in that direction.

How do we define cities/towns/urban?

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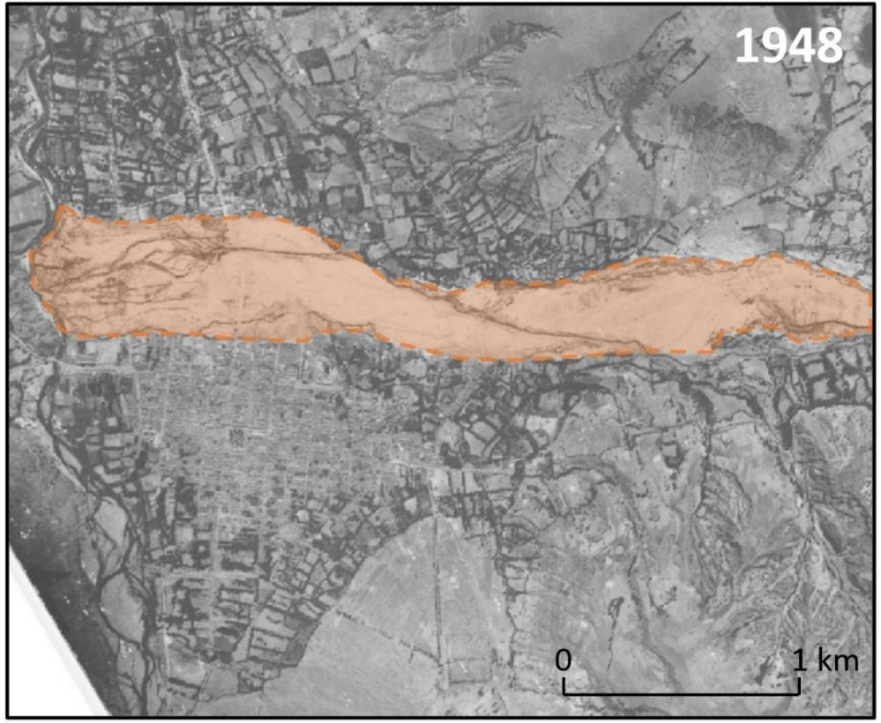
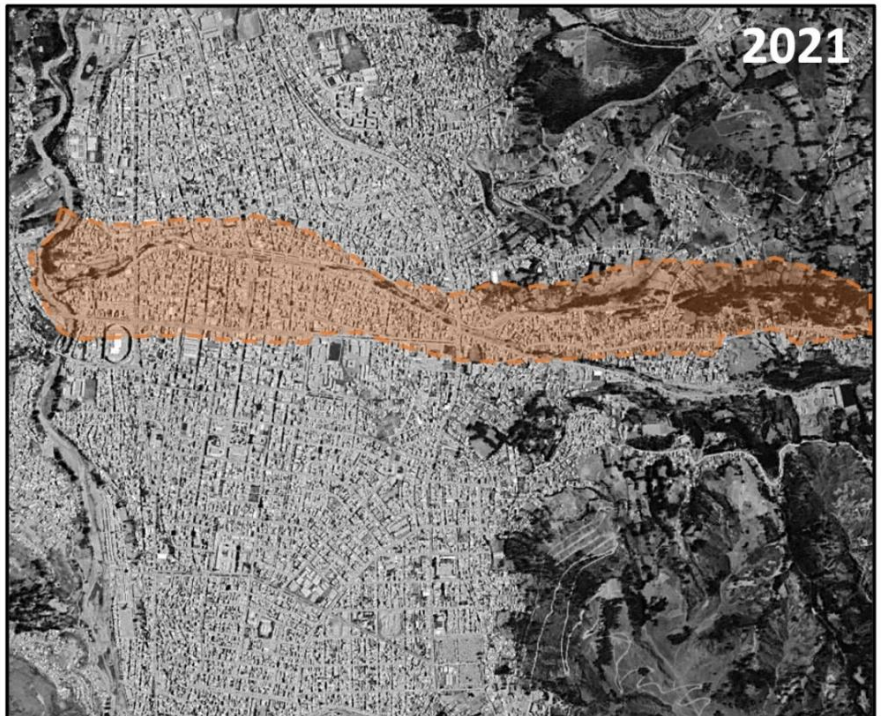


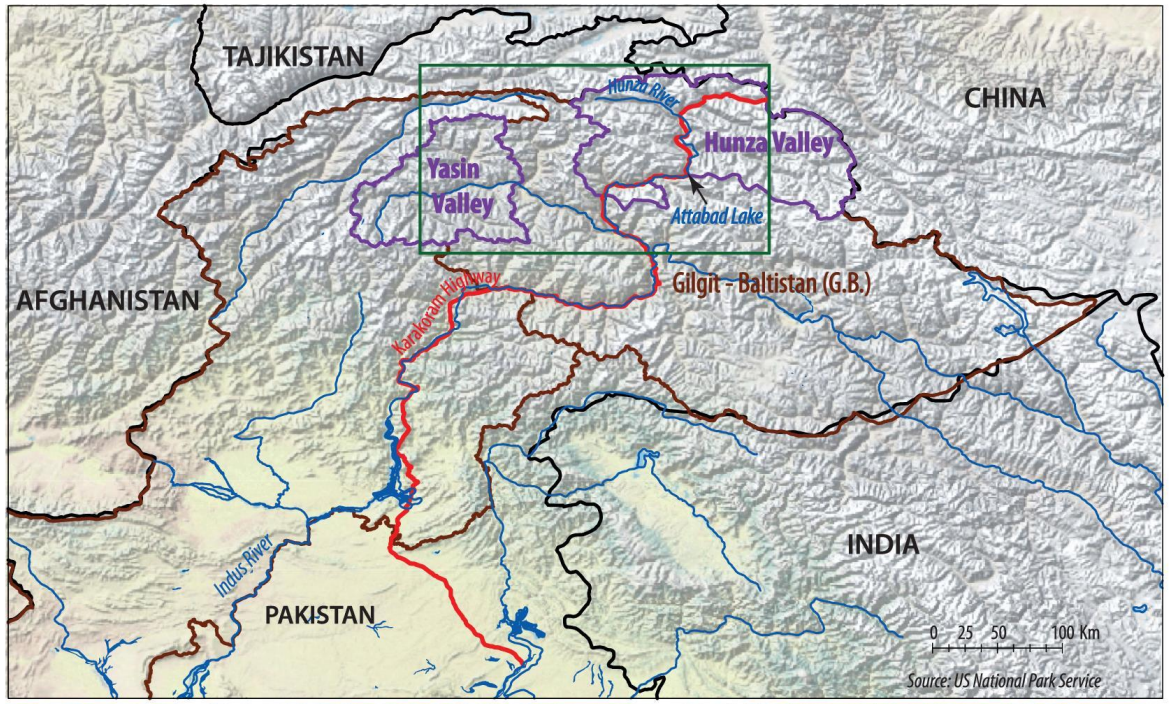


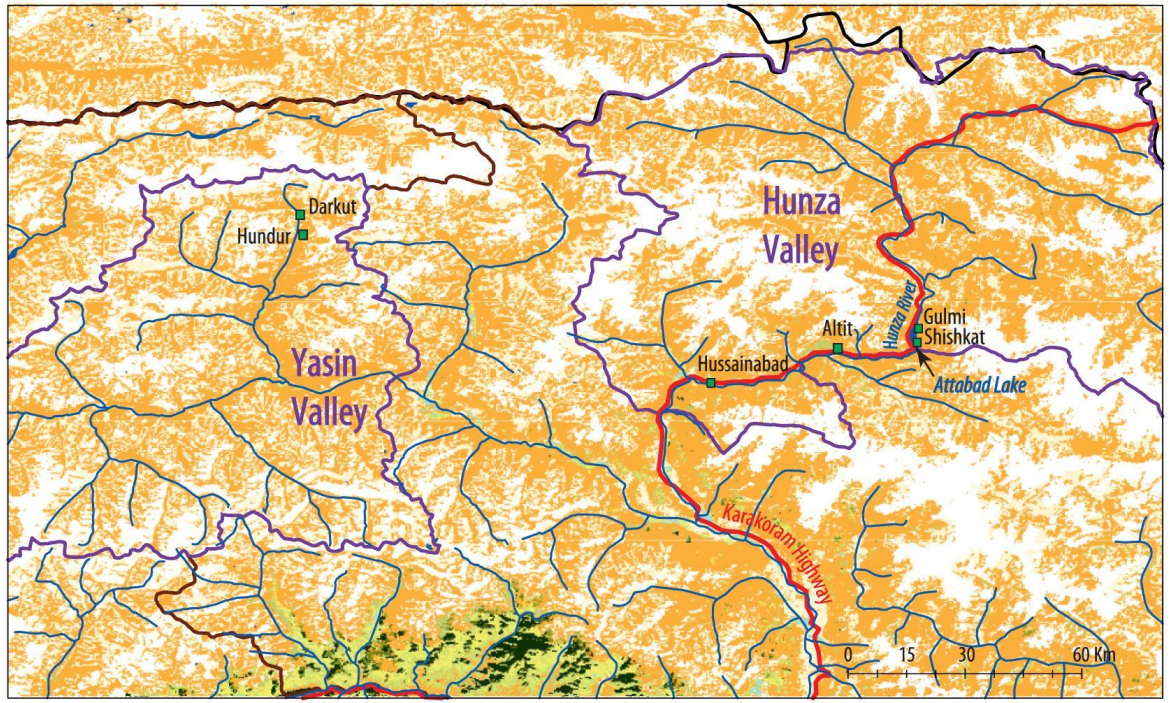
Emmer, Steiner and Kalvoda (2024)

Migration and climate risks

Do people move away because of (climate) risks in mountains?

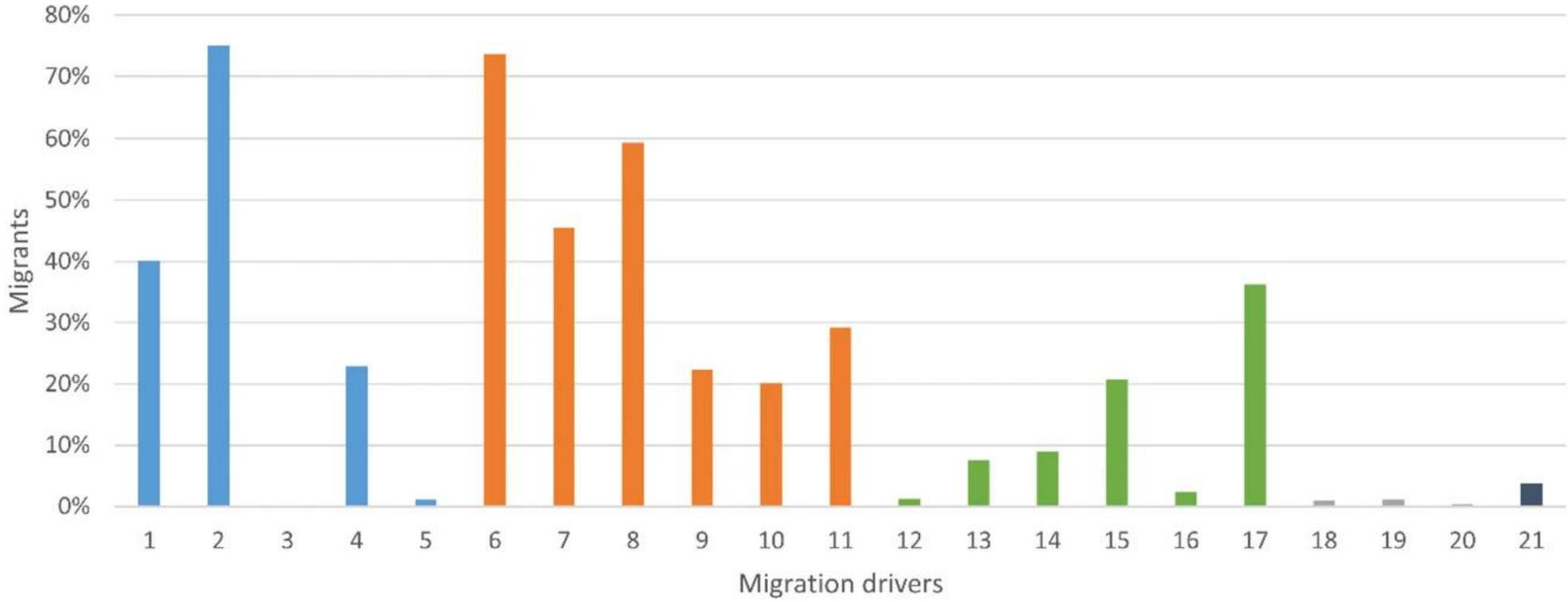
Why do people move? Why do people stay?





Gioli et al., 2014

Reasons to migrate



Social
1. better education at destination
2. better healthcare at destination
3. marriage
4. family connections at destination
5. discrimination at the place of origin

Economic
6. employment opportunity
7. unemployment at place of origin
8. higher income at destination
9. lack of agricultural land
10. crop failure
11. not enough income

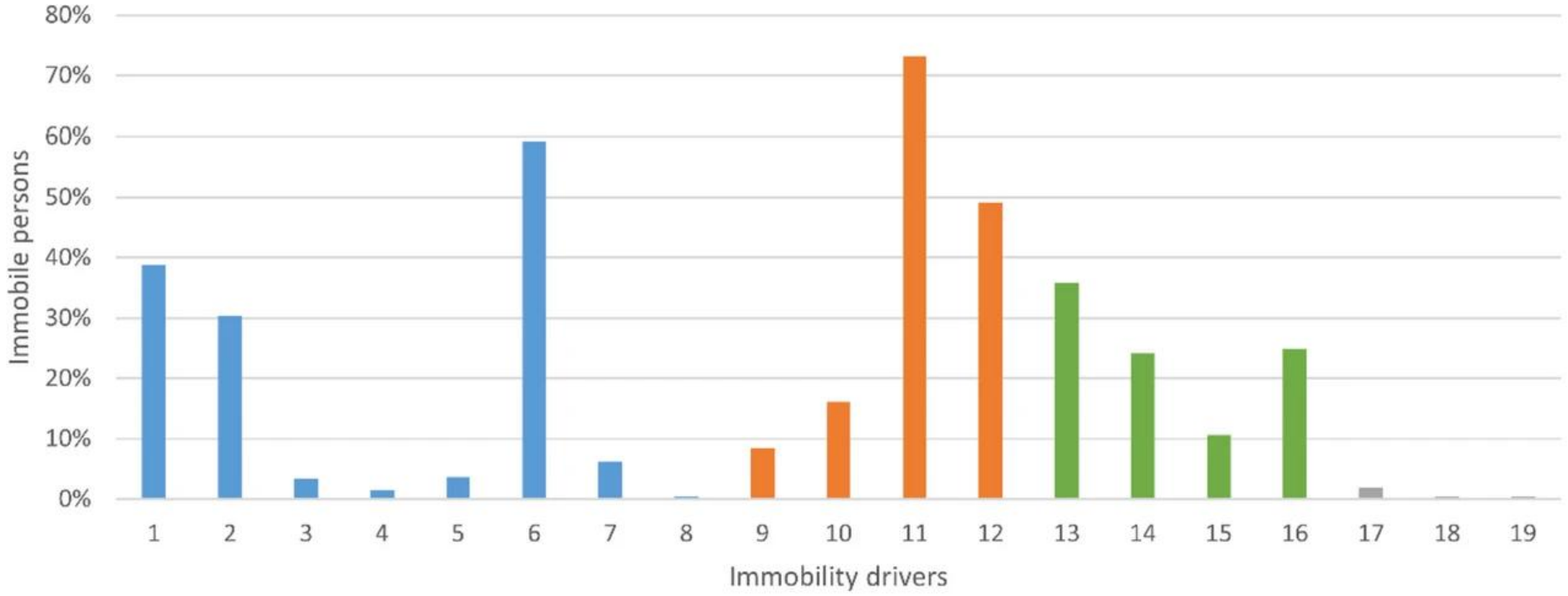
Environmental
12. water shortage or drought like conditions
13. land degradation
14. floods
15. landslides/ avalanche
16. cyclone/ storm
17. heavy snowfall/ rainfall

Political
18. conflict with family/ community
19. lack of political freedom
20. government incentives

other reasons

Khan et al., 2024

Reasons to (involuntarily) stay



Social
1. lack of education
2. health reasons
3. disability
4. gender
5. Age
6. household care
7. lack of family connections
8. discrimination at destination

Economic
9. lack of transportation costs
10. no employment opportunity at destination
11. no financial resources
12. resource for basic needs

Environmental
13. impact of extreme events on income
14. loss of economic assets
15. collapse of transportation network
16. labor for land rehabilitation

Political
17. legal & administrative barrier
18. conflict at destination
19. lack of political freedom

Khan et al., 2024

Hydropower

Germany: 3.5% of demand

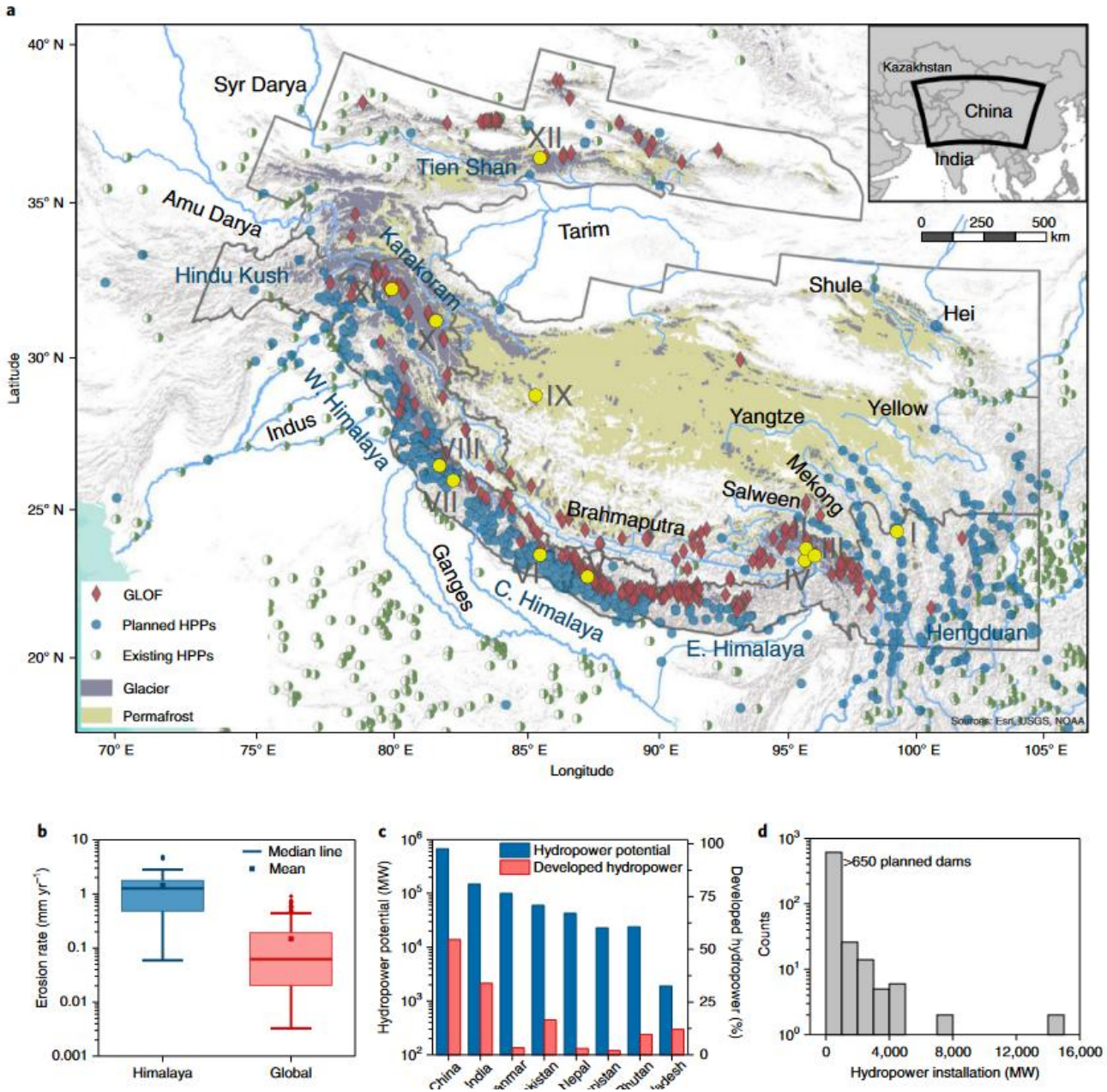
Austria: 58% of total generation

Switzerland: 62% of total generation

New Zealand: 55% of total generation

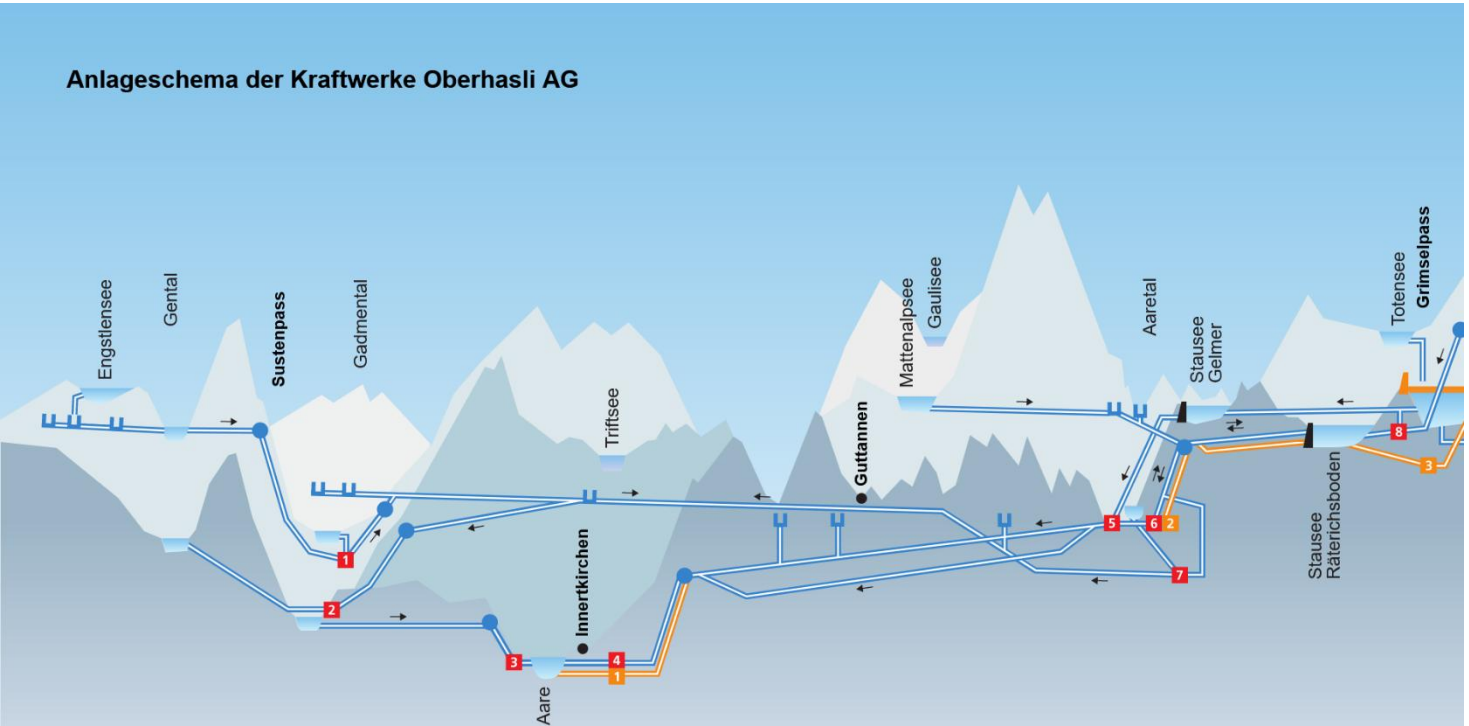
Bhutan: 99.5% of total generation

Nepal: 75% of total generation



Li et al., 2022

Hydropower Infrastructure



- KRAFTWERKE
  - A Grimsel 1
  - B Grimsel 2
  - C Handeck 1
  - D Handeck 2
  - E Handeck 3
  - F Innertkirchen 1
  - G Innertkirchen 2
  - H Fuhren
  - I Hopflauenen
- SPEICHERSEEN
  1. Oberaarsee
  2. Grimselsee
  3. Räterichsbodensee
  4. Gelmersee
  5. Totensee
  6. Trübtensee
  7. Engstlensee
  8. Mattenalpsee
- BAHNEN
  1. Meiringen-Innertkirchen-Bahn (MIB)
  2. Gelmerbahn
  3. Triftbahn
  4. Tällibahn
  5. Luftseilbahn Handeck-Gerstenegg
  6. Oberaarbahn
  7. Sommerlochbahn
- GRIMSELHOTELS
  - Handeck · Kinder- und Familienhotel
  - Grimsel Hospiz · Historisches Alpinhotel
  - Oberaar · Restaurant und Berghaus
  - Bäregg · Ferien- und Alpinhütte
  - Tälli · Restaurant und Berghaus
- FIRMENSITZ

Konzessionsgebiet
Wasserführende Stollen
Zugangsstollen

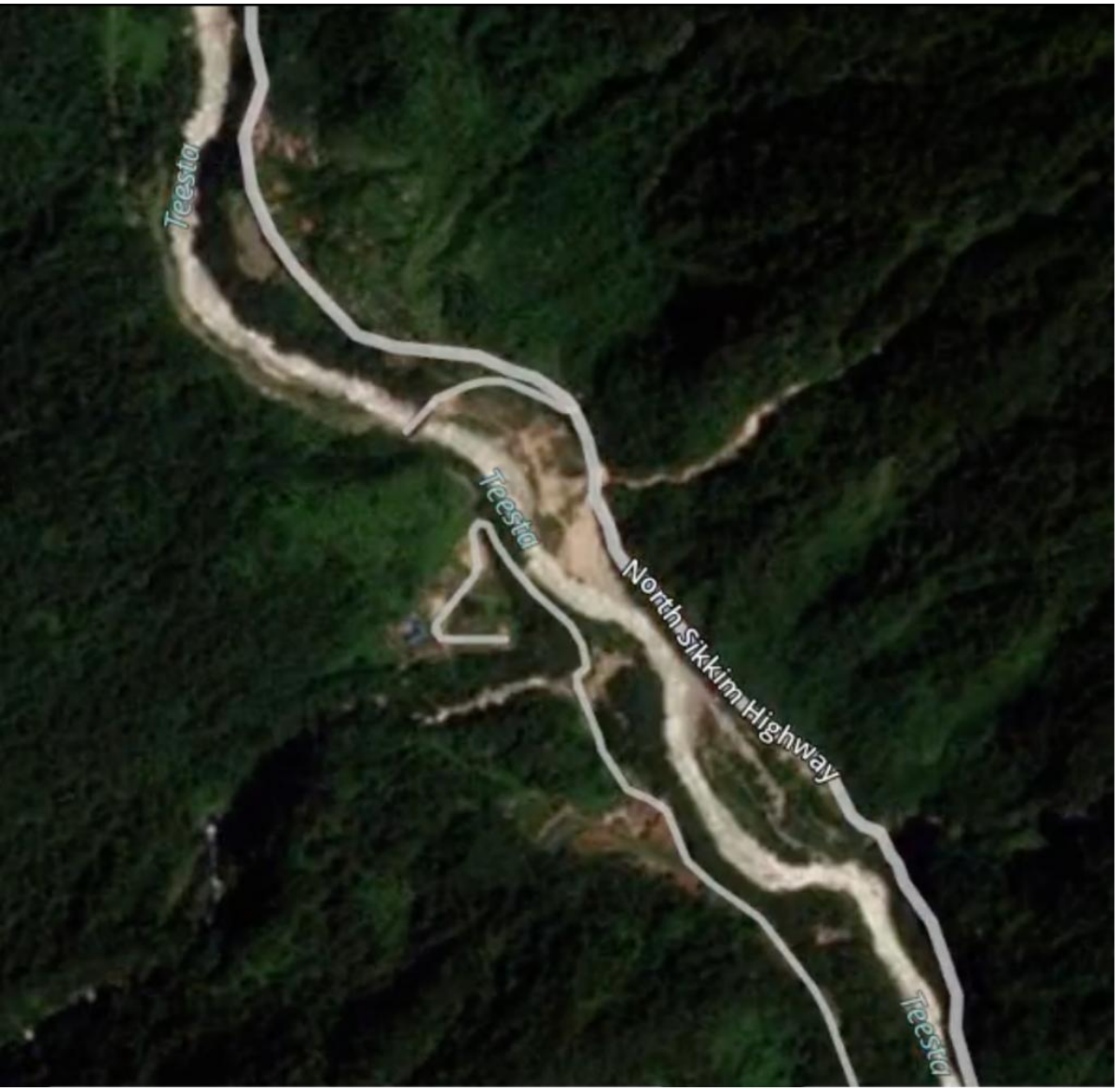
South Lonak GLOF – October 2023

- 100 people died
- Well known lake that was expected to cause problems

Hydropower infrastructure could help risk reduction if operated in accordance

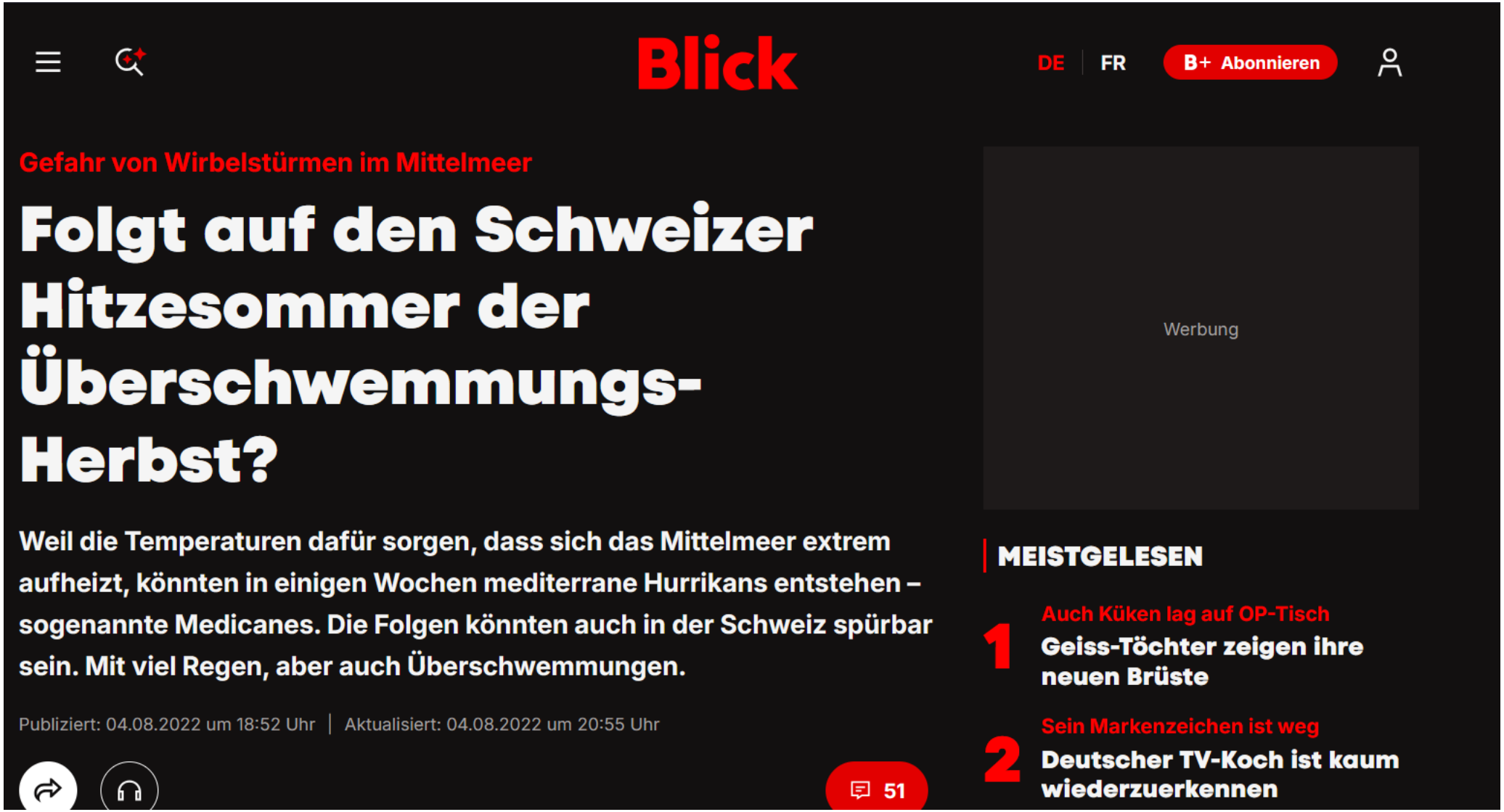
In this case, it increased the impact, as the water was first retained and then breached with even higher force

04-10-2023
Earlier
TEESTA 3 DAM
BEFORE & AFTER


Teesta
Teesta
North Sikkim Highway
Teesta

Teesta
Teesta
North Sikkim Highway
Teesta

Framing of risks



Framing of risks

Reaktion auf Unwetter

«Alles okay hier» – Tessin und Wallis wollen den Tourismus retten

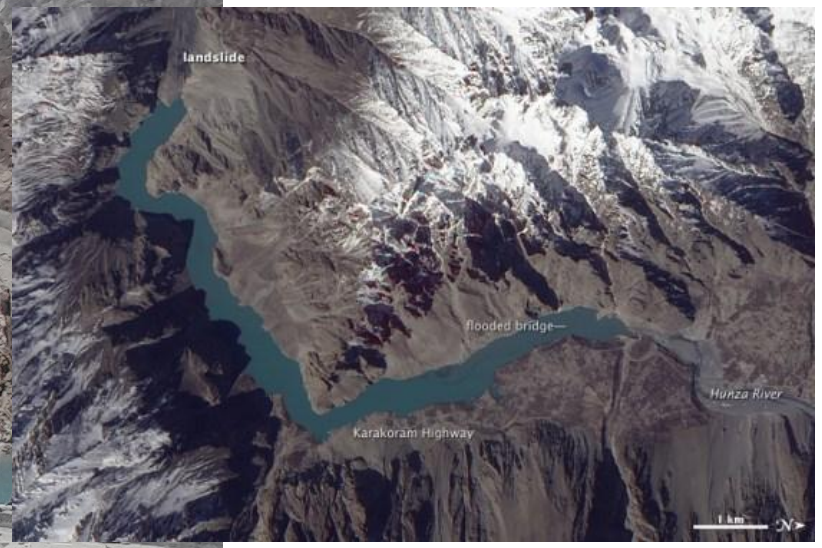
Das Extremwetter der vergangenen Wochen traf die Südkantone hart. Lokale Touristiker versprühen Zuversicht und wollen Reisende von Stornierungen abhalten.

Christopher Gilb
Publiziert: 02.07.2024, 17:55

3

In Saas-Grund sorgte das Unwetter für schlimme Verwüstungen.
Foto: Jean-Christophe Bott (Keystone)

Tourism

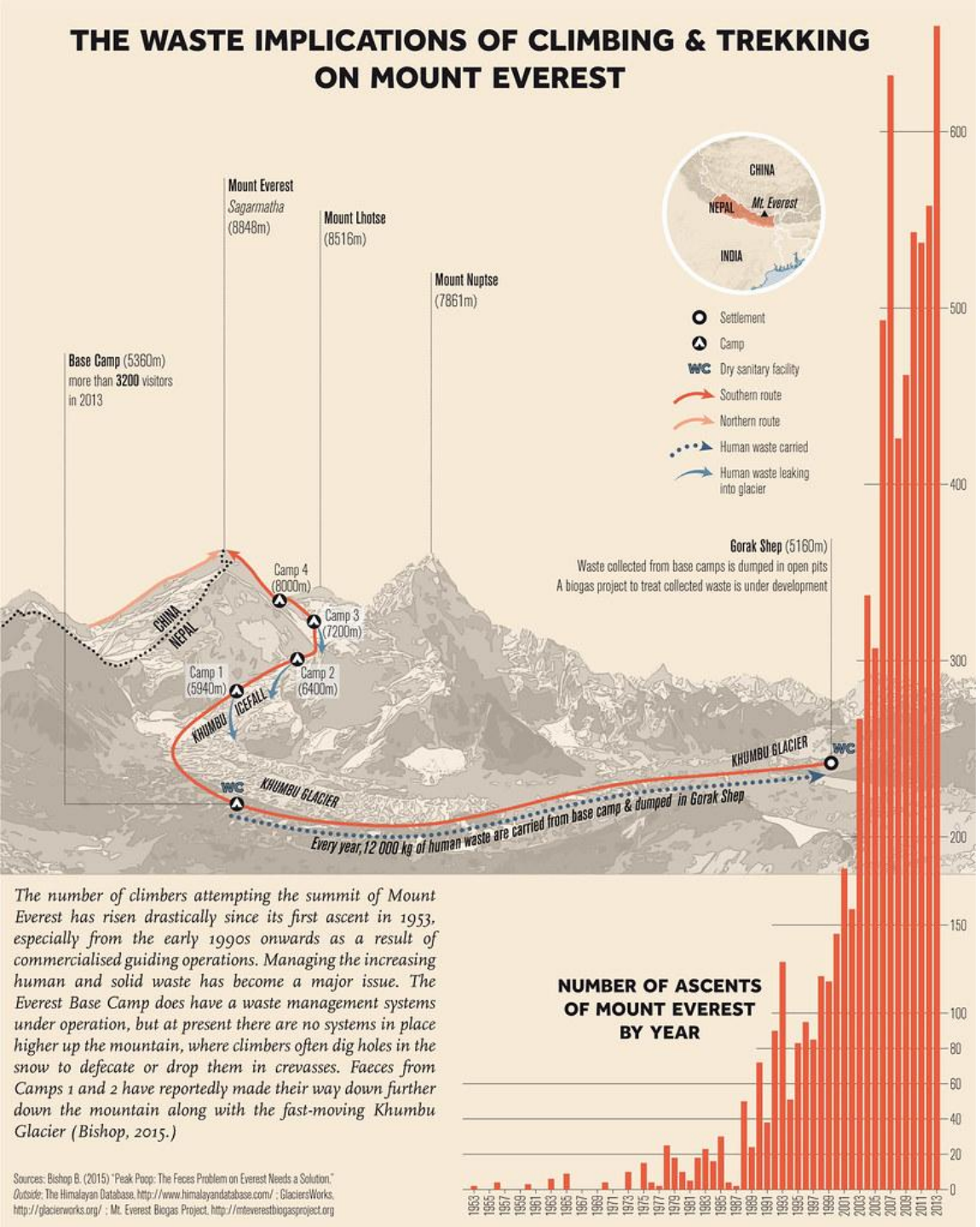


Attabad Lake, Pakistan

Tourism

Risk for tourism

Tourism as risk



Focus has been long on *risk for foreign climbers*

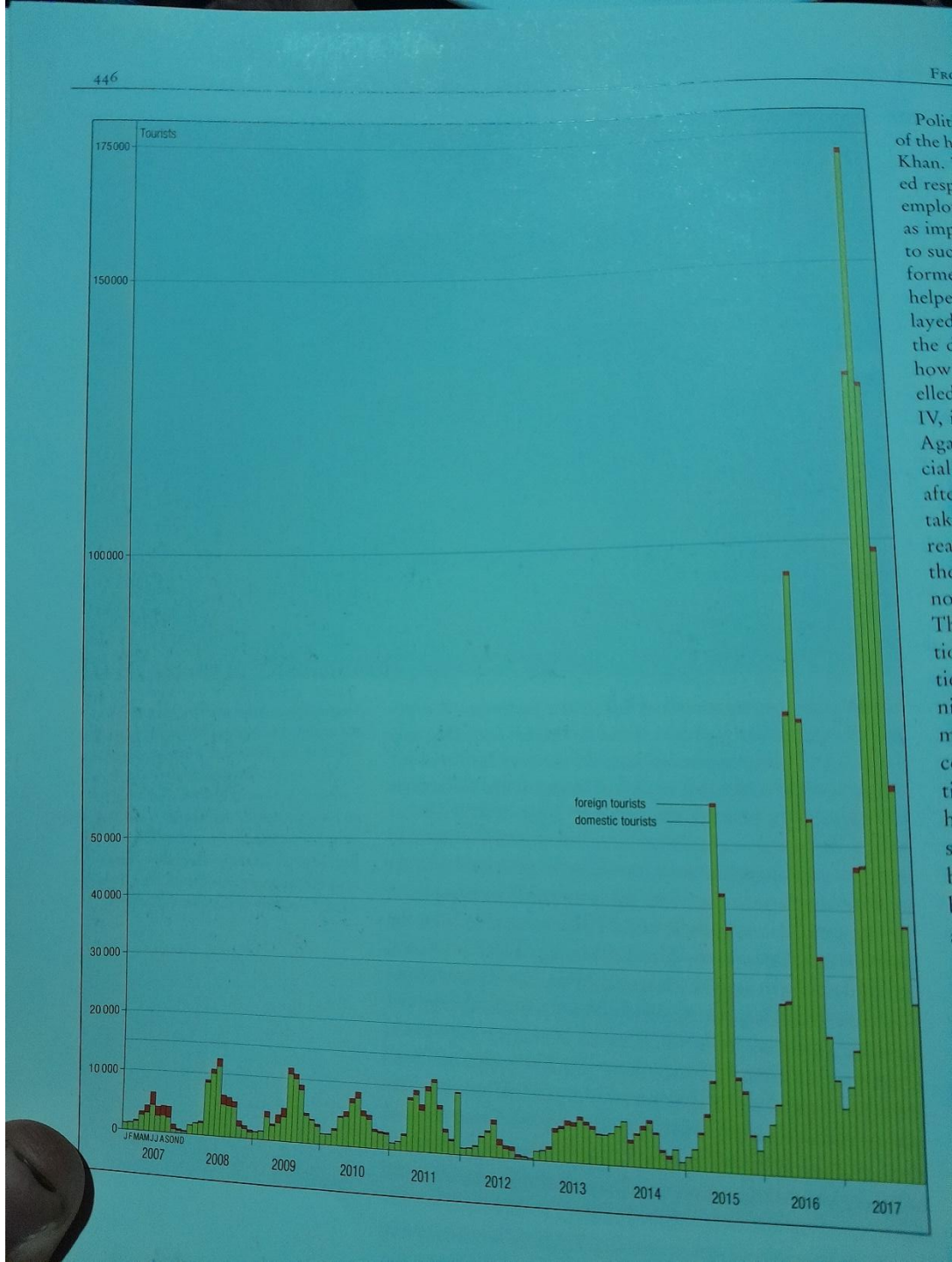
Slowly shifting to *risks posed by tourists* (overtourism)

https://www.grida.no/resources/5028

Myths of mountain tourism's relevance

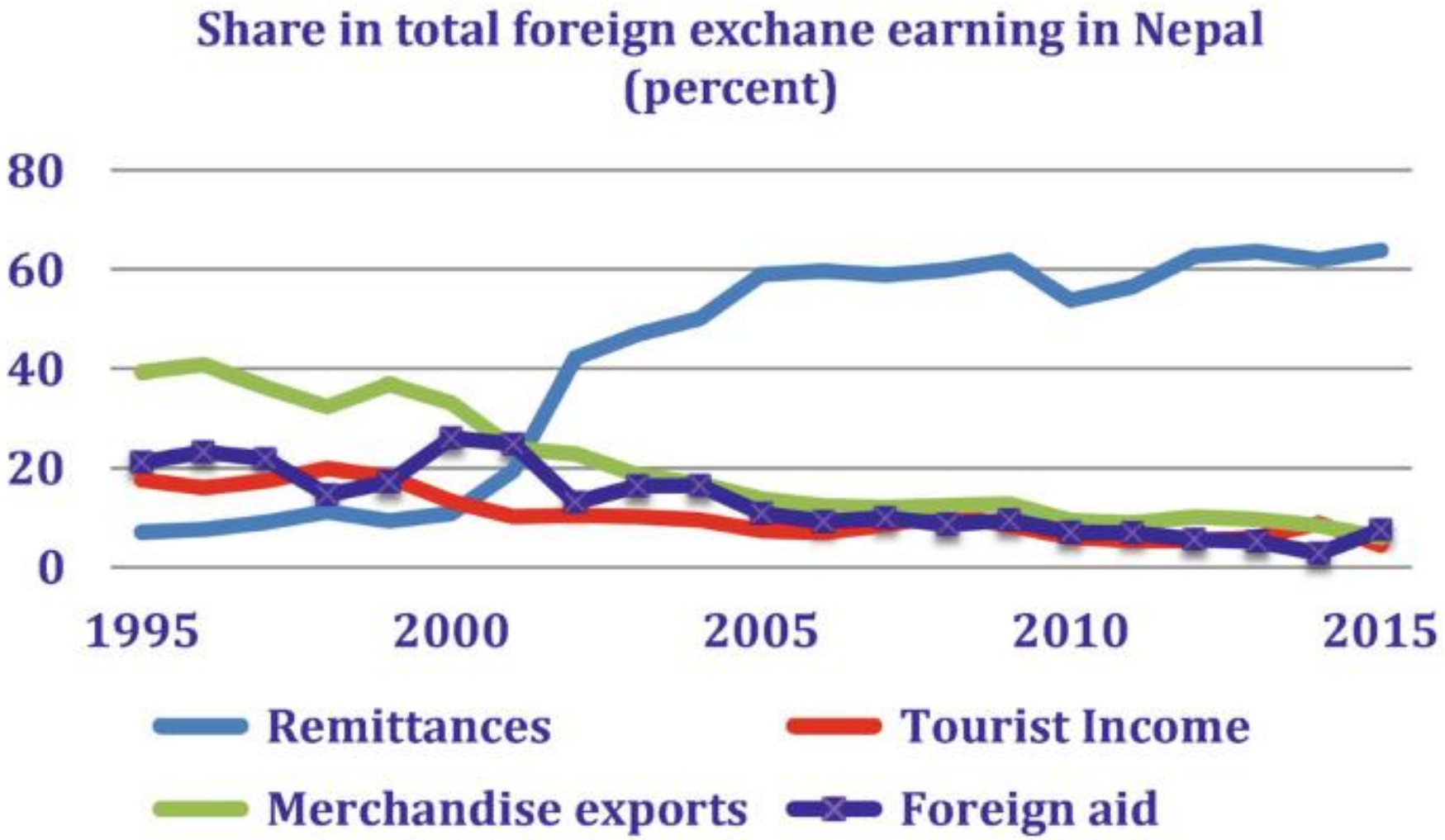
How relevant is it for the local economy?

How relevant are foreign tourists?



Total tourists Gilgit Baltistan, Pakistan, from Kreutzmann, 2020

Tourism earnings vs. remittances

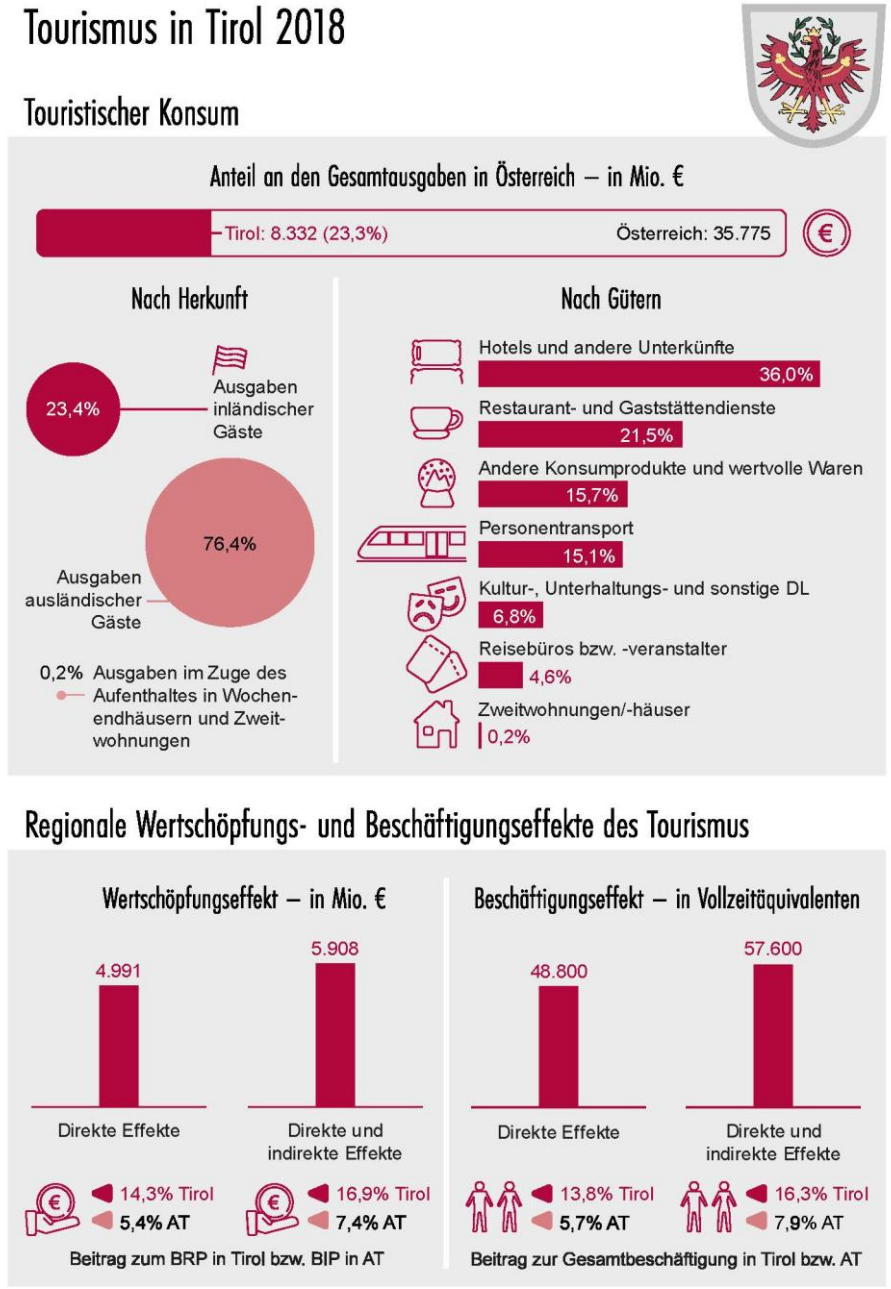


What is the role of remittances/migrant labour with respect to climate risks?

Sharma, 2019

Tourism relevance in the Alps

Similar challenges of adapting to a changing climate (no more skiing in future?, more hazards?)



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10:30 - 11:00	Slow onset hazards
11:00 - 11:15	Break
11:15 - 12:00	Multihazards, compound events and cascading risks
12:00 – 13:00	Lunch Break
13:00 - 13:30	Urbanisation, Hydropower and Tourism
13:30 – 14:00	Breakout Groups – Discuss
14:00 - 14:30	In class discussion (see Moodle)

In class discussion

We want to cover:

- Communicating mountain risks
- How do existing initiatives address mountain risks?

In class discussion

Watch any of the clips provided (max. 10 minutes) and then discuss in the breakout rooms:

a) Is the **language** used in these clips **comprehensible** to you? Would it be to anyone, for example people living in mountain environments?

b) To what degree do these initiatives or assessments address **experiences from people** living with risks in mountains and to what degree do they simply feel academic? Is there a reasonable balance?

c) Who among the people speaking addresses the issues of **long term risks**, i.e. not just the immediate impacts of a single event? Do you feel like 'risk' in its entirety is captured or are there crucial aspects missing?

d) What is **crucially missing** in the clips for you to be convinced of what they are trying to get across?

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